Financial Derivatives

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Table of contents

Preamble **10**

1 Introduction to Financial Derivatives **11**
1.1 What Are Derivatives? . . . 11
1.1.1 The Significance of Derivatives . . . 12
1.2 Derivative Markets . . . 12
1.2.1 OTC Market Evolution . . . 13
1.3 Unconditional vs. Conditional Derivatives . . . 13
1.4 Types of Derivatives . . . 14
1.4.1 Forward Contract . . . 14
1.4.2 Futures Contract . . . 14
1.4.3 Swap Contract . . . 15
1.4.4 Options . . . 15
1.5 Applications of Derivatives . . . 16
1.5.1 Hedging . . . 16
1.5.2 Arbitrage . . . 16
1.5.3 Speculation . . . 17
1.6 Criticisms and Misuses of Derivatives . . . 17
1.6.1 Risk Mitigation and Regulatory Measures . . . 17
1.7 Practice Questions and Problems . . . 18
1.7.1 Time Value of Money . . . 18
1.7.2 Theoretical Foundations of Derivatives . . . 18
1.7.3 Practical Applications of Forwards and Futures . . . 18
1.7.4 Practical Applications of Options . . . 19
1.7.5 Hedging Strategies . . . 19

2 Forwards and Futures **20**
2.1 Forwards and Futures Characteristics . . . 20
2.2 Payoff From a Forward Contract . . . 22
2.3 Exchange Markets (Futures) . . . 23
2.3.1 Market Quotes and Trading Activity . . . 24
2.3.2 Daily Settlement and Margins . . . 25
2.4 OTC Markets (Forwards) . . . 25
2.4.1 Collateral Requirements . . . 26
2.4.2 Clearing Mechanisms . . . 26

2.5 Hedging with Futures and Managing Basis Risk . . . 27
2.5.1 Understanding Basis Risk . . . 27
2.5.2 Hedging Examples . . . 28
2.6 Cross Hedging and Tailing Adjustments . . . 29
2.6.1 Optimal Hedge Ratio and Number of Contracts . . . 29
2.6.2 Daily Settlement and Tailing Adjustments . . . 30
2.7 Stock Index Futures . . . 31
2.7.1 Additional Hedging Considerations . . . 32
2.8 Practice Questions and Problems . . . 33
2.8.1 Fundamentals of Futures Trading . . . 33
2.8.2 Open Interest and Trading Volume . . . 33
2.8.3 Margin Mechanics in Futures Trading . . . 34
2.8.4 Basics of Hedging with Futures . . . 34
2.8.5 Advanced Hedging Scenarios and Strategies . . . 35
2.8.6 Practical Concerns and Risk Management . . . 36
2.8.7 Hedging with Stock and Commodity Futures . . . 36

3 Determination of Forward and Futures Prices 38
3.1 Pricing and Valuation of Forward/Futures Contracts . . . 38
3.2 Forward Price for an Investment Asset . . . 40
3.2.1 Arbitrage Example . . . 40
3.2.2 Forward Price Formula . . . 41
3.3 Valuing a Forward Contract . . . 41
3.4 Forward vs. Futures Prices . . . 42
3.5 Forward Price for Specific Assets . . . 42
3.5.1 Stock Index . . . 42
3.5.2 Exchange Rates . . . 43
3.5.3 Commodities . . . 44
3.5.4 The Cost of Carry . . . 44
3.6 Futures Prices and Expected Spot Prices . . . 45
3.7 Forward Rate Agreement (FRA) . . . 45
3.8 Practice Questions and Problems . . . 47

4 Swaps 50
4.1 Nature of Swaps . . . 50
4.2 Interest Rate Swaps . . . 51
4.2.1 Typical Applications of Interest Rate Swaps . . . 52
4.3 The Comparative-Advantage Argument . . . 52
4.3.1 Case Study: AAACorp and BBBCorp . . . 52
4.3.2 Criticism of the Comparative-Advantage Argument . . . 53
4.4 Valuation of Interest Rate Swaps . . . 53
4.4.1 Valuation Framework . . . 54
4.4.2 Swap Agreement Overview . . . 54

4.4.3 Step 1: Determine Floating Rates . . . 54
4.4.4 Step 2: Convert to Semi-Annual Compounding . . . 55
4.4.5 Step 3: Calculate and Discount Cash Flows . . . 56
4.4.6 Detailed Calculation (0.7-year exchange) . . . 56
4.4.7 Result . . . 56
4.5 Currency Swaps . . . 57
4.5.1 Comparative Advantage in Currency Swaps . . . 58
4.6 Valuation of Fixed-for-Fixed Currency Swaps . . . 59
4.6.1 Swap Agreement Overview . . . 59
4.6.2 Step 1: Calculate Forward Exchange Rates . . . 59
4.6.3 Step 2: Determine Cash Flows and Convert to USD . . . 60
4.6.4 Step 3: Discount Net Cash Flows . . . 61
4.6.5 Result . . . 61
4.7 Other Currency Swaps . . . 61
4.7.1 Fixed-for-Floating Currency Swaps . . . 61
4.7.2 Floating-for-Floating Currency Swaps . . . 62
4.8 Other Types of Swaps . . . 63
4.9 Practice Questions and Problems . . . 64

5 Options 66
5.1 Understanding Options . . . 66
5.2 Option Payoffs and Profits . . . 69
5.2.1 Understanding Option Payoffs . . . 69
5.2.2 Calculating Option Profit . . . 70
5.3 Dividend Adjustments and Related Assets . . . 71
5.3.1 Adjustments for Dividends and Stock Splits . . . 71
5.3.2 Related Assets . . . 71
5.4 Understanding the Dynamics of Option Pricing . . . 72
5.4.1 American vs. European Options . . . 73
5.5 Upper and Lower Bounds for Option Prices . . . 73
5.5.1 American Options and Early Exercise . . . 74
5.5.2 American Put Options (Non-Dividend Paying Stocks) . . . 75
5.6 Put-Call Parity . . . 75
5.6.1 Put-Call Parity for Non-Dividend Paying Stocks . . . 75
5.6.2 Extension to American Options . . . 76
5.7 Effect of Dividends on Options . . . 76
5.7.1 Adjusted Lower Bound Valuations . . . 77
5.7.2 Adjusted Put-Call Parity with Dividends . . . 77
5.8 Practice Questions and Problems . . . 77
5.8.1 Option Profitability and Exercise Conditions . . . 77
5.8.2 Margin Requirements, Market Choices, and Contract Adjustments . . . 78
5.8.3 Option Pricing Bounds . . . 78
5.8.4 Early Exercise and Put-Call Parity . . . 79

6 Options Trading Strategies and Hedging **80**
6.1 Principal Protected Note . . . 80
6.2 Combining Underlying and Options . . . 81
6.2.1 Protective Put . . . 82
6.2.2 Covered Call . . . 82
6.3 Option Spreads . . . 83
6.3.1 Bull Spread . . . 84
6.3.2 Bear Spread . . . 85
6.3.3 Butterfly Spread . . . 87
6.3.4 Key Features of Option Spreads . . . 90
6.4 Option Combinations . . . 91
6.4.1 Straddles . . . 91
6.4.2 Strangles . . . 93
6.4.3 Strips and Straps . . . 95
6.4.4 Key Characteristics . . . 95
6.5 Practice Questions and Problems . . . 96

7 Reading Week **97**

8 Option Pricing - Binomial Trees **98**
8.1 A One-Step Binomial Model . . . 98
8.1.1 A Simple Binomial Model for a Call Option . . . 99
8.1.2 General Binomial Option Pricing Framework . . . 99
8.1.3 Key Binomial Option Pricing Formulas . . . 101
8.2 Risk-Neutral Valuation Framework . . . 102
8.2.1 Core Concept . . . 102
8.2.2 Why the Real-World Expected Return Doesn't Matter . . . 103
8.3 Two-Step Binomial Trees . . . 103
8.3.1 Valuing a Call Option . . . 103
8.3.2 Valuing a Put Option . . . 104
8.3.3 Impact of the American Feature on a Put Option . . . 104
8.4 Choosing u and d for Binomial Models . . . 104
8.4.1 Cox-Ross-Rubinstein (CRR) Approach . . . 105
8.5 The Binomial Tree Model Summary . . . 105
8.5.1 Core Formulas . . . 105
8.5.2 Adjusting for Different Asset Types . . . 106
8.6 Practice Questions and Problems . . . 107

9 Option Pricing - Black-Scholes Model **109**
9.1 Black-Scholes-Merton Model . . . 109
9.1.1 Model Assumptions . . . 110
9.1.2 Black-Scholes-Merton Pricing Formulas . . . 110
9.1.3 The $N(x)$ Function . . . 111

9.2 Interpretation of $N(d_1)$ and $N(d_2)$ 112
9.2.1 Interpretation 1: Probabilistic Meaning 112
9.2.2 Interpretation 2: Weighted Expectation Form 112
9.3 Properties of the Black-Scholes Formula 113
9.3.1 When $S_0 \to \infty$ 113
9.3.2 When $S_0 \to 0$ 113
9.4 Risk-Neutral Valuation 114
9.4.1 Key Principles 114
9.4.2 How to Apply Risk-Neutral Valuation 114
9.5 The Effect of Dividends 115
9.5.1 Valuing European Options on Dividend-Paying Stocks 115
9.5.2 American Calls and Early Exercise with Dividends 115
9.5.3 Black's Approximation for American Calls with Dividends 116
9.6 Volatility in Financial Markets 116
9.6.1 Historical Volatility 116
9.6.2 Implied Volatility (IV) 117
9.6.3 Role of Implied Volatility in Option Trading 117
9.7 Practice Questions and Problems 117

10 Options on Indices, Currencies, and Futures 120
10.1 Options on Stock Indices 120
10.1.1 Using Index Options for Portfolio Insurance 121
10.2 Valuation of Stock Index Options 123
10.2.1 Lower Bounds and Put-Call Parity 123
10.2.2 Black-Scholes with Dividend Yield 124
10.2.3 Valuation Using Forward Prices 124
10.2.4 Implied Forward Prices and Dividend Yields 125
10.3 Currency Options 125
10.3.1 Range Forward Contracts 125
10.3.2 Black-Scholes for Currency Options 126
10.3.3 Valuation Using Forward Prices 126
10.4 Options on Futures and Black's Model 126
10.4.1 Mechanics of Futures Options 126
10.4.2 Futures Option Payoffs (Immediate Closeout) 127
10.4.3 Why Use Futures Options? 127
10.4.4 Put-Call Parity and Lower Bounds for European Futures Options 128
10.4.5 Black's Model for Futures Options 129
10.4.6 Futures vs. Spot Option Valuation 129
10.5 Practice Questions and Problems 129
10.5.1 Index Options 129
10.5.2 Currency Options 130
10.5.3 Futures Options 131
10.5.4 Strategic Considerations 132

11 The Greek Letters 134
11.1 Case Study . . . 134
11.1.1 Naked Position . . . 135
11.1.2 Covered Position . . . 135
11.1.3 Dynamic Hedging: Stop-Loss Strategy . . . 135
11.2 The Greek Letters . . . 136
11.2.1 Description of Key Greeks . . . 137
11.3 Understanding Delta (Δ) . . . 137
11.3.1 Delta Hedging: Principles and Practices . . . 139
11.3.2 Hedging Cost Scenarios from Case Study . . . 141
11.4 Understanding Gamma (Γ) . . . 142
11.4.1 Gamma and Hedging Dynamics . . . 143
11.5 Understanding Theta (Θ) . . . 145
11.6 Understanding Vega (ν) . . . 148
11.7 Understanding Rho (ρ) . . . 149
11.8 Strategic Management of Delta, Gamma, and Vega . . . 150
11.8.1 Hedging in Practice . . . 151
11.9 Practice Questions and Problems . . . 152

12 Structured Products I 154
12.1 Introduction to Structured Products . . . 154
12.1.1 Advantages of Investing in Structured Products . . . 154
12.1.2 European Structured Investment Products Association (EUSIPA) . . . 155
12.2 Definition and Nature of Structured Products (Investment Certificates) . . . 155
12.2.1 Credit Risk Associated with Structured Products . . . 156
12.2.2 Categorization of Structured Products . . . 156
12.2.3 Investors and Sellers of Structured Products . . . 157
12.2.4 Platforms to Find Structured Products . . . 158
12.2.5 Additional References . . . 158
12.3 Barrier Options . . . 159
12.3.1 Types of Barrier Options . . . 159
12.3.2 Pricing Dynamics . . . 160
12.3.3 Barrier Styles . . . 160
12.4 Capital Protection Investment Products . . . 161
12.4.1 Uncapped Capital Protection . . . 162
12.4.2 Capital Protection Products Modifications . . . 164
12.4.3 General Recommendations . . . 168
12.5 Behavior of Capital Guarantee Products - Case Study . . . 169
12.5.1 Impact of Volatility and Interest Rate Variations . . . 171
12.5.2 Conclusive Insights . . . 175
12.6 Participation Investment Products . . . 176
12.6.1 Tracker Certificate . . . 176
12.6.2 Bonus Certificate . . . 177

12.6.3 Twin-Win Certificate . . . 177
12.6.4 Outperformance Certificate . . . 178
12.6.5 Capped Outperformance Certificate . . . 178
12.7 Behavior of Participation Products . . . 179
12.7.1 Bonus Certificate - Case Study . . . 179
12.8 Practice Questions and Problems . . . 182

13 Structured Products II **185**
13.1 Yield Enhancement Investment Products . . . 185
13.1.1 Reverse Convertibles . . . 186
13.1.2 Worst-of Barrier Reverse Convertible . . . 186
13.1.3 Discount Certificates . . . 187
13.1.4 Strategic Variations and Recommendations . . . 187
13.2 Behavior of Yield Enhancement Products . . . 188
13.2.1 Knock-In Reverse Convertibles - Case Study . . . 188
13.2.2 Variations in Spot Price . . . 189
13.2.3 Impact of Implied Volatility Changes . . . 189
13.2.4 Understanding Implied Correlation . . . 191
13.3 Leverage Products . . . 192
13.3.1 Warrants . . . 192
13.3.2 Mini-Futures . . . 196
13.3.3 Constant Leverage Certificates . . . 197
13.3.4 Additional Links . . . 198
13.4 Common Special Features of Structured Products . . . 198
13.4.1 Autocall and Callable Options . . . 199
13.4.2 Settlement Types in Equity-Based Products . . . 201
13.4.3 Issue Minimum/Maximum Size and Liquidity Considerations . . . 202
13.5 Practice Questions and Problems . . . 204

Appendices **209**

A Time Value of Money **209**
A.1 Notation and Terminology . . . 209
A.1.1 Basic Notation and Terminology . . . 209
A.1.2 Simple Interest . . . 210
A.1.3 Discrete Compound Interest . . . 210
A.1.4 Continuous Compound Interest . . . 210
A.1.5 Effective Rate of Interest . . . 210
A.2 Future Value Examples . . . 211
A.2.1 Example 1 . . . 211
A.2.2 Example 2 . . . 211
A.2.3 Example 3 . . . 212

A.2.4 Example 4 214
A.2.5 Example 5 215
A.2.6 Example 6 216
A.2.7 Example 7 217
A.2.8 Example 8 218
A.3 Present Value Examples 219
A.3.1 Example 1 219
A.3.2 Example 2 220
A.3.3 Example 3 220
A.3.4 Example 4 221
A.4 Try It Yourself 221
A.4.1 Exercise 1 221
A.4.2 Exercise 2 221
A.4.3 Exercise 3 222
A.4.4 Exercise 4 222
A.4.5 Exercise 5 222
A.4.6 Exercise 6 222
A.4.7 Exercise 7 223
A.4.8 Exercise 8 223
A.4.9 Exercise 9 223
A.4.10 Exercise 10 223

B Interest Rates **225**
B.1 Types of Rates 225
B.1.1 Treasury Rate 225
B.1.2 The U.S. Fed Funds Rate 225
B.1.3 Repo Rate 226
B.1.4 LIBOR (ICE LIBOR) 226
B.2 Alternative Reference Rates 226
B.2.1 SOFR (Secured Overnight Financing Rate) 226
B.2.2 SONIA (Sterling Overnight Index Average) 227
B.2.3 €STR (or ESTER, Euro Short-Term Rate) 227
B.3 OIS Rate 227
B.3.1 The Risk-Free Rate 228
B.4 Time Value of Money 228
B.4.1 Compounding Frequency 228
B.4.2 Continuous Compounding 228
B.4.3 Conversion Formulas 229

Preamble

The course Financial Derivatives delves into the intricacies of derivative instruments, focusing on both fixed and conditional contracts, including forwards, futures, swaps, and options. It explores the utilization of derivatives within financial markets, particularly for hedging against various risks. Students will gain a comprehensive understanding of the essence and valuation of financial derivatives, along with detailed insights into the exchange trading mechanisms for futures and options.

Upon completing this course, the student will be able to:

- define financial derivatives and understand their specific characteristics
- explain the differences between fixed future contracts and options
- price financial derivatives
- suggest their use

References:

- HULL, John. Options, futures, and other derivatives. Global edition. Harlow: Pearson, 2018. ISBN 9781292212890.
- PIRIE, Wendy L. Derivatives. Hoboken: Wiley, 2017. xix, 597. ISBN 9781119381815.
- BLÜMKE, Andreas. How to invest in structured products : a guide for investors and investment advisors. Chichester: Wiley, 2009. xvi, 374. ISBN 9780470746790.

1 Introduction to Financial Derivatives

References

- HULL, John. Options, futures, and other derivatives. Ninth edition. Harlow: Pearson, 2018. ISBN 978-1-292-21289-0.
  - Chapter 1 - Introduction
  - Chapter 4 - Interest Rates
- PIRIE, Wendy L. Derivatives. Hoboken: Wiley, 2017. CFA institute investment series. ISBN 978-1-119-38181-5.
  - Chapter 1 - Derivative Markets and Instruments

Learning Outcomes:

- Define and explain the concept and significance of derivatives in financial markets.
- Describe the structure and function of derivative markets, including exchange-traded and OTC markets.
- Differentiate between unconditional (forward commitments) and conditional (contingent claims) derivatives.
- Identify and briefly describe the main types of derivatives: forwards, futures, swaps, and options.
- Understand the applications of derivatives in hedging, speculation, and arbitrage.
- Recognize common criticisms and potential misuses of derivatives, and understand key regulatory safeguards.

1.1 What Are Derivatives?

- A **derivative** is a financial contract whose value is derived from the price of an **underlying asset**.
- The **underlying asset** can be a stock, currency, interest rate, commodity, bond, index, or even non-financial elements like weather conditions or insurance claims.
- The market price of the underlying is called the **cash price** or **spot price**.

1.1.1 The Significance of Derivatives

Derivatives play a critical role in modern financial markets by offering several key benefits:

- **Risk Management:** Allow parties to hedge and transfer various types of financial risks.
- **Investment Strategies:** Enable the construction of complex strategies and returns beyond simple stock or bond investments.
- **Market Expectations:** Reflect future price expectations and offer valuable market information.
- **Cost Efficiency:** Reduce transaction costs compared to trading the underlying asset.
- **Capital Efficiency:** Require lower initial capital outlay due to leverage.
- **Short Selling:** Simplify taking short positions compared to shorting the underlying directly.
- **Liquidity and Market Efficiency:** Improve liquidity and enhance the functioning of the underlying asset markets.

1.2 Derivative Markets

The global derivatives market is one of the largest financial markets. Derivatives are traded in two main venues:

- **organized exchanges** and
- **over-the-counter (OTC)** markets.

Feature	Exchange-Traded Derivatives	OTC Derivatives
Contract Standardization	Standardized terms (size, expiry, asset)	Fully customizable
Clearing Mechanism	Guaranteed by a clearing house	No central clearing (except post-2008 for some)
Counterparty Risk	Low (due to clearing house)	Higher credit risk
Transparency	High (regulated, prices public)	Lower (private negotiations)
Flexibility	Limited	High
Capital and Margin Requirements	Margin required	Negotiated, often lower pre-2008

1.2.1 OTC Market Evolution

- **Prior to 2008:** Largely unregulated, dominated by banks as market makers; transactions governed by master agreements, some cleared through central counterparties (CCPs).
- **Since 2008:** Regulatory reforms introduced:
  - Standardized OTC transactions must be cleared through CCPs.
  - All trades must be reported to a central repository.
  - Aim: Reduce systemic risk and enhance transparency.

The Lehman Bankruptcy

- **Event:** Lehman Brothers filed for bankruptcy on September 15, 2008—the largest in U.S. history.
- **Role in OTC Market:** Heavily involved in high-risk derivatives.
- **Cause:** Inability to refinance short-term debt.

At the time of its bankruptcy, Lehman Brothers had an extensive network of transactions, with hundreds of thousands outstanding across approximately 8,000 counterparties. The process of unwinding these transactions has posed significant challenges for both the Lehman liquidators and the involved counterparties, illustrating the complex and interconnected nature of modern financial markets.

1.3 Unconditional vs. Conditional Derivatives

Both unconditional (Forward Commitments) and conditional derivatives (Contingent Claims) are essential financial instruments that derive their value from the performance of an underlying asset, playing pivotal roles in global financial markets for hedging, speculation, and arbitrage.

Unconditional Derivatives (Forward Commitments)

Unconditional derivatives create a binding **obligation** to buy or sell an asset at a predetermined future date and price. They include:

- Forwards
- Futures
- Swaps

Conditional Derivatives (Contingent Claims)

Conditional derivatives offer the holder **the right, but not the obligation**, to buy or sell an asset under specified conditions. The primary form is:

- Options

1.4 Types of Derivatives

1.4.1 Forward Contract

Definition

A **forward contract** is a customized, over-the-counter derivative agreement between two parties, where the buyer agrees to purchase, and the seller agrees to sell, an underlying asset at a predetermined future date and price established at the contract's inception (forward price).

- **Long Position**: The party committing to purchase the asset.
- **Short Position**: The party committing to sell the asset.

1.4.1.1 Key Points

- **Popularity in Foreign Exchange**: Forward contracts are frequently used for hedging in the foreign exchange markets.
- **OTC Markets**: Typically involves at least one financial institution, allowing for customization.
- **Market Dynamics and Pricing**: Forward prices reflect the market's consensus on future price movements, adjusted for the time value of money, influenced by factors such as the underlying asset's current price, interest rates, and expected volatility.

1.4.2 Futures Contract

Definition

A **futures contract** is a standardized derivative, similar to a forward contract but traded on futures exchanges, like the CME Group or Intercontinental Exchange, facilitating the buying and selling of underlying assets at future dates.

Forwards	Futures
Customized terms, traded over-the-counter.	Standardized terms, traded on regulated exchanges.
Counterparty risk, with less regulatory oversight.	Mitigated counterparty risk through clearinghouses.
Settlement occurs at contract maturity.	Daily mark-to-market settlement.

1.4.3 Swap Contract

Definition

A **swap** is an OTC derivative where two parties exchange cash flow series over time. It can be viewed as a series of forward contracts. Swaps address multi-period risks and are commonly used to manage interest rate, currency, or commodity exposure.

- **Key Usage**: Interest rate swaps exchange fixed for floating interest rate payments to manage interest rate risk, while currency swaps exchange cash flows in different currencies to hedge currency risk.

1.4.4 Options

Definition

Options are versatile financial derivatives allowing the holder to buy (call option) or sell (put option) an underlying asset at a predetermined price (strike price) within a specific timeframe. The buyer of the option pays a premium to the seller (writer) for this right, without the obligation to execute the transaction.

1.4.4.1 Types of Options

- **Call Options**: Grant the holder the right to **purchase** the underlying asset at the strike price. Investors buy calls when they anticipate the underlying asset's price will increase.
- **Put Options**: Provide the holder the right to **sell** the underlying asset at the strike price. Puts are purchased when an investor expects the underlying asset's price to decline.

1.4.4.2 Exercise Styles

- **American Options**: Can be exercised at any point up to and including the expiration date, offering maximum flexibility to the holder.
- **European Options**: Can only be exercised on the expiration date itself, limiting the timing of execution to this single moment.

1.5 Applications of Derivatives

1.5.1 Hedging

Hedging aims to **reduce risk** by protecting against adverse price movements. It is widely used by businesses, investors, and financial institutions to manage exposure.

- **Forward Contracts**: Lock in a future price, eliminating uncertainty.
- **Options**: Provide insurance-like protection while allowing upside potential; the cost is the premium paid.

> **i** Examples
>
> 1. **Currency Hedge:** A U.S. company expects to pay GBP 10 million in 3 months. It uses a forward contract to secure a fixed exchange rate, avoiding currency risk.
> 2. **Stock Hedge:** An investor holds 1,000 Microsoft shares at \$28 each. To protect against price drops, they buy put options with a \$27.50 strike price, limiting losses while keeping upside potential. A two-month put option costs \$1 per share.

1.5.2 Arbitrage

Arbitrage exploits **price differences** across markets to earn **risk-free profits**. It relies on the **Law of One Price**: identical assets should have the same price.

- **Buy low in one market, sell high in another**.
- Arbitrage aligns prices and improves market efficiency.

> **i** Example
>
> A stock trades at GBP 100 in London and USD 150 in New York, with the exchange rate at 1.5300 GBP/USD.
>
> - Buy in New York for USD 150.
> - Sell in London for GBP 100.

- Convert GBP 100 to USD 153 for a **risk-free profit of USD 3**.

1.5.3 Speculation

Speculators seek **profit from price movements**, accepting higher risk for potential reward.

- **Futures**: High risk and reward; obligates buying/selling at a future date.
- **Options**: Lower risk; limited loss (premium), but potential for large gains.

Examples

1. **Buying Shares:** An investor with $2,000 buys stock directly, gaining full exposure to price movements.
2. **Buying Call Options:** The investor buys call options, controlling more stock with less money; loss is limited to the premium.
3. **Futures:** The investor takes a futures position betting on a price increase; potential for large gains but also large losses.

1.6 Criticisms and Misuses of Derivatives

While derivatives are vital for risk management and price discovery, they have been criticized for their potential misuse and the risks they can pose to the financial system. Key concerns include:

1. **Excessive Speculation:** Derivatives can encourage speculative trading akin to gambling, leading to large losses when markets move unexpectedly.
2. **Systemic Risk:** Large-scale or complex derivative positions can amplify financial instability. Failures can spread through interconnected markets, as seen in the 2008 crisis.
3. **Complexity:** Many derivatives are difficult to understand and value, even for experienced investors, increasing the risk of mispricing and misuse.
4. **Blurring of Roles:** Market participants may shift between hedging, speculation, and arbitrage, leading to excessive risk-taking under the guise of risk management.
5. **Need for Regulation:** Without adequate oversight, derivatives can be misused, increasing market instability. Strong controls ensure derivatives serve their intended purpose.

1.6.1 Risk Mitigation and Regulatory Measures

- **Central Clearing Counterparties (CCPs):** Reduce counterparty risk and increase transparency by acting as intermediaries.

- **Margin Requirements:** Ensure traders hold sufficient capital to cover potential losses.
- **Post-2008 Reforms:** Regulations like the **Dodd-Frank Act (U.S.)** and **EMIR (EU)** enhance market transparency, reduce systemic risk, and promote responsible use of derivatives.

1.7 Practice Questions and Problems

1.7.1 Time Value of Money

1. A bank quotes an interest rate of 7% per annum with quarterly compounding. How much you will earn from $100 investment after (a) 1 year and (b) 3 years? What is the equivalent rate with (a) continuous compounding and (b) annual compounding? Verify your results.

1.7.2 Theoretical Foundations of Derivatives

1. Explain carefully the difference between hedging, speculation, and arbitrage.
2. What is the difference between the over-the-counter market and the exchange-traded market?
3. "Options and futures are zero-sum games." What do you think is meant by this statement?
4. What is the difference between a long forward position and a short forward position?
5. What is the difference between entering into a long forward contract when the forward price is $50 and taking a long position in a call option with a strike price of $50?
6. Explain carefully the difference between selling a call option and buying a put option.
7. When first issued, a stock provides funds for a company. Is the same true of an exchange-traded stock option? Discuss.

1.7.3 Practical Applications of Forwards and Futures

1. A trader enters into a short cotton futures contract when the futures price is 50 cents per pound. The contract is for the delivery of 50,000 pounds. How much does the trader gain or lose if the cotton price at the end of the contract is (a) 48.20 cents per pound; (b) 51.30 cents per pound?
2. An investor enters into a short forward contract to sell 100,000 British pounds for US dollars at an exchange rate of 1.5000 US dollars per pound. How much does the investor gain or lose if the exchange rate at the end of the contract is (a) 1.4900 and (b) 1.5200?

1.7.4 Practical Applications of Options

1. A trader buys a call option with a strike price of $30 for $3. Does the trader ever exercise the option and lose money on the trade. Explain.
2. Suppose that you write a put contract with a strike price of $40 and an expiration date in three months. The current stock price is $41 and the contract is on 100 shares. What have you committed yourself to? How much could you gain or lose?
3. Suppose you own 5,000 shares that are worth $25 each. How can put options be used to provide you with insurance against a decline in the value of your holding over the next four months?
4. You would like to speculate on a rise in the price of a certain stock. The current stock price is $29, and a three-month call with a strike of $30 costs $2.90. You have $5,800 to invest. Identify two alternative strategies, one involving an investment in the stock and the other involving investment in the option. What are the potential gains and losses from each?

1.7.5 Hedging Strategies

1. Explain why a futures contract can be used for either speculation or hedging.
2. A US company expects to have to pay 1 million Canadian dollars in six months. Explain how the exchange rate risk can be hedged using (a) a forward contract and (b) an option.
3. The CME Group offers a futures contract on long-term Treasury bonds. Characterize the investors likely to use this contract.

2 Forwards and Futures

References

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  - Chapter 1 - Derivative Markets and Instruments

Learning Outcomes:

- Understand forwards and futures, including their characteristics, payoff structures, and key differences.
- Gain knowledge of exchange and over-the-counter (OTC) markets, focusing on their functionalities and distinctions.
- Learn the basics of hedging with futures, managing basis risk, and applying cross hedging techniques.
- Explore the role of stock index futures in portfolio risk management and speculation.

2.1 Forwards and Futures Characteristics

Definition

Forwards and futures are agreements where two parties **commit to** exchange an asset at a set price on a future date.

Forward/Futures Price:

- The agreed-upon price is fixed when the contract starts.
- Different expiration dates can lead to different prices based on market expectations.

Positions:

- **Long Position:** The buyer promises to purchase the asset.
  - Benefits if the asset's price goes up.
- **Short Position:** The seller promises to sell the asset.
  - Benefits if the asset's price goes down.

Risk Exposure:

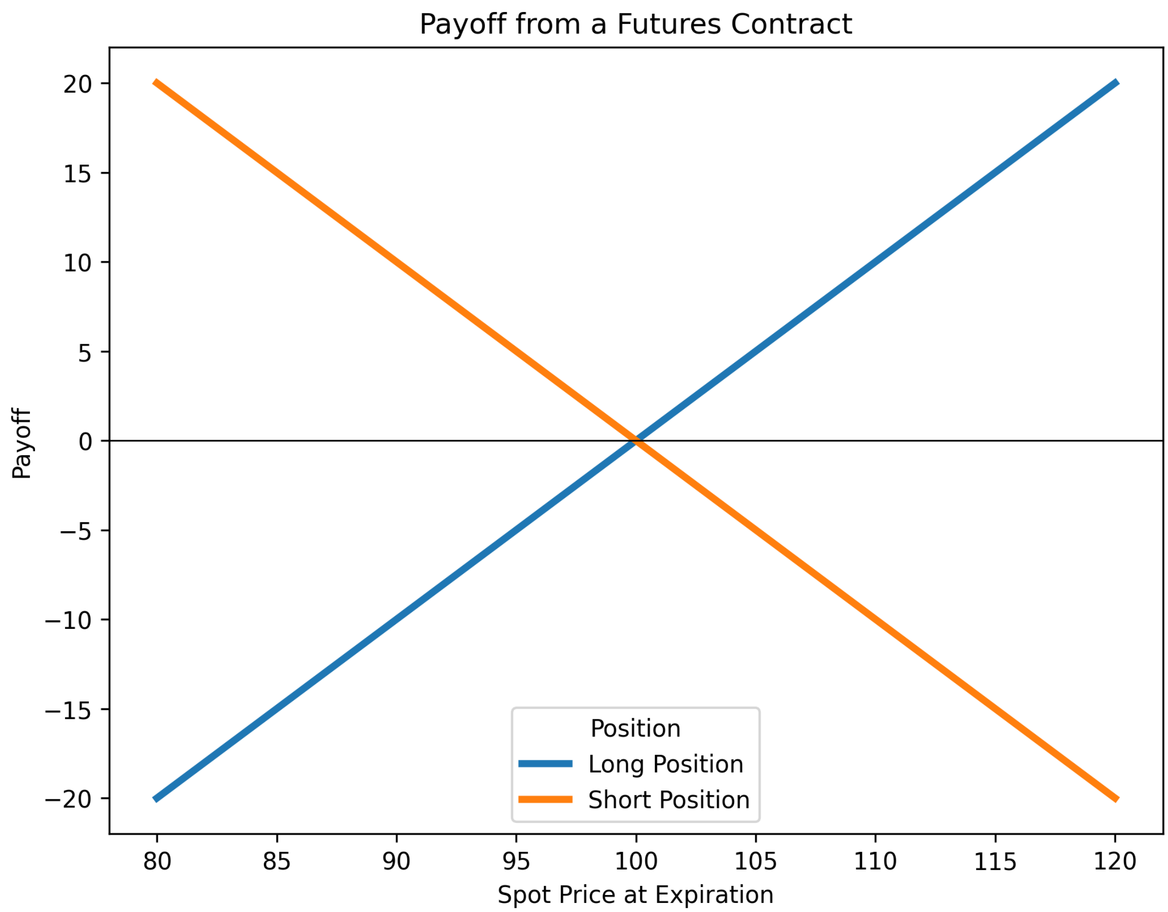
- **Long Position:** Loss can be large if the asset's price drops significantly.
- **Short Position:** Loss can be unlimited if the asset's price rises dramatically.

Contract Details:

- **Asset Specification:** What asset is being traded and any quality standards.
- **Delivery Terms:** Where and when the asset will be delivered.

Settlement Methods:

- **Futures Contracts:** Settled daily (marking to market).
- **Forward Contracts:** Settled at the end of the contract term by comparing the agreed price with the final market price.



2.2 Payoff From a Forward Contract

- **Start of Contract:** Time $= 0$, with an initial spot price S_0.
- **End of Contract:** Time $= T$, with a final spot price S_T.
- **Forward Price:** The fixed price agreed upon, F_0.

Long Position (Buyer):

- Profit if S_T is higher than F_0; loss if it is lower.

$$\text{Payoff}_{\text{Long}} = S_T - F_0$$

Short Position (Seller):

- Profit if S_T is lower than F_0; loss if it is higher.

$$\text{Payoff}_{\text{Short}} = -(S_T - F_0)$$

Example: Gold Purchase Agreement

Barbara Nix agreed to buy 1 kilogram of gold at a price of $38,000 per kilo from Metals Inc. in 90 days. After 90 days, the spot price of gold reached $38,500 per kilo. What is the payoff for each party?

- **For the Long Party (Barbara Nix):** The gain is $500, calculated as

$$S_T - F_0 = \$38,500 - \$38,000 = \$500$$

- **For the Short Party (Metals Inc.):** The loss is $500

$$-(S_T - F_0) = -(\$38,500 - \$38,000) = -\$500$$

2.3 Exchange Markets (Futures)

Exchange markets provide a regulated, **standardized** environment for trading futures contracts. They enable participants to hedge risks or speculate on market movements.

- **Regulation:**
  The Commodity Futures Trading Commission (CFTC) in the U.S. oversees these markets to protect participants and ensure fair trading.
- **Contract Specifications:**
  Futures contracts are standardized, clearly defining the asset, quality, delivery location, and date. For example, see CME Group's Gold Futures.
- **Settlement:**
  Most futures contracts are closed out before maturity, avoiding physical delivery through offsetting trades.
  - **Physical Delivery:** When required, contracts specify the deliverable asset, delivery location, and timeframe. Typically, the short (seller) sets the delivery details.
  - **Cash Settlement:** Some contracts, such as stock index futures, settle in cash rather than through the physical delivery of assets.

2.3.1 Market Quotes and Trading Activity

- **Settlement Price:** The closing price used for daily settlements.
- **Open Interest:** The total number of outstanding contracts (either long or short), reflecting market liquidity.
- **Trading Volume:** The total number of contracts traded during a day, regardless of changes in open interest.

Example of Open Interest and Trading Volume

Imagine a wheat futures market that starts the day with **1,000 open contracts**. During the day, the following transactions occur:

1. **Transaction 1:** A new buyer and a new seller agree on a contract.
   - **Open Interest:** Increases by 1 (total: 1,001).
   - **Trading Volume:** +1 contract.
2. **Transaction 2:** An existing long position is sold to a new participant.
   - **Open Interest:** Remains at 1,001 (one party is simply replaced).
   - **Trading Volume:** +1 contract.
3. **Transaction 3:** Two new participants (a buyer and a seller) enter with 5 new contracts.
   - **Open Interest:** Increases by 5 (total: 1,006).
   - **Trading Volume:** +5 contracts.
4. **Transaction 4:** An existing buyer and seller close their positions with an offsetting contract.
   - **Open Interest:** Decreases by 1 (total: 1,005).
   - **Trading Volume:** +1 contract.

Final Totals:

- **Open Interest:** 1,005 contracts (active positions at the end of the day).
- **Trading Volume:** 8 contracts traded throughout the day.

This example shows that **open interest** reflects the number of active contracts, while **trading volume** is the total number of contracts exchanged during the day.

2.3.2 Daily Settlement and Margins

- **Mark-to-Market:**
  The clearinghouse adjusts each trader's margin account daily to reflect gains or losses based on the settlement price.
- **Margin Requirements:**
  - **Initial Margin:** The deposit required to open a position.
  - **Maintenance Margin:** The minimum balance that must be maintained. Falling below this triggers a **margin call** (additional funds must be deposited to meet the initial margin requirement).

Example: Margin and Margin Call

Alex enters a crude oil futures contract valued at \$60,000.

- **Initial Margin (5%):** \$3,000
- **Maintenance Margin (3%):** \$1,800

If the contract value drops to \$58,000, Alex loses \$2,000, reducing his margin account to \$1,000—below the maintenance margin. He then receives a margin call and must deposit an additional \$2,000 to restore his account to \$3,000.
If Alex fails to meet the margin call, his position may be closed out by the broker to limit further losses.

Some Useful Links

- Margin: Know What's Needed
- The Benefits of Futures Margins
- Understanding Futures Margin

2.4 OTC Markets (Forwards)

Over-the-Counter (OTC) markets allow forward contracts and other derivatives to be traded directly between two parties without an exchange. This setup offers **flexibility and customization** but traditionally lacks the transparency and regulation found in exchange-traded markets.

Pre-Crisis Era:

- OTC markets operated with minimal oversight, contributing to systemic risk before the 2007–2008 financial crisis.

Post-Crisis Reforms:

- In response to the crisis, regulations such as the Dodd-Frank Act (U.S.) and EMIR (EU) were introduced.

- These reforms require reporting OTC transactions and often mandate central clearing to enhance transparency and reduce risk.

2.4.1 Collateral Requirements

To manage counterparty risk in OTC markets, collateral is required:

- **Initial Margin:**
  - An upfront deposit to cover potential losses immediately after a default.
- **Variation Margin:**
  - Additional funds required to adjust for daily price changes, ensuring the collateral always matches the current exposure.

2.4.2 Clearing Mechanisms

Bilateral Clearing:

- Each transaction is settled directly between the two parties.

- This method allows customization but lacks centralized risk management.

Central Clearing:

- A Central Counterparty (CCP) steps in between the buyer and seller, reducing counterparty risk by standardizing margin requirements and daily mark-to-market adjustments.
- Although OTC contracts are customized, central clearing helps bring some uniformity to risk management and settlement processes.

2.5 Hedging with Futures and Managing Basis Risk

Hedging with futures is a strategy used to lock in prices for assets you plan to buy or sell in the future, helping to manage price volatility in commodities, currencies, interest rates, and more.

- **Long Futures Hedge:**
  Used when you expect to purchase an asset. Locking in a price now protects against future price increases.
- **Short Futures Hedge:**
  Used when you plan to sell an asset. Locking in a price now protects against future price decreases.

Advantages of hedging:

- **Business Focus:** Reduces exposure to market risks, allowing companies to concentrate on their core activities.
- **Cost Predictability:** Helps with budgeting by stabilizing costs and revenues.
- **Risk Management:** Mitigates risks from volatile market factors.

Disadvantages of hedging:

- **Shareholder Autonomy:** Some argue that shareholders can manage risks through personal diversification.
- **Competitive Dynamics:** Hedging may add extra risk if competitors do not hedge.
- **Complexity:** Hedges can be difficult to implement and explain, especially if outcomes differ from expectations.

2.5.1 Understanding Basis Risk

> Definition
>
> Basis risk is the uncertainty that the difference between the current (spot) price and the futures price will not behave as expected when the hedge is closed.

This risk arises from:

1. **Asset Mismatch:** The asset being hedged may not exactly match the underlying asset of the futures contract.
2. **Timing Uncertainty:** Variations in the timing of the actual purchase or sale can affect the hedge.
3. **Early Contract Closure:** Exiting a futures contract before its delivery date can alter the expected price difference.

Mitigation Strategies:

- **Delivery Month Selection:** Choose a contract with a delivery month as close as possible to, but after, your expected transaction date.
- **Cross Hedging:** If a direct futures contract isn't available, use a contract with a price that closely correlates with the asset's price.

2.5.2 Hedging Examples

Long Hedge Example for Asset Purchase

- Initial Futures Price (F_1): 88.0
- Futures Price at Purchase (F_2): 89.1
- Spot Price at Purchase (S_2): 90.0
- Basis at Purchase (b_2): $S_2 - F_2 = 0.9$

Outcome:

- Asset cost: \$90.0
- Gain from futures: $F_2 - F_1 = 1.1$
- **Net cost:** $\$90.0 - \$1.1 = \$88.9$

Short Hedge Example for Asset Sale

- Initial Futures Price (F_1): 0.98
- Futures Price at Sale (F_2): 0.925
- Spot Price at Sale (S_2): 0.92
- Basis at Sale (b_2): $S_2 - F_2 = -0.005$

Outcome:

- Asset sale price: \$0.92
- Gain from futures: $F_1 - F_2 = 0.055$

- **Net sale price:** $\$0.92 + \$0.055 = \$0.975$

2.6 Cross Hedging and Tailing Adjustments

Definition

Cross hedging is used when no futures market exists for the asset you want to hedge. Instead, you use a futures contract on a correlated asset. The effectiveness of a cross hedge depends on how closely the prices of the two assets move together.

2.6.1 Optimal Hedge Ratio and Number of Contracts

The **optimal hedge ratio** (h^*) without daily settlement adjustments is given by:

$$h^* = \rho \frac{\sigma_S}{\sigma_F}$$

- ρ is the correlation coefficient between changes in the spot price (ΔS) and the futures price (ΔF).
- σ_S is the standard deviation of the changes in the spot price.
- σ_F is the standard deviation of the changes in the futures price.

Once h^* is determined, the **optimal number of futures contracts** (N^*) needed is calculated as:

$$N^* = h^* \times \frac{Q_A}{Q_F}$$

- Q_A is the size of the position being hedged (in units).
- Q_F is the size of one futures contract (in units).

Example: Hedging Jet Fuel with Heating Oil Futures

An airline plans to purchase 2 million gallons of jet fuel in one month and decides to hedge using heating oil futures.

- $\sigma_F = 0.0313$, $\sigma_S = 0.0263$, and $\rho = 0.928$

- Compute the hedge ratio:

$$h^* = 0.928 \times \frac{0.0263}{0.0313} \approx 0.78$$

- One heating oil futures contract covers 42,000 gallons.

- Optimal number of contracts:

$$N^* = 0.78 \times \frac{2,000,000}{42,000} \approx 37$$

Thus, the airline should use about 37 heating oil futures contracts.

2.6.2 Daily Settlement and Tailing Adjustments

Futures contracts are settled daily, which can affect the hedge's performance. To account for these daily changes, we use an **adjusted optimal hedge ratio** ($\hat{h}$):

$$\hat{h} = \hat{\rho}\frac{\hat{\sigma_S}}{\hat{\sigma_F}}$$

- $\hat{\rho}$ is the correlation between the daily percentage changes in spot and futures prices.
- $\hat{\sigma_S}$ and $\hat{\sigma_F}$ are the standard deviations of these daily percentage changes.

Using $\hat{h}$, the adjusted **optimal number of futures contracts** is:

$$N^* = \hat{h} \times \frac{V_A}{V_F}$$

- V_A is the total value of the position being hedged (spot price × quantity).
- V_F is the value of one futures contract (futures price × contract size).

Example: Hedging with Tailing Adjustments

A transportation company, TransCo, needs to purchase 500,000 gallons of diesel in one month. With no diesel futures available, it hedges using heating oil futures (each covering 42,000 gallons).

- Diesel spot price: \$3.00 per gallon

- Heating oil futures price: \$2.90 per gallon

- Correlation of daily changes, $\hat{\rho}$: 0.9

- Daily volatility: $\widehat{\sigma_S} = 1.5\%$ for diesel, $\widehat{\sigma_F} = 1.8\%$ for heating oil

Step 1: Calculate the adjusted hedge ratio

$$\hat{h} = 0.9 \times \frac{0.015}{0.018} = 0.75$$

Step 2: Determine the values

$$V_A = \$3.00 \times 500,000 = \$1,500,000$$

$$V_F = \$2.90 \times 42,000 = \$121,800$$

Step 3: Calculate the optimal number of contracts

$$N^* = 0.75 \times \frac{1,500,000}{121,800} \approx 9.23$$

Since partial contracts aren't possible, TransCo would hedge with 9 contracts to avoid over-hedging.

2.7 Stock Index Futures

Stock index futures are powerful tools for managing portfolio risk. They allow investors to hedge against market fluctuations or adjust their portfolio's sensitivity to market movements without altering its actual composition.

To determine the number of futures contracts needed, use:

$$N^* = (\beta^* - \beta) \times \frac{V_A}{V_F}$$

- V_A: Portfolio value.
- β: Current portfolio beta.
- β^*: Target beta (if fully hedging, $\beta^* = 0$).
- V_F: Value of one futures contract (e.g., futures price multiplied by the contract multiplier).
- A negative N^* means shorting futures; a positive value means going long.

This approach allows investors to either fully hedge their portfolio ($\beta^* = 0$) or fine-tune their market exposure based on strategic objectives.

Example: Hedging a Portfolio

A portfolio valued at $5 million has a beta of 1.5. Using S&P 500 futures, where each contract is worth $250,000, to fully hedge ($\beta^* = 0$):

$$N^* = (0 - 1.5) \times \frac{5,000,000}{250,000} = -30$$

Shorting 30 futures contracts would hedge the portfolio against market fluctuations.

Example: Adjusting Portfolio Beta

A portfolio manager wants to lower a $10 million portfolio's beta from 1.2 to 0.8. Using an index futures contract priced at 3,000 points with a multiplier of $250 (thus, $V_F = 250 \times 3,000 = \$750,000$):

$$N^* = (0.8 - 1.2) \times \frac{10,000,000}{750,000} = -0.4 \times 13.33 \approx -5.33$$

Rounding to -5 indicates shorting 5 contracts to reduce the portfolio beta.

2.7.1 Additional Hedging Considerations

2.7.1.1 Why Hedge with Stock Index Futures?

- **Market Timing:** Allows an exit from market exposure without selling underlying assets, avoiding transaction costs and capital gains taxes.
- **Risk Management:** Offers precise control over market risk.
- **Performance Focus:** Helps ensure returns are driven by the performance of selected assets rather than broad market movements.

Tip

Imagine your portfolio's stocks have an average beta of 1.0, aligning their performance with the market. Yet, you're confident these stocks will surpass market returns in any scenario. By hedging, you secure returns at the risk-free rate plus any outperformance of your stocks over the market. This strategy minimizes market volatility's impact, ensuring your portfolio's gains are primarily due to your stock selection skills.

2.7.1.2 Stack and Roll Strategy

This strategy involves rolling futures contracts forward to maintain continuous hedging:

- **Process:**
  - Initiate futures contracts for a specific time horizon.
  - Before maturity, close out the contracts and replace them with new ones.
- **Risks:**
  - Liquidity issues and the possibility of realizing losses on the hedge while gains in the underlying assets remain unrealized.

> ⚠ Warning
>
> The Metallgesellschaft (MG) case in the early 1990s is a reminder that rolling hedges can lead to severe liquidity issues. MG's strategy of hedging long-term exposure with short-dated futures resulted in massive margin calls and a \$1.33 billion loss when oil prices dropped, forcing the closure of hedge positions.

2.8 Practice Questions and Problems

2.8.1 Fundamentals of Futures Trading

1. What are the most important aspects of the design of a new futures contract?
2. The party with a short position in a futures contract sometimes has options as to the precise asset that will be delivered, where delivery will take place, when delivery will take place, and so on. Do these options increase or decrease the futures price? Explain your reasoning.
3. What do you think would happen if an exchange started trading a contract in which the quality of the underlying asset was incompletely specified?
4. "Speculation in futures markets is pure gambling. It is not in the public interest to allow speculators to trade on a futures exchange." Discuss this viewpoint.

2.8.2 Open Interest and Trading Volume

1. Distinguish between the terms open interest and trading volume.
2. "When a futures contract is traded on the floor of the exchange, it may be the case that the open interest increases by one, stays the same, or decreases by one." Explain this statement.
3. Why does the open interest usually decline during the month preceding the delivery month?

4. On a particular day, there were 2,000 trades in a particular futures contract. This means that there were 2000 buyers (going long) and 2000 sellers (going short). Of the 2,000 buyers, 1,400 were closing out positions and 600 were entering into new positions. Of the 2,000 sellers, 1,200 were closing out positions and 800 were entering into new positions. What is the impact of the day's trading on open interest?

2.8.3 Margin Mechanics in Futures Trading

1. Explain how margin accounts protect investors against the possibility of default.
2. Suppose that you enter into a short futures contract to sell July silver for $17.20 per ounce. The size of the contract is 5,000 ounces. The initial margin is $4,000, and the maintenance margin is $3,000. What change in the futures price will lead to a margin call? What happens if you do not meet the margin call?

> **Solution**
>
> $S = \$17.40$

3. A trader buys two July futures contracts on frozen orange juice. Each contract is for the delivery of 15,000 pounds. The current futures price is 160 cents per pound, the initial margin is $6,000 per contract, and the maintenance margin is $4,500 per contract. What price change would lead to a margin call? Under what circumstances could $2,000 be withdrawn from the margin account?

> **Solution**
>
> $S = 166.67$ cents

4. A company enters into a short futures contract to sell 5,000 bushels of wheat for 750 cents per bushel. The initial margin is $3,000 and the maintenance margin is $2,000. What price change would lead to a margin call? Under what circumstances could $1,500 be withdrawn from the margin account?

> **Solution**
>
> Margin call: $S = 770$ cents; Withdraw \$1500: $S = 720$ cents

2.8.4 Basics of Hedging with Futures

1. Under what circumstances are (a) a short hedge and (b) a long hedge appropriate?
2. Explain what is meant by basis risk when futures contracts are used for hedging.

3. Explain what is meant by a perfect hedge. Does a perfect hedge always lead to a better outcome than an imperfect hedge? Explain your answer.
4. Does a perfect hedge always succeed in locking in the current spot price of an asset for a future transaction? Explain your answer.
5. "For an asset where futures prices for contracts on the asset are usually less than spot prices, long hedges are likely to be particularly attractive." Explain this statement.

2.8.5 Advanced Hedging Scenarios and Strategies

1. Sixty futures contracts are used to hedge an exposure to the price of silver. Each futures contract is on 5,000 ounces of silver. At the time the hedge is closed out, the basis is \$0.20 per ounce. What is the effect of the basis on the hedger's financial position if (a) the trader is hedging the purchase of silver and (b) the trader is hedging the sale of silver?
2. In the corn futures contract, the following delivery months are available: March, May, July, September, and December. State the contract that should be used for hedging when the expiration of the hedge is in a) June, b) July, and c) January.
3. Suppose that the standard deviation of quarterly changes in the prices of a commodity is \$0.65, the standard deviation of quarterly changes in a futures price on the commodity is \$0.81, and the coefficient of correlation between the two changes is 0.8. What is the optimal hedge ratio for a three-month contract? What does it mean?

> **Solution**
>
> $h = 64.2\%$

4. The standard deviation of monthly changes in the spot price of live cattle is 1.2 (in cents per pound). The standard deviation of monthly changes in the futures price of live cattle for the closest contract is 1.4. The correlation between the futures price changes and the spot price changes is 0.7. It is now October 15. A beef producer is committed to purchasing 200,000 pounds of live cattle on November 15. The producer wants to use the December live-cattle futures contracts to hedge its risk. Each contract is for the delivery of 40,000 pounds of cattle. What strategy should the beef producer follow?

> **Solution**
>
> Long 3 contracts.

2.8.6 Practical Concerns and Risk Management

1. Give three reasons why the treasurer of a company might not hedge the company's exposure to a particular risk.
2. A corn farmer argues "I do not use futures contracts for hedging. My real risk is not the price of corn. It is that my whole crop gets wiped out by the weather." Discuss this viewpoint. Should the farmer estimate his or her expected production of corn and hedge to try to lock in a price for expected production?
3. Imagine you are the treasurer of a Japanese company exporting electronic equipment to the United States. Discuss how you would design a foreign exchange hedging strategy and the arguments you would use to sell the strategy to your fellow executives.
4. A futures contract is used for hedging. Explain why the daily settlement of the contract can give rise to cash flow problems.

2.8.7 Hedging with Stock and Commodity Futures

1. A company has a $20 million portfolio with a beta of 1.2. It would like to use futures contracts on a stock index to hedge its risk. The index futures is currently standing at 1080, and each contract is for delivery of $250 times the index. What is the hedge that minimizes risk? What should the company do if it wants to reduce the beta of the portfolio to 0.6? What should the company do if it wants to increase the beta of the portfolio to 1.5?

> Solution
>
> - short 89 contracts
> - short 44 contracts to reduce beta
> - long 22 contracts to increase beta

2. On July 1, an investor holds 50,000 shares of a certain stock. The market price is $30 per share. He decides to use the September Mini S&P 500 futures contract. The index is currently 1,500 and one contract is for delivery of $50 times the index. The beta of the stock is 1.3. What strategy should the investor follow? Under what circumstances will it be profitable?

> Solution
>
> Short 26 contracts.

3. It is now June. A company knows that it will sell 5,000 barrels of crude oil in September. It uses the October CME Group futures contract to hedge the price it will receive. Each

contract is on 1,000 barrels of “light sweet crude”. What position should it take? What price risks is it still exposed to after taking the position?

3 Determination of Forward and Futures Prices

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  - Chapter 2 - Basics of Derivative Pricing and Valuation
  - Chapter 3 - Pricing and Valuation of Forward Commitments

Learning Outcomes:

- Understand the principles of pricing and valuation for forward and futures contracts.
- Apply pricing models to specific assets, including commodities and financial instruments.
- Analyze the relationship between futures prices and expected spot prices.
- Understand the structure and valuation of Forward Rate Agreements (FRAs).

3.1 Pricing and Valuation of Forward/Futures Contracts

- **Pricing** determines the fair market rate at contract **initiation**, ensuring neutrality for both parties.
- **Valuation** assesses the contract's value **after initiation**, fluctuating based on market conditions.

Example: Forward Price vs. Value

- On November 2, an investor takes a **long position** in a December Gold futures contract at **$1,250/oz**. At initiation, the contract's value is **$0** (fair market agreement).

- On November 3, if the market price drops to **$1,225/oz**, the contract's value turns **negative**, reflecting a loss for the investor.

- **Risk Aversion:** Investors require compensation for taking on additional risk, reflecting the principle that higher risk should be rewarded with higher potential returns.
- **Risk-Neutral Pricing:** Under this approach, it's assumed that investors are indifferent to risk. The pricing of derivatives through arbitrage opportunities ensures that the portfolio, combining the derivative and its underlying asset, yields the risk-free rate of return.
- **Arbitrage-Free Pricing:** This methodology prices derivatives based on the assumption that the market operates under risk-neutral conditions and is free of arbitrage opportunities, ensuring no risk-free profits can be made from market inefficiencies.
- **Law of One Price:** asserts that identical cash flows must have the same price, irrespective of future outcomes.
- **Arbitrage Principles:**

  1. Do not use your own money.
  2. Do not take any price risk.

- **Short Selling:** Short selling involves selling borrowed securities, aiming to repurchase them at a lower price. Sellers must cover dividends and borrowing fees.

Example: Short Selling

Short 100 shares at **$100 each** and close the position at **$90**, while a **$3/share** dividend is issued:

- **Profit Calculation:**

$$100 \times (100 - 90 - 3) = 700$$

- If instead you buy 100 shares, a price drop results in a loss:

$$100 \times (90 - 100 + 3) = -700$$

- **Investment Assets:** Held for potential value appreciation over time (e.g., gold, stocks, bonds).
- **Consumption Assets:** Used in production or consumption (e.g., oil, copper).

3.2 Forward Price for an Investment Asset

Assumptions:

1. No transaction costs in trading.
2. All net trading profits are taxed equally.
3. Borrowing and lending occur at the same **risk-free interest rate**.
4. Arbitrage opportunities are **immediately** exploited.

Notation:

- S_0: Current spot price of the asset.
- F_0: Today's futures or forward price of the asset.
- T: Time to contract maturity (in years).
- r: Annualized risk-free interest rate over T.

3.2.1 Arbitrage Example

Consider an asset that provides with **no income**, priced at **$40**, with a **5% annual interest rate** and a **3-month forward contract**.

3.2.1.1 Case 1: Forward Price = $43

- **Now**: Borrow $40, buy the asset, and enter a forward contract to sell at $43.
- **In 3 months**: Sell at $43, repay the loan ($40.50 with interest).
- **Profit**: $2.50.

3.2.1.2 Case 2: Forward Price = $39

- **Now**: Short sell the asset for $40, invest the proceeds at 5% interest, and enter a forward contract to buy at $39.
- **In 3 months**: Buy back at $39, close the short position, and receive $40.50 from the investment.
- **Profit**: $1.50.

3.2.2 Forward Price Formula

For an **investment asset** with **no income**, the no-arbitrage forward price is:

$$F_0 = S_0 e^{rT}$$

> Example: Eliminate arbitrage
>
> Using $S_0 = 40$, $T = 0.25$ (3 months), and $r = 0.05$ (5%):
>
> $$F_0 = 40e^{0.05\times 0.25} = 40.50$$
>
> This aligns with the **theoretical fair price**, ensuring no arbitrage.

- If the asset generates a **known income** I (expressed as the present value of the income), the formula adjusts to:

$$F_0 = (S_0 - I)e^{rT}$$

- If the asset provides a **yield** q (expressed as a continuously compounded annualized rate):

$$F_0 = S_0 e^{(r-q)T}$$

3.3 Valuing a Forward Contract

At initiation, a forward contract has **zero value**, ensuring fairness for both parties (excluding bid-offer spreads). Over time, its value can become **positive or negative** as market conditions change.

Let:

- K = **Agreed delivery price** in the contract.
- F_0 = **Forward price** for a contract negotiated today.

The value of a forward contract depends on the difference between the agreed **delivery price** and the current **forward price**, discounted at the risk-free rate:

- **Long position (buying the asset):**

$$V_{\text{long}} = (F_0 - K)e^{-rT}$$

- **Short position (selling the asset):**

$$V_{\text{short}} = -(F_0 - K)e^{-rT}$$

These formulas show how the **risk-free rate** r and **time to maturity** T impact the present value of expected gains or losses.

3.4 Forward vs. Futures Prices

Although **forward** and **futures** contracts both involve agreeing to buy/sell an asset at a future date, their prices may diverge due to **interest rate volatility**.

- If interest rates and asset prices are **positively correlated** $\rightarrow$ **Futures price** $>$ **Forward price**
  - Daily settlement allows gains to be reinvested at **higher rates**.
- If interest rates and asset prices are **negatively correlated** $\rightarrow$ **Futures price** $<$ **Forward price**
  - No daily settlement avoids reinvesting at **lower rates**.

One notable example is **Eurodollar futures**, where pricing anomalies arise due to unique market characteristics.

3.5 Forward Price for Specific Assets

3.5.1 Stock Index

A **stock index** is an **investment asset** that effectively pays a **dividend yield**, similar to the income generated by holding its underlying stocks.

The relationship between the **futures price** (F_0) and the **spot price** (S_0) is:

$$F_0 = S_0 e^{(r-q)T}$$

where, q represents the average dividend yield of the portfolio reflected by the index over the contract's life.

Warning

This formula assumes the index represents a **tradable investment asset**. However, a purely numerical index (e.g., **Nikkei 225** without its underlying assets) does not qualify as a true investment asset.

Index Arbitrage

Arbitrageurs exploit price discrepancies between **futures** and **spot prices** adjusted for dividend yield and the risk-free rate:

- If $F_0 > S_0 e^{(r-q)T} \rightarrow$ Buy the underlying stocks, sell futures.
- If $F_0 < S_0 e^{(r-q)T} \rightarrow$ Buy futures, short-sell the index's underlying stocks.

High-frequency trading algorithms often execute these strategies. However, **real-world frictions** (e.g., execution delays, transaction costs) may prevent perfect arbitrage.

3.5.2 Exchange Rates

A **foreign currency** behaves like a security yielding **interest**, where the yield is the **foreign risk-free rate** (r_f). The forward exchange rate is given by:

$$F_0 = S_0 e^{(r-r_f)T}$$

Exchange Rate Arbitrage

Consider two equivalent strategies for converting **1,000 units** of a foreign currency into dollars by time T:

1. **Invest in the foreign currency** at rate r_f, then convert at the forward rate F_0.
2. **Convert immediately** at spot rate S_0, then invest in dollars at rate r.

No-arbitrage implies both must yield the same result, leading to:

$$1,000 \times e^{r_f T} \times F_0 = 1,000 \times S_0 \times e^{rT}$$

Solving for F_0 confirms the forward rate formula:

$$F_0 = S_0 e^{(r-r_f)T}$$

3.5.3 Commodities

For **consumption assets** (e.g., oil, wheat, copper), **storage costs** act as **negative income**, affecting forward pricing:

$$F_0 \leq S_0 e^{(r+u)T}$$

where:

- u = **Storage cost per unit time** as a percentage of the asset's value.

Alternatively, if U represents the **present value** of storage costs:

$$F_0 \leq (S_0 + U)e^{rT}$$

These formulas accommodate the costs associated with holding and storing physical commodities, from agricultural products to metals, affecting their forward pricing.

3.5.4 The Cost of Carry

The **cost of carry** (c) represents the total cost of holding an asset, including:

- Storage costs
- Financing costs
- Income earned (e.g., dividends, yields)

For **investment assets**, the forward price follows:

$$F_0 = S_0 e^{cT}$$

For **consumption assets**, storage and financing costs may create **lower price bounds**:

$$F_0 \leq S_0 e^{cT}$$

A **convenience yield** (y) represents the **benefit of physically holding** a consumption asset rather than a derivative. The forward price adjusts to:

$$F_0 = S_0 e^{(c-y)T}$$

where y reflects **scarcity, supply-chain security, and non-financial benefits** of physical possession.

3.6 Futures Prices and Expected Spot Prices

- **Contango** ($F_t > S_t$): Futures prices are higher than the current or expected future spot price.
  - Occurs when carrying costs (storage, interest) exceed convenience yield.
  - Common in gold, silver, natural gas, and agricultural commodities.
  - Contango is more common across most futures markets due to carrying costs.
- **Backwardation** ($F_t < S_t$): Futures prices are lower than the expected future spot price.
  - Happens when convenience yield is high (scarcity or strong demand for immediate delivery).
  - Common in crude oil, perishable goods, and equity index futures.
- **Futures prices are poor predictors of future spot prices.** Empirical evidence shows biases due to risk premia and market inefficiencies. Speculative demand, liquidity constraints, and irrational expectations can further distort pricing.
- While futures prices provide useful signals, they should not be relied upon as perfect forecasts of future spot prices.

> Tip
>
> - What is Contango and Backwardation - CME Institute

3.7 Forward Rate Agreement (FRA)

An **FRA** is a financial contract where parties exchange a **fixed interest rate** for a **floating rate** (e.g., LIBOR, SOFR, SONIA) at a future date, based on a specified **notional principal**.

- No principal exchange occurs—only the difference in interest payments is settled.
- The **initial value** of an FRA is **zero** since the fixed rate equals the forward rate at inception.
- As the forward rate changes over time, the FRA's value fluctuates.

Example: FRA

Party A and Party B agree to exchange a fixed rate of 3% for the three-month SOFR on a $100 million notional in two years (compounded quarterly).

- **Party A** pays floating SOFR and receives a fixed 3%.
- **Party B** takes the opposite position.

If in two years the SOFR rate is 3.5%, Party A receives a payment from Party B:

$$100,000,000 \times (0.035 - 0.030) \times 0.25 = 125,000$$

This amount is discounted for three months at **3.5%**, since payments are made at the two-year mark.

- A **zero rate** is the interest rate for an investment lasting n years, with interest and principal paid at maturity (no intermediate payments).
- A **forward rate** represents the future interest rate implied by current zero rates for future periods. It reflects the cost of borrowing/lending at a future time.

For a forward rate between T_1 and T_2, given zero rate R_1 and R_2:

$$R_F = \frac{R_2 T_2 - R_1 T_1}{T_2 - T_1}$$

Example: Forward Rate Calculation

Year	Zero Rate (% per annum)	Forward Rate for nth Year (% per annum)
1	3.0	
2	4.0	5.0
3	4.6	5.8
4	5.0	6.2
5	5.3	6.5

For **year 4**, given $T_1 = 3$, $T_2 = 4$, $R_1 = 4.6\%$, and $R_2 = 5.0\%$:

$$R_F = \frac{0.05 \times 4 - 0.046 \times 3}{4 - 3} = 6.2\%$$

This forward rate (**6.2%**) is higher than the **zero rate (5.0%)**, reflecting an **upward-sloping yield curve**.

The **value of an FRA** is determined by:

- The difference between the **agreed fixed rate** (R_K) and the **current forward rate** (R_F).
- The **contract period** (t).
- Discounting the difference to **present value**.

3.8 Practice Questions and Problems

1. Explain what happens when an investor shorts a certain share.
2. What is the difference between the forward price and the value of a forward contract?
3. Explain carefully why the futures price of gold can be calculated from its spot price and other observable variables whereas the futures price of copper cannot.
4. Explain carefully the meaning of the terms convenience yield and cost of carry. What is the relationship between futures price, spot price, convenience yield, and cost of carry?
5. What is the cost of carry for (a) a non-dividend-paying stock, (b) a stock index, (c) a commodity with storage costs, and (d) a foreign currency?
6. Suppose that you enter into a three-month forward contract on a non-dividend-paying stock when the stock price is \$108 and the risk-free interest rate (with continuous compounding) is 4% per annum. What is the forward price?

Solution

$F_0 = 109.085$

7. A four-months long forward contract on a non-dividend-paying stock is entered into when the stock price is \$150 and the risk-free rate of interest is 5.7% per annum with continuous compounding.
    - a) What are the forward price and the initial value of the forward contract?

- b) Two months later, the price of the stock is \$168 and the risk-free interest rate is still 5.7%. What are the forward price and the value of the forward contract?

Solution

Forward price $F_0 = 152.88$; initial value 0; final value 169.6

8. The risk-free rate of interest is 4.1% per annum with continuous compounding, and the dividend yield on a stock index is 6.2% per annum. The current value of the index is 2445. What is the one-month futures price?

Solution

$F_0 = 2440.725$

9. A stock index currently stands at 725. The risk-free interest rate is 7.6% per annum (with continuous compounding) and the dividend yield on the index is 1.8% per annum. What should the futures price for a three-month contract be?

Solution

$F_0 = 735.589$

10. An index is 550. The three-month risk-free rate is 4.60% per annum and the dividend yield over the next three months is 5.80% per annum. The six-month risk-free rate is 5.34% per annum and the dividend yield over the next six months is 4.93% per annum. Estimate the futures price of the index for three-month and six-month contracts. All interest rates and dividend yields are continuously compounded.

Solution

3-month $F_0 = 548.352$; 6-month $F_0 = 551.129$

11. The spot price of silver is \$11 per ounce. The storage costs are \$0.25 per ounce payable quarterly in advance. Assuming that interest rates are 1.80% per annum for all maturities, calculate the futures price of silver for delivery in nine months.

Solution

$F_0 = 11.91$

12. The spot price of oil is \$39 per barrel and the cost of storing a barrel of oil for one year is \$1.2, payable at the end of the year. The risk-free interest rate is 8.60% per annum, continuously compounded. What is an upper bound for the one-year futures price of oil?

> **Solution**
>
> $F_0 = 43.70$

13. Suppose that the risk-free interest rate is 3.00% per annum with continuous compounding and that the dividend yield on a stock index is 0.60% per annum. The index is standing at 3646, and the futures price for a contract deliverable in ten months is 3701. What arbitrage opportunities does this create?

> **Solution**
>
> $F_0 = 3719.654$

14. The eight-month interest rates in Switzerland and the United States are, respectively, 3.60% and 5.40% per annum with continuous compounding. The spot price of the Swiss franc is \$0.9. The futures price for a contract deliverable in two months is also \$0.9. What arbitrage opportunities does this create?

> **Solution**
>
> $F_0 = 0.911$

15. When a known future cash outflow in a foreign currency is hedged by a company using a forward contract, there is no foreign exchange risk. When it is hedged using futures contracts, the daily settlement process does leave the company exposed to some risk. Explain the nature of this risk. Assume that the forward price equals the futures price. In particular, consider whether the company is better off using a futures contract or a forward contract when

    - a) The value of the foreign currency falls rapidly during the life of the contract
    - b) The value of the foreign currency rises rapidly during the life of the contract
    - c) The value of the foreign currency first rises and then falls back to its initial value
    - d) The value of the foreign currency first falls and then rises back to its initial value

4 Swaps

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Learning Outcomes:

- Understand the fundamental principles and nature of swaps, including their purpose and functionality in financial markets.
- Analyze the mechanics and applications of interest rate swaps, identifying the roles of comparative advantage and valuation methods.
- Explore currency swaps, including their types, uses, and the valuation process for fixed-for-fixed currency swaps.
- Examine additional swap arrangements outside of interest rate and currency swaps, recognizing their unique features and applications.

4.1 Nature of Swaps

Definiton

A swap is an over-the-counter (OTC) contract where two parties agree to exchange cash flows at specified intervals based on predetermined terms. Due to its bilateral nature and customization, it carries default risk.

- **Comparison with Forward Contracts**: While forward contracts involve a single cash flow exchange at a future date, swaps facilitate multiple exchanges over time, making them more flexible and useful for various financial strategies.

- **Conceptual View**: A swap can be seen as a series of bundled forward contracts. This perspective highlights its role in hedging and financing, where cash flow timing and valuation are crucial.
- **Initial Market Value**: At inception, a swap has a market value of zero, reflecting a balanced structure where some forward components may hold positive value while others hold negative value. Over time, market fluctuations and underlying asset performance alter the swap's value.

4.2 Interest Rate Swaps

This section explores a **Plain Vanilla Interest Rate Swap**, using a transaction between **Apple and Citigroup** as an example. The swap spans three years and demonstrates the core mechanics of interest rate swaps.

- Apple agrees to pay Citigroup a **fixed 3% annual interest rate**, with quarterly payments based on a **notional principal of $100 million**.
- In return, Citigroup pays Apple interest at the **three-month LIBOR rate**, calculated on the same principal.
- Apple is the fixed-rate payer, while Citigroup is the floating-rate payer.

The table below illustrates potential cash flow outcomes, maintaining the **$100 million notional principal**. The net cash flow for Apple varies due to LIBOR fluctuations, sometimes resulting in payments and other times in receipts.

Date	SOFR Rate (%)	Floating Cash Flow Received ($'000s)	Fixed Cash Flow Paid ($'000s)	Net Cash Flow ($'000s)
June 8, 2022	2.20	550	750	-200
Sept. 8, 2022	2.60	650	750	-100
Dec. 8, 2022	2.80	700	750	-50
Mar. 8, 2023	3.10	775	750	+25
June 8, 2023	3.30	825	750	+75
Sept. 8, 2023	3.40	850	750	+100

Date	SOFR Rate (%)	Floating Cash Flow Received ($'000s)	Fixed Cash Flow Paid ($'000s)	Net Cash Flow ($'000s)
Dec. 8, 2023	3.60	900	750	+150
Mar. 8, 2024	3.80	950	750	+200

4.2.1 Typical Applications of Interest Rate Swaps

Interest rate swaps are widely used for managing interest rate risk, allowing firms to **adjust their exposure** by switching between fixed and floating rates based on financial strategy and market outlook.

- Apple Converts a Floating-Rate Liability to Fixed:
- Intel Converts a Fixed-Rate Liability to Floating:
- Apple Converts a Fixed-Rate Asset to Floating:
- Intel Converts a Floating-Rate Asset to Fixed:

4.3 The Comparative-Advantage Argument

The **comparative-advantage argument** explains why companies engage in interest rate swaps to **optimize borrowing costs**. It highlights how firms can **leverage differences in borrowing rates** to achieve more favorable financing conditions.

4.3.1 Case Study: AAACorp and BBBCorp

Consider two hypothetical companies:

- **AAACorp** prefers **floating-rate borrowing**.
- **BBBCorp** prefers **fixed-rate borrowing**.

Their respective borrowing costs are:

Entity	Fixed Rate	Floating Rate
AAACorp	4.0%	Floating - 0.1%

Entity	Fixed Rate	Floating Rate
BBBCorp	5.2%	Floating + 0.6%

- **Direct Swap Mechanism:** AAACorp borrows at its preferred floating rate, while BBBCorp borrows at its preferred fixed rate. They swap obligations, effectively allowing each company to access the preferred rate indirectly.
- **Indirect Swap via a Financial Institution:** A financial institution can act as an intermediary, structuring the swap between AAACorp and BBBCorp. This setup reduces negotiation complexities and often enhances efficiency.

4.3.2 Criticism of the Comparative-Advantage Argument

While this argument supports the rationale for interest rate swaps, it has limitations:

Mismatch in Rate Terms:

- Fixed rates (e.g., 4.0% for AAACorp and 5.2% for BBBCorp) apply to **5-year loans**.
- Floating rates (e.g., Floating - 0.1% for AAACorp and Floating + 0.6% for BBBCorp) are based on **6-month terms**.
- This disparity complicates direct comparisons of borrowing advantages.

Future Rate Uncertainty:

- BBBCorp's **future borrowing costs** depend on the spread over the floating rate at each reset.
- If interest rates **fluctuate significantly**, BBBCorp may face unexpected financial exposure.

4.4 Valuation of Interest Rate Swaps

Interest rate swaps are used to manage interest rate exposure, and their value fluctuates as market conditions change. At inception, a swap's value is **approximately zero**, reflecting an initial balance between fixed and floating cash flows. Over time, valuation techniques—particularly those using **Forward Rate Agreements (FRAs)**—help determine its market value.

4.4.1 Valuation Framework

The swap valuation process consists of three key steps:

1. **Calculate Floating Forward Rates:** Use the risk-free zero curve (derived from OIS rates) to estimate future floating interest rates that determine floating-rate payments.
2. **Determine Swap Cash Flows:** Apply calculated forward rates (converted to semi-annual compounding) to estimate future cash flows for both fixed and floating legs of the swap.
3. **Discount Cash Flows to Present Value:** Discount net future cash flows using the risk-free zero rates to obtain the present value of the swap.

Example: Swap Valuation

4.4.2 Swap Agreement Overview

- **Structure**: A financial institution pays a fixed rate of 3% per annum (semi-annual payments) and receives the SOFR six-month reference rate on a notional principal of $100 million.
- **Remaining Term**: 1.2 years, with cash flow exchanges at **0.2, 0.7, and 1.2 years**.
- **Risk-Free Zero Rates** (continuously compounded, derived from OIS):
  - 0.2 years: 2.8%
  - 0.7 years: 3.2%
  - 1.2 years: 3.4%
- **Historical SOFR Rate**: The continuously compounded risk-free rate observed for the last 0.3 years (which covers 60% of the days determining the exchange at 0.2 years) is **2.3%**.

4.4.3 Step 1: Determine Floating Rates

Exchange at 0.2 years (blended rate):
The first floating rate is not a true forward rate. Part of the period has already elapsed (60%), so the rate is a weighted average of the already-observed rate and the zero rate for the remaining portion:

$$R_{0.2} = 0.6 \times 2.3\% + 0.4 \times 2.8\% = 2.50\% \quad \text{(cont. comp.)}$$

Exchange at 0.7 years (forward rate for 0.2 → 0.7 years):
Using the formula for forward rates from the zero curve:

$$R_{0.7} = \frac{R_2 T_2 - R_1 T_1}{T_2 - T_1} = \frac{0.032 \times 0.7 - 0.028 \times 0.2}{0.7 - 0.2} = \frac{0.0224 - 0.0056}{0.5} = 3.36\% \quad \text{(cont. comp.)}$$

Exchange at 1.2 years (forward rate for 0.7 → 1.2 years):

$$R_{1.2} = \frac{0.034 \times 1.2 - 0.032 \times 0.7}{1.2 - 0.7} = \frac{0.0408 - 0.0224}{0.5} = 3.68\% \quad \text{(cont. comp.)}$$

4.4.4 Step 2: Convert to Semi-Annual Compounding

Since swap payments are made semi-annually, we convert from continuous to semi-annual compounding using $R_s = 2(e^{R_c/2} - 1)$:

Exchange at (years)	Continuous Compounding	Semi-Annual Compounding
0.2	2.500%	2.516%
0.7	3.360%	3.388%
1.2	3.680%	3.714%

4.4.5 Step 3: Calculate and Discount Cash Flows

Each semi-annual **fixed payment** is:

$$-0.5 \times 0.03 \times 100 = -1.500 \text{ million}$$

Each **floating payment** uses the semi-annually compounded forward rate:

$$+0.5 \times R_s \times 100$$

The **discount factor** for time T uses the continuously compounded zero rate R_T:

$$DF(T) = e^{-R_T \times T}$$

Time (years)	Fixed CF ($M)	Floating CF ($M)	Net CF ($M)	Discount Factor	PV of Net CF ($M)
0.2	−1.500	+1.258	−0.242	0.9944	−0.241
0.7	−1.500	+1.694	+0.194	0.9778	+0.190
1.2	−1.500	+1.857	+0.357	0.9600	+0.343
Total					**+0.292**

4.4.6 Detailed Calculation (0.7-year exchange)

- **Fixed Cash Flow**: $-0.5 \times 0.03 \times 100 = -1.500$ million
- **Floating Cash Flow**: $+0.5 \times 0.03388 \times 100 = +1.694$ million
- **Net Cash Flow**: $-1.500 + 1.694 = +0.194$ million
- **Discount Factor**: $e^{-0.032 \times 0.7} = 0.9778$
- **Present Value**: $0.194 \times 0.9778 = +0.190$ million

4.4.7 Result

The swap value (from the perspective of the floating-rate receiver) is **+$0.292 million**. *Note: This valuation uses simplifications — it ignores holiday calendars and day count conventions, which would affect precise cash flows in practice.*

4.5 Currency Swaps

Currency swaps are financial instruments that enable **the exchange of principal and interest payments in different currencies** between two parties. These contracts help firms manage currency exposure, reduce borrowing costs, and optimize investments across global markets.

In a **fixed-for-fixed currency swap**, parties exchange fixed interest rate payments in one currency for fixed payments in another currency, along with the principal exchange at the start and end of the contract.

Currency swaps are widely used for:

1. **Liability Management:** Firms **convert liabilities** from one currency to another to match their currency exposure and reduce exchange rate risk.
2. **Investment Optimization:** Investors **swap investment returns** between currencies to access favorable interest rates or hedge against currency fluctuations.

Case Study: British Petroleum and Barclays Swap Agreement

4.5.0.1 Agreement Overview

- **Parties Involved**: British Petroleum and Barclays.
- **Term**: 5 years.
- **Structure**:
  - British Petroleum pays a fixed 3% interest rate in USD.
  - British Petroleum receives a fixed 4% interest rate in GBP.
- **Principal Exchange**:
  - USD 15 million exchanged for GBP 10 million at inception.
  - Principal amounts are re-exchanged at the end of the swap.
- **Interest Payments**: Made annually, based on the fixed rates in each currency.

4.5.0.2 Yearly Cash Flow Summary

The table below illustrates the cash flows for British Petroleum exchanged over the swap's duration:

Date	USD Cash Flow (millions)	GBP Cash Flow (millions)
Feb 1, 2022	+15.00	-10.00
Feb 1, 2023	-0.45	+0.40
Feb 1, 2024	-0.45	+0.40
Feb 1, 2025	-0.45	+0.40
Feb 1, 2026	-0.45	+0.40
Feb 1, 2027	-15.45	+10.40

4.5.1 Comparative Advantage in Currency Swaps

A **comparative advantage** in currency swaps arises when companies face different borrowing costs in various currencies due to **taxation, credit ratings, or market conditions**.

- **Efficiency**: Currency swaps enable companies to **lower borrowing costs** and manage currency risk.
- **Strategic Use**: They help **convert liabilities and investments** across different currencies.
- **Comparative Advantage**: Firms leverage **differential borrowing costs** to gain financial benefits.

Case Study: General Electric & Qantas Airways

- **Scenario**:
  - General Electric (GE) wants to borrow in Australian dollars (AUD).
  - Qantas Airways wants to borrow in U.S. dollars (USD).
- **Borrowing Costs (after tax adjustments)**:

	USD	AUD
General Electric	5.0%	7.6%
Qantas Airways	7.0%	8.0%

Since **General Electric has a lower borrowing cost in USD** and **Qantas Airways has a lower borrowing cost in AUD**, they can **swap their borrowing obligations** to access cheaper rates than they would get individually.

4.6 Valuation of Fixed-for-Fixed Currency Swaps

Fixed-for-fixed currency swaps involve **exchanging fixed interest payments in two different currencies** over a set period, alongside an initial and final principal exchange. These swaps can be viewed as a series of forward foreign exchange contracts, where future exchange rates determine the valuation of expected cash flows.

The valuation process involves:

1. **Calculate Forward Exchange Rates:** Derive forward rates from the spot rate and the interest rate differential between the two currencies.
2. **Convert Foreign Cash Flows:** Use forward rates to express all cash flows in a single (domestic) currency and compute net cash flows.
3. **Discount to Present Value:** Discount net cash flows at the domestic risk-free rate.

Example: Valuation of a Currency Swap

4.6.1 Swap Agreement Overview

- **Structure**: One party receives 3% per annum on JPY 1,200 million and pays 4% per annum on USD 10 million, with annual payments and principal exchange at maturity.
- **Remaining Term**: 3 years, with cash flow exchanges at 1, 2, and 3 years.
- **Risk-Free Rates** (continuously compounded):
  - USD: 2.5%
  - JPY: 1.5%
- **Current Spot Exchange Rate**: 110 JPY per USD (i.e., $S_0 = 1/110 = 0.009091$ USD per JPY).

4.6.2 Step 1: Calculate Forward Exchange Rates

Forward exchange rates (USD per JPY) are derived from the interest rate differential:

$$F_t = S_0 \times e^{(r_{USD} - r_{JPY}) \times t}$$

Time (years)	Calculation	Forward Rate (USD/JPY)
1	$0.009091 \times e^{(0.025-0.015)\times 1}$	0.009182
2	$0.009091 \times e^{(0.025-0.015)\times 2}$	0.009275
3	$0.009091 \times e^{(0.025-0.015)\times 3}$	0.009368

Since $r_{USD} > r_{JPY}$, the forward rates show the JPY gradually appreciating against the USD (each JPY buys more USD over time).

4.6.3 Step 2: Determine Cash Flows and Convert to USD

Annual cash flows:

- **USD leg (paid):** $0.04 \times 10 = -0.4$ million per year; at maturity -10.4 million (including principal).
- **JPY leg (received):** $0.03 \times 1{,}200 = +36$ million JPY per year; at maturity $+1{,}236$ million JPY (including principal).

Convert JPY cash flows to USD using forward rates, then compute net cash flows:

Time (years)	USD CF ($M)	JPY CF (¥M)	Fwd Rate (USD/JPY)	JPY CF in USD ($M)	Net CF ($M)
1	−0.400	+36	0.009182	+0.3306	−0.0694
2	−0.400	+36	0.009275	+0.3339	−0.0661
3	−10.400	+1,236	0.009368	+11.5786	+1.1786

For example, at Year 1: $36 \times 0.009182 = 0.3306$ million USD, and the net cash flow is $0.3306 - 0.4 = -0.0694$ million.

4.6.4 Step 3: Discount Net Cash Flows

Discount at the USD risk-free rate (2.5%, continuously compounded):

Time (years)	Net CF ($M)	Discount Factor $e^{-0.025 \times t}$	PV of Net CF ($M)
1	-0.0694	0.9753	-0.0677
2	-0.0661	0.9512	-0.0629
3	$+1.1786$	0.9277	$+1.0934$
Total			$\mathbf{+0.9629}$

4.6.5 Result

The swap value (from the perspective of the JPY receiver / USD payer) is **+$0.9629 million**. The positive value arises primarily from the principal exchange at maturity — the JPY 1,200 million received is worth more in USD terms (at the forward rate) than the USD 10 million paid.
Note: This valuation assumes continuous compounding throughout and ignores day count conventions and holiday calendars.

4.7 Other Currency Swaps

4.7.1 Fixed-for-Floating Currency Swaps

A fixed-for-floating currency swap combines elements of a fixed-for-fixed currency swap and a fixed-for-floating interest rate swap. In this arrangement, one party pays a fixed interest rate in one currency while receiving a floating interest rate in another currency.

Example

Consider a swap agreement where:

- A party receives a fixed 3% interest rate on USD 10 million.
- The same party pays a floating interest rate on GBP 7 million.
- Payments are semiannual over 10 years.

This swap can be decomposed into two components:

1. **Fixed-for-Fixed Currency Swap**:
   - **Receives** 3% fixed on USD 10 million.
   - **Pays** 4% fixed on GBP 7 million.
2. **Interest Rate Swap (GBP)**:
   - **Receives** 4% fixed on GBP 7 million.
   - **Pays** floating on GBP 7 million.

Verification: The 4% fixed GBP flows cancel between the two components, leaving the party receiving 3% fixed USD and paying floating GBP — exactly the original swap.

4.7.2 Floating-for-Floating Currency Swaps

A floating-for-floating currency swap involves exchanging floating interest payments in two different currencies. It can be decomposed into a fixed-for-fixed currency swap combined with two interest rate swaps.

Example

A party exchanges:

- Receives floating USD interest on USD 10 million.
- Pays floating sterling interest on GBP 7 million.

This structure can be decomposed into three components:

1. **Fixed-for-Fixed Currency Swap**:
   - **Receives** 3% fixed on USD 10 million.
   - **Pays** 4% fixed on GBP 7 million.
2. **Interest Rate Swap #1 (GBP)**:
   - **Receives** 4% fixed on GBP 7 million.
   - **Pays** floating on GBP 7 million.

3. **Interest Rate Swap #2 (USD)**:
    - **Pays** 3% fixed on USD 10 million.
    - **Receives** floating on USD 10 million.

Verification: The 4% fixed GBP cancels between components 1 and 2. The 3% fixed USD cancels between components 1 and 3. What remains is receiving floating USD and paying floating GBP — exactly the original swap.

4.8 Other Types of Swaps

Beyond currency and interest rate swaps, financial markets offer specialized swaps tailored for **risk management, investment strategies, and arbitrage opportunities**.

- **Amortizing/Step-Up Swap**: The notional principal gradually increases or decreases to match an underlying asset or liability.
- **Compounding Swap**: Interest payments are reinvested, compounding over the swap's duration.
- **Constant Maturity Swap (CMS)**: The interest rate resets periodically based on the rate of a constant maturity instrument (e.g., a 10-year Treasury note).
- **LIBOR-in-Arrears Swap**: The interest rate is determined at the end of the payment period instead of the beginning, increasing uncertainty.
- **Accrual Swap**: Interest accrues only when a benchmark rate (or index) meets specific conditions.
- **Equity Swap**: Exchanges equity returns (e.g., stock performance) for either fixed or floating interest payments.
- **Cross-Currency Interest Rate Swap**: A variation of currency swaps where one or both legs involve floating rates in different currencies.
- **Diff Swap**: An interest rate differential swap, where payments depend on the difference between two reference rates.
- **Commodity Swap**: One party pays a fixed price for a commodity, while the other pays the floating market price over time.
- **Variance Swap**: A contract based on future volatility, where payments depend on the variance of an underlying asset's price.

4.9 Practice Questions and Problems

1. A bank finds that its assets are not matched with its liabilities. It is taking floating-rate deposits and making fixed-rate loans. How can swaps be used to offset the risk?
2. Explain the difference between the credit risk and the market risk in a financial contract.
3. Explain why a bank is subject to credit risk when it enters into two offsetting swap contracts.
4. Why is the expected loss from a default on a swap less than the expected loss from the default on a loan to the same counterparty with the same principal?
5. A corporate treasurer tells you that he has just negotiated a five-year loan at a competitive fixed rate of interest of 5.2%. The treasurer explains that he achieved the 5.2% rate by borrowing at six-month LIBOR plus 150 basis points and swapping LIBOR for 3.7%. He goes on to say that this was possible because his company has a comparative advantage in the floating-rate market. What has the treasurer overlooked?
6. Companies A and B have been offered the following rates per annum on a $20 million five-year loan:

	Fixed Rate	Floating Rate
Company A	5.0%	Floating + 0.1%
Company B	6.4%	Floating + 0.6%

Company A requires a floating-rate loan; company B requires a fixed-rate loan. Design a swap that will net a bank, acting as intermediary, 0.1% per annum and that will appear equally attractive to both companies.

7. Company X wishes to borrow U.S. dollars at a fixed rate of interest. Company Y wishes to borrow Japanese yen at a fixed rate of interest. The amounts required by the two companies are roughly the same at the current exchange rate. The companies have been quoted the following interest rates, which have been adjusted for the impact of taxes:

	Yen	Dollars
Company X	5.0%	9.6%
Company Y	6.5%	10.0%

Design a swap that will net a bank, acting as intermediary, 50 basis points per annum. Make the swap equally attractive to the two companies and ensure that all foreign exchange risk is assumed by the bank.

8. Companies X and Y have been offered the following rates per annum on a \$5 million 10-year investment:

	Fixed Rate	Floating Rate
Company X	8.0%	Floating
Company Y	8.8%	Floating

Company X requires a fixed-rate investment; company Y requires a floating-rate investment. Design a swap that will net a bank, acting as intermediary, 0.2% per annum and will appear equally attractive to X and Y.

9. A \$100 million interest rate swap has a remaining life of 10 months. Under the terms of the swap, six-month LIBOR is exchanged for 7% per annum (compounded semiannually). The average of the bid-offer rate being exchanged for six-month LIBOR in swaps of all maturities is currently 5% per annum with continuous compounding (5.063% with semi-annual compounding). The six-month LIBOR rate was 4.6% per annum two months ago. What is the current value of the swap to the party paying floating? What is its value to the party paying fixed?

10. A currency swap has a remaining life of 15 months. It involves exchanging interest at 10% on 20 million GBP for interest at 6% on 30 million USD once a year. The term structure of interest rates in both the United Kingdom and the United States is currently flat, and if the swap were negotiated today the interest rates exchanged would be 4% in dollars and 7% in sterling. All interest rates are quoted with annual compounding (the continuously compounded interest rates in sterling and dollars are 6.766% per annum and 3.922% per annum). The current exchange rate (dollars per pound sterling) is 1.5500. What is the value of the swap to the party paying sterling? What is the value of the swap to the party paying dollars?

5 Options

References

- HULL, John. Options, futures, and other derivatives. Ninth edition. Harlow: Pearson, 2018. ISBN 978-1-292-21289-0.
  - Chapter 10. Mechanics of Options Markets
  - Chapter 11. Properties of Stock Options
- PIRIE, Wendy L. Derivatives. Hoboken: Wiley, 2017. CFA institute investment series. ISBN 978-1-119-38181-5.
  - Chapter 1 - Derivative Markets and Instruments

Learning Outcomes:

- Comprehend the fundamentals and definitions of options, including the mechanics behind option payoffs and profits.
- Identify and understand other option characteristics, adjustments, and the relationship with related assets.
- Grasp the principles underlying option pricing dynamics, along with the factors influencing option values.
- Analyze the upper and lower bounds for option prices and the concept of put-call parity in option trading strategies.
- Evaluate the specific considerations for American options, including the impact of early exercise decisions and dividends on option valuation.

5.1 Understanding Options

Definition

Options are financial derivatives that give the buyer the **right, but not the obligation**, to buy (call option) or sell (put option) an asset at a predetermined price (strike price) on or before a specific date. The seller (writer) receives a premium in exchange for granting this right.

Types of options:

- **Call Option**: Gives the holder the right **to buy an asset** at the strike price before expiration. Buyers expect the asset's price to rise.
- **Put Option**: Gives the holder the right **to sell an asset** at the strike price before expiration. Buyers expect the asset's price to fall.

Execution Styles:

- **American Option**: Can be exercised **at any time before expiration**, offering more flexibility.
- **European Option**: Can only be exercised **on the expiration date**, making it more predictable but less flexible.

Options can be written on various assets, including:

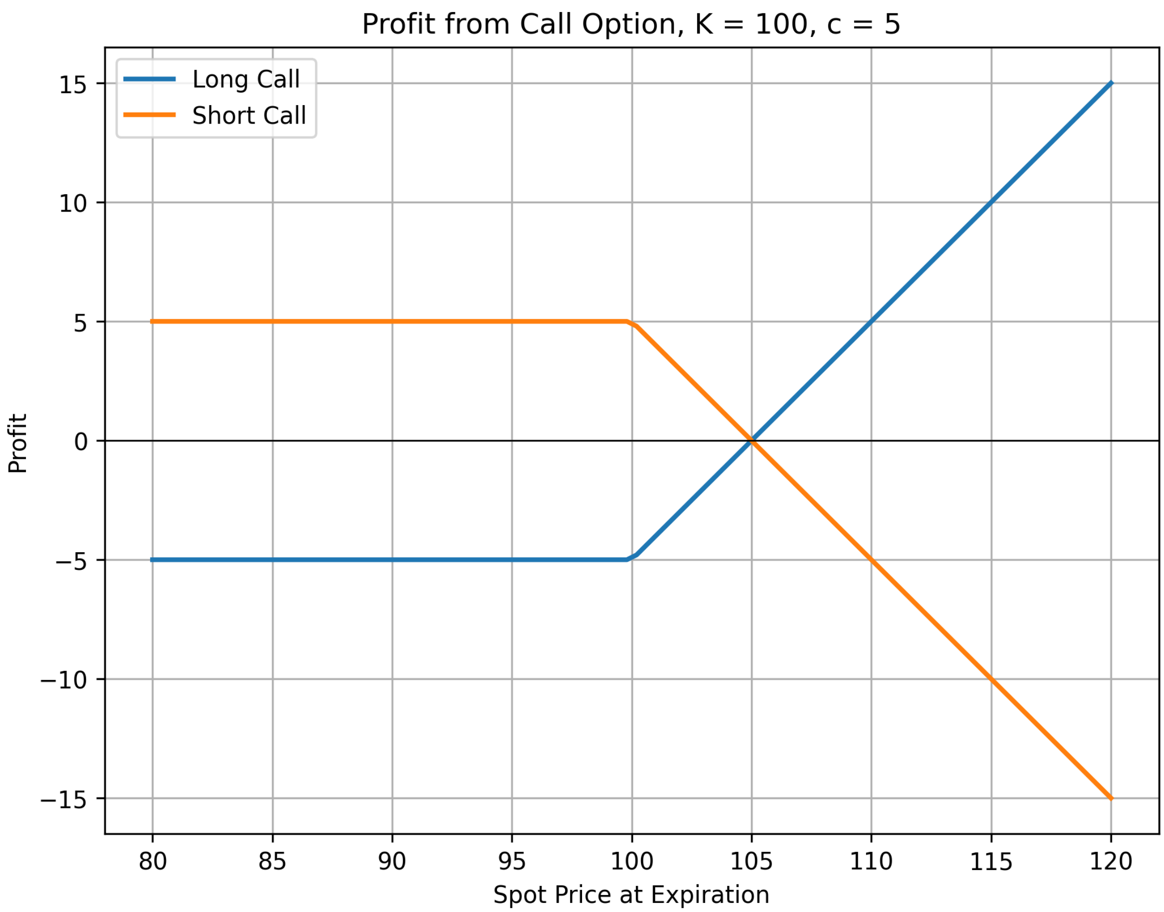
- Stocks
- ETFs and ETPs
- Foreign currency
- Stock indices
- Futures

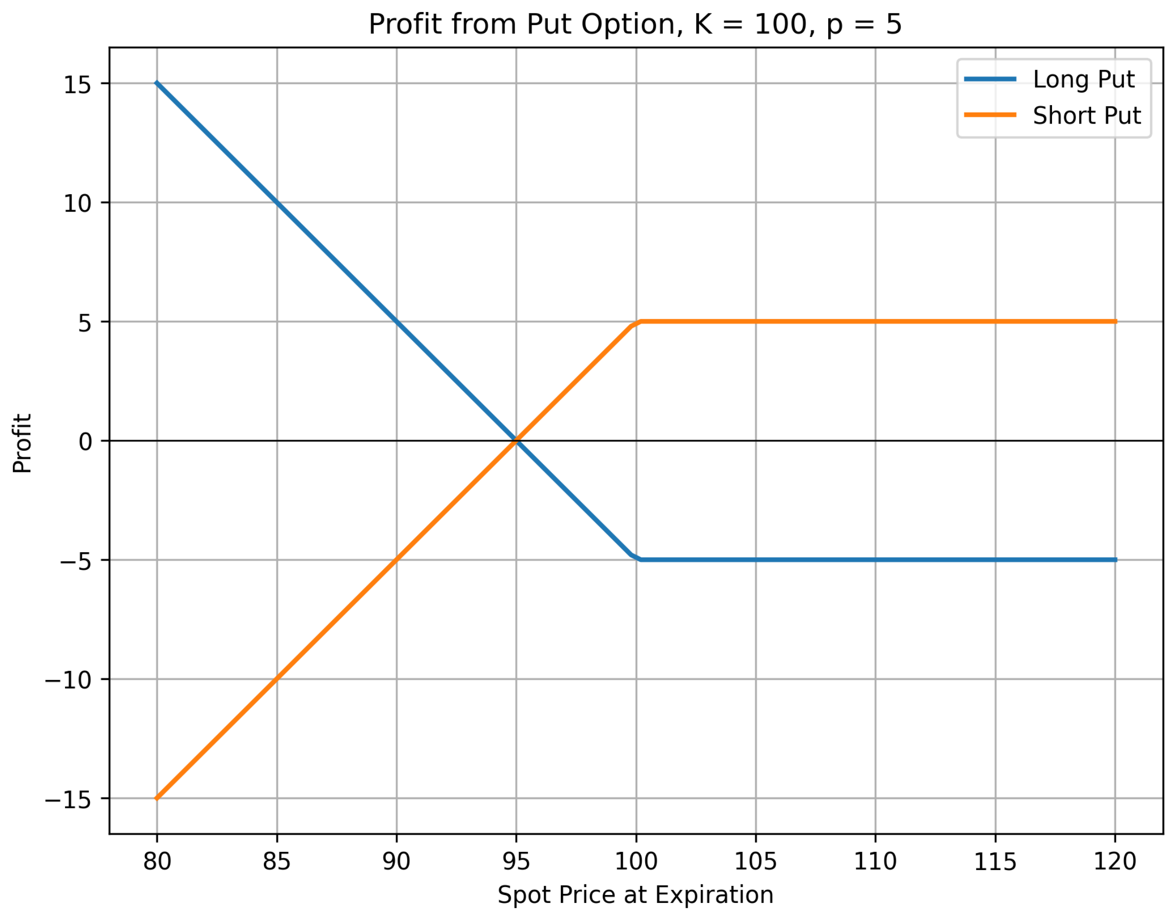
Margin account:

- Buying options **does not require margin** as the buyer's risk is limited to the premium paid.
- Selling options **requires margin** due to the obligation to fulfill the contract.

Option Value Components:

- **Intrinsic Value**: The profit from immediate exercise.
- **Time Value**: The extra value reflecting potential future price changes.


Profit from Call Option, K = 100, c = 5
Long Call
Short Call
15
10
5
0
−5
−10
−15
Profit
80
85
90
95
100
105
110
115
120
Spot Price at Expiration



5.2 Option Payoffs and Profits

5.2.1 Understanding Option Payoffs

The option buyer's payoff depends on the relationship between the price of the underlying asset at expiration (S_T) and strike price of the option (K).

Moneyness determines whether exercising the option would be profitable:

- **In the Money (ITM)**: Exercising the option yields a positive payoff.
- **At the Money (ATM)**: The option's strike price equals the asset price ($S_T = K$). There is no intrinsic value.
- **Out of the Money (OTM)**: Exercising the option results in zero payoff.

Moneyness is crucial for pricing and trading decisions, as ITM options have intrinsic value, while ATM and OTM options are purely speculative.

- **Call Option Payoff**:
$$c_T = \max(0, S_T - K)$$
- **Put Option Payoff**:
$$p_T = \max(0, K - S_T)$$

> **Example: Option Payoff**
>
> Given an underlying asset price at expiration S_T of \$28 and a strike price K of \$25, the payoffs for call and put options are calculated as follows:
>
> - **Call Buyer Payoff**: $c_T = \max(0, \$28 - \$25) = \$3$
> - **Put Buyer Payoff**: $p_T = \max(0, \$25 - \$28) = \$0$
>
> The call is **in the money** with a payoff of \$3, while the put is **out of the money** with no payoff.

5.2.2 Calculating Option Profit

To determine profit, the option premium (c_0 for calls, p_0 for puts) must be considered:

- **Call Option Profit**:
$$\Pi_{call} = \max(0, S_T - K) - c_0$$
- **Put Option Profit**:
$$\Pi_{put} = \max(0, K - S_T) - p_0$$

> **Example: Option Profit**
>
> Considering CBX stock options with a strike price $K = \$30$, where the call and put premiums are \$1 and \$2 respectively, and the stock price at expiration S_T is \$27.50:
>
> - **Call Option Profit**: $\Pi_{call} = \max(0, \$27.5 - \$30) - \$1 = -\$1$
> - **Put Option Profit**: $\Pi_{put} = \max(0, \$30 - \$27.5) - \$2 = \$0.5$
>
> The call buyer loses \$1, while the put buyer profits \$0.5.

5.3 Dividend Adjustments and Related Assets

5.3.1 Adjustments for Dividends and Stock Splits

- **Cash Dividends**: Typically do not affect options, as they are factored into pricing.
- **Stock Splits & Stock Dividends**: Adjustments ensure option value remains unchanged.

For an n**-for-**m **stock split**:

- New Strike Price:

$$K' = \frac{m}{n} \times K$$

- New Number of Options:

$$N' = \frac{n}{m} \times N$$

Example: Option Adjustments

5.3.1.1 2-for-1 Stock Split

A call option for 100 shares at \$20 per share:

- **New Strike Price**: $\frac{1}{2} \times 20 = 10$
- **New Number of Options**: $\frac{2}{1} \times 100 = 200$

5.3.1.2 5% Stock Dividend

A call option for 100 shares at \$20 per share:

- **New Strike Price**: $\frac{1}{1.05} \times 20 = 19.05$
- **New Number of Options**: $\frac{1.05}{1} \times 100 = 105$

5.3.2 Related Assets

5.3.2.1 Warrants

- Issued by corporations, granting the right to buy shares at a fixed price.
- Exercising warrants leads to new stock issuance.

5.3.2.2 Employee Stock Options

- Compensation-based options, often issued at-the-money.
- Exercising creates new company shares.

5.3.2.3 Convertible Bonds

- Bonds that can be converted into equity at predetermined terms.
- Often callable, allowing issuers to force conversion.

5.4 Understanding the Dynamics of Option Pricing

The value of stock options depends on several key factors, each influencing option prices under the **ceteris paribus** (all else equal) assumption. The table below summarizes the impact of these variables on European and American options:

Variable	European Call	European Put	American Call	American Put
Current Stock Price (S_0)	+	−	+	−
Strike Price (K)	−	+	−	+
Time to Expiration (T)	?	?	+	+
Volatility (σ)	+	+	+	+
Risk-free Rate (r)	+	−	+	−
Amount of Future Dividends (D)	−	+	−	+

- c, p: Prices of European call and put options.
- C, P: Prices of American call and put options.
- S_0: Current stock price.
- S_T: Stock price at expiration.
- K: Strike price.
- T: Time to expiration.
- σ: Stock price volatility.
- D: Present value of future dividends.
- r: Risk-free interest rate (continuous compounding).

5.4.1 American vs. European Options

American options, which can be exercised anytime before expiration, are **always worth at least as much** as their European counterparts:

$$C \geq c, \quad P \geq p$$

This is due to the added flexibility of early exercise, making American options inherently more valuable or, at minimum, equal in price to European options.

5.5 Upper and Lower Bounds for Option Prices

5.5.0.1 Upper Bound - Call Option

The price of a call option, whether American or European, cannot exceed the current stock price:

$$c \leq S_0, \quad C \leq S_0$$

This prevents arbitrage, as no rational investor would pay more for a call than the stock itself.

5.5.0.2 Upper Bound - Put Option

- **American put options**: Capped at the strike price since the holder can sell at K:

$$P \leq K$$

- **European put options**: Limited by the present value of K, as they can only be exercised at expiration:

$$p \leq Ke^{-rT}$$

5.5.0.3 Lower Bound - European Call Option

The minimum price of a European call on a non-dividend paying stock is:

$$c \geq \max(S_0 - Ke^{-rT}, 0)$$

Example: European Call Lower Bound

Given: $S_0 = 20$, $K = 18$, $T = 1$, $r = 10\%$, $D = 0$

$$\text{Lower bound} = 20 - 18e^{-0.1} = 3.71$$

If the call price is \$3, an arbitrageur could **short the stock, buy the call, and invest at 10%**.

5.5.0.4 Lower Bound - European Put Option

For puts on non-dividend paying stock, the lower bound is:

$$p \geq \max(Ke^{-rT} - S_0, 0)$$

Example: European Put Lower Bound

Given: $S_0 = 37$, $K = 40$, $T = 0.5$, $r = 5\%$, $D = 0$

$$40e^{-0.05\times0.5} - 37 = 2.01$$

If the put price is \$1, an arbitrageur could **borrow \$38 for six months to buy both the put and stock**.

5.5.1 American Options and Early Exercise

5.5.1.1 American Call Options (Non-Dividend Paying Stocks)

It is never optimal to exercise early because:

- **Preserving capital**: Delaying exercise defers payment of K.
- **Insurance benefit**: Holding the call provides downside protection.
- **Maximizing value**: Selling the option retains its time value.

Thus, their price bounds are the same as European calls:

$$C \geq \max(S_0 - Ke^{-rT}, 0)$$

5.5.2 American Put Options (Non-Dividend Paying Stocks)

Early exercise may be optimal, especially when deep in the money:

- **Immediate value realization**: $K - S_0$ can be collected now.
- **Time value of money**: Receiving proceeds sooner is preferable.
- **Stock price limitations**: Since $S_0 \geq 0$, the put's value is capped.

Thus, the lower bound for American puts is adjusted:

$$P \geq \max(K - S_0, 0)$$

5.6 Put-Call Parity

Put-call parity establishes a fundamental relationship between European call and put options with the same strike price and expiration. It ensures pricing consistency in the absence of arbitrage opportunities.

5.6.1 Put-Call Parity for Non-Dividend Paying Stocks

Consider two equivalent portfolios:

- **Portfolio A**: A European call option (c) and a zero-coupon bond paying K at expiration.
- **Portfolio B**: A European put option (p) and the underlying stock (S_0).

At expiration (T), both portfolios yield the same payoff:

		$S_T > K$ (Above Strike)	$S_T < K$ (Below Strike)
Portfolio A	Call option	$S_T - K$	0
	Zero-coupon bond	K	K
	Total	S_T	K
Portfolio B	Put Option	0	$K - S_T$
	Share	S_T	S_T
	Total	S_T	K

Since both portfolios must have the same present value, we get the **put-call parity equation**:

$$c + Ke^{-rT} = p + S_0$$

Arbitrage Opportunities via Put-Call Parity

Consider an example where $S_0 = \$31$, $r = 10\%$, the call option price $c = \$3$, and the strike price $K = \$30$. The table outlines potential arbitrage strategies based on discrepancies in put option pricing, illustrating the mechanism for securing risk-free profits by leveraging the put-call parity principle.

Three-month put price = \$2.25	Three-month put price = \$1
Action now:	*Action now:*
Buy call for \$3	Borrow \$29 for 3 months
Short put to realize \$2.25	Short call to realize \$3
Short the stock to realize \$31	Buy put for \$1
Invest \$30.25 for 3 months	Buy the stock for \$31
...	
Action in 3 months if $S_T > 30$:	*Action in 3 months if* $S_T > 30$:
Receive \$31.02 from investment	Call exercised: sell stock for \$30
Exercise call to buy stock for \$30	Use \$29.73 to repay loan
Net profit = \$1.02	Net profit = \$0.27
...	
Action in 3 months if $S_T < 30$:	*Action in 3 months if* $S_T < 30$:
Receive \$31.02 from investment	Exercise put to sell stock for \$30
Put exercised: buy stock for \$30	Use \$29.73 to repay loan
Net profit = \$1.02	Net profit = \$0.27

5.6.2 Extension to American Options

Put-call parity strictly applies to **European** options. However, for **American** options, which allow early exercise, the pricing relationship is bounded by:

$$S_0 - K \leq C - P \leq S_0 - Ke^{-rT}$$

This range accounts for the added flexibility of early exercise, particularly for American put options, which may be exercised early if deep in the money.

5.7 Effect of Dividends on Options

Dividends impact option valuation and exercise strategies, particularly for **American call options**. The present value of expected dividends over the option's life is denoted as D.

Understanding the impact of dividends is crucial for accurate option pricing and strategic decision-making.

- **Dividends reduce call option value** due to lost dividend income.
- **Dividends increase put option value** as they lower the stock price.
- For **American call options**, early exercise is typically suboptimal. However, if a dividend is expected, it may be beneficial to **exercise just before the ex-dividend date** to capture the dividend payout. Apart from this scenario, early exercise remains suboptimal.

5.7.1 Adjusted Lower Bound Valuations

Dividends modify the lower bounds of option prices:

- **Call Options:** The lower bound decreases since early exercise forfeits dividends:

$$c \geq S_0 - D - Ke^{-rT}$$

- **Put Options:** The lower bound increases, as dividends reduce the stock price:

$$p \geq D + Ke^{-rT} - S_0$$

5.7.2 Adjusted Put-Call Parity with Dividends

Dividends also modify the **put-call parity** relationship:

- **European Options with Dividends**:

$$c + D + Ke^{-rT} = p + S_0$$

- **American Options with Dividends**:

$$S_0 - D - K < C - P < S_0 - Ke^{-rT}$$

5.8 Practice Questions and Problems

5.8.1 Option Profitability and Exercise Conditions

1. Suppose that a European call option to buy a share for \$100.00 costs \$5.00 and is held until maturity. Under what circumstances will the holder of the option make a profit? Under what circumstances will the option be exercised? Draw a diagram illustrating how the profit from a long position in the option depends on the stock price at maturity of the option.

2. An investor sells a European call on a share for \$4. The stock price is \$47 and the strike price is \$50. Under what circumstances does the investor make a profit? Under what circumstances will the option be exercised? Draw a diagram showing the variation of the investor's profit with the stock price at the maturity of the option.

3. An investor buys a European put on a share for \$3. The stock price is \$42 and the strike price is \$40. Under what circumstances does the investor make a profit? Under what circumstances will the option be exercised? Draw a diagram showing the variation of the investor's profit with the stock price at the maturity of the option.

4. Suppose that a European put option to sell a share for \$60 costs \$8 and is held until maturity. Under what circumstances will the seller of the option (the party with the short position) make a profit? Under what circumstances will the option be exercised? Draw a diagram illustrating how the profit from a short position in the option depends on the stock price at maturity of the option.

5.8.2 Margin Requirements, Market Choices, and Contract Adjustments

5. Explain why margin accounts are required when clients write options but not when they buy options.

6. A corporate treasurer is designing a hedging program involving foreign currency options. What are the pros and cons of using (a) the NASDAQ OMX and (b) the over-the-counter market for trading?

7. The treasurer of a corporation is trying to choose between options and forward contracts to hedge the corporation's foreign exchange risk. Discuss the advantages and disadvantages of each.

8. Consider an exchange-traded call option contract to buy 500 shares with a strike price of \$40 and maturity in four months. Explain how the terms of the option contract change when there is

    1. A 10% stock dividend
    2. A 10% cash dividend
    3. A 4-for-1 stock split

5.8.3 Option Pricing Bounds

9. Explain why an American option is always worth at least as much as a European option on the same asset with the same strike price and exercise date.

10. Explain why an American option is always worth at least as much as its intrinsic value.

11. What is a lower bound for the price of a four-month call option on a non-dividend-paying stock when the stock price is \$28, the strike price is \$25, and the risk-free interest rate is 8% per annum?

12. What is a lower bound for the price of a one-month European put option on a non-dividend paying stock when the stock price is \$12, the strike price is \$15, and the risk-free interest rate is 6% per annum?

5.8.4 Early Exercise and Put-Call Parity

13. Give at least two reasons that the early exercise of an American call option on a non-dividend-paying stock is not optimal.

14. The early exercise of an American put is a trade-off between the time value of money and the insurance value of a put. Explain this statement.

15. The price of a non-dividend paying stock is \$19 and the price of a three-month European call option on the stock with a strike price of \$20 is \$1. The risk-free rate is 4% per annum. What is the price of a three-month European put option with a strike price of \$20?

16. List and explain the six factors affecting stock option prices.

6 Options Trading Strategies and Hedging

References

- HULL, John. Options, futures, and other derivatives. Ninth edition. Harlow: Pearson, 2018. ISBN 978-1-292-21289-0.
  - Chapter 12 - Trading Strategies Involving Options
- PIRIE, Wendy L. Derivatives. Hoboken: Wiley, 2017. CFA institute investment series. ISBN 978-1-119-38181-5.
  - Chapter 5 - Derivatives Strategies

Additonal sources on option trading strategies:

- tastylive
- options playbook
- investopedia

Learning Outcomes:

- Understand the structure and purpose of Principal Protected Notes, including how they safeguard the principal amount while offering potential investment gains.
- Analyze strategies that combine positions in the underlying asset with options to manage risk and enhance potential returns, focusing on protective puts and covered calls.
- Explore various option spread strategies, such as bull spreads, bear spreads, calendar spreads, and butterfly spreads, to understand their risk/reward profiles and market outlook implications.
- Examine option combination strategies like straddles, strangles, strips, and straps, highlighting their use in volatile markets to capitalize on significant price movements in either direction.

6.1 Principal Protected Note

Principal Protected Note (PPN) allow investors to participate in high-reward opportunities while safeguarding their principal. This is achieved through a combination of a zero-coupon bond and a derivative, typically a call option.

Example: PPN Structure

Consider a $1,000 PPN structured as follows:

1. **Zero-Coupon Bond Component**: A 3-year zero-coupon bond with a face value of $1,000 ensures principal protection. Assuming a continuously compounded interest rate of 6%, its present value is:

$$PV = 1,000 \times e^{-0.06\times 3} = 835.27$$

   This confirms that an initial investment of $835.27 will grow to $1,000 in 3 years, ensuring capital protection.

2. **Call Option Component**: The remaining $164.73 ($1,000 - $835.27) is used to buy a 3-year at-the-money call option on a stock portfolio, providing upside potential.

Key Factors Affecting PPN Feasibility:

- **Dividends**: High dividend yields reduce the appeal of the call option component.
- **Interest Rates**: Higher rates lower the cost of the zero-coupon bond, making principal protection cheaper.
- **Volatility**: Increased volatility raises the option price but enhances upside potential.

PPNs can be tailored to different investor needs through:

- **Strike Price Adjustments**: Setting options out of the money for higher potential returns.
- **Return Caps**: Imposing limits on maximum gains to reduce option costs.
- **Structural Features**: Incorporating knock-outs, averaging mechanisms, and other innovations to refine risk-return profiles.

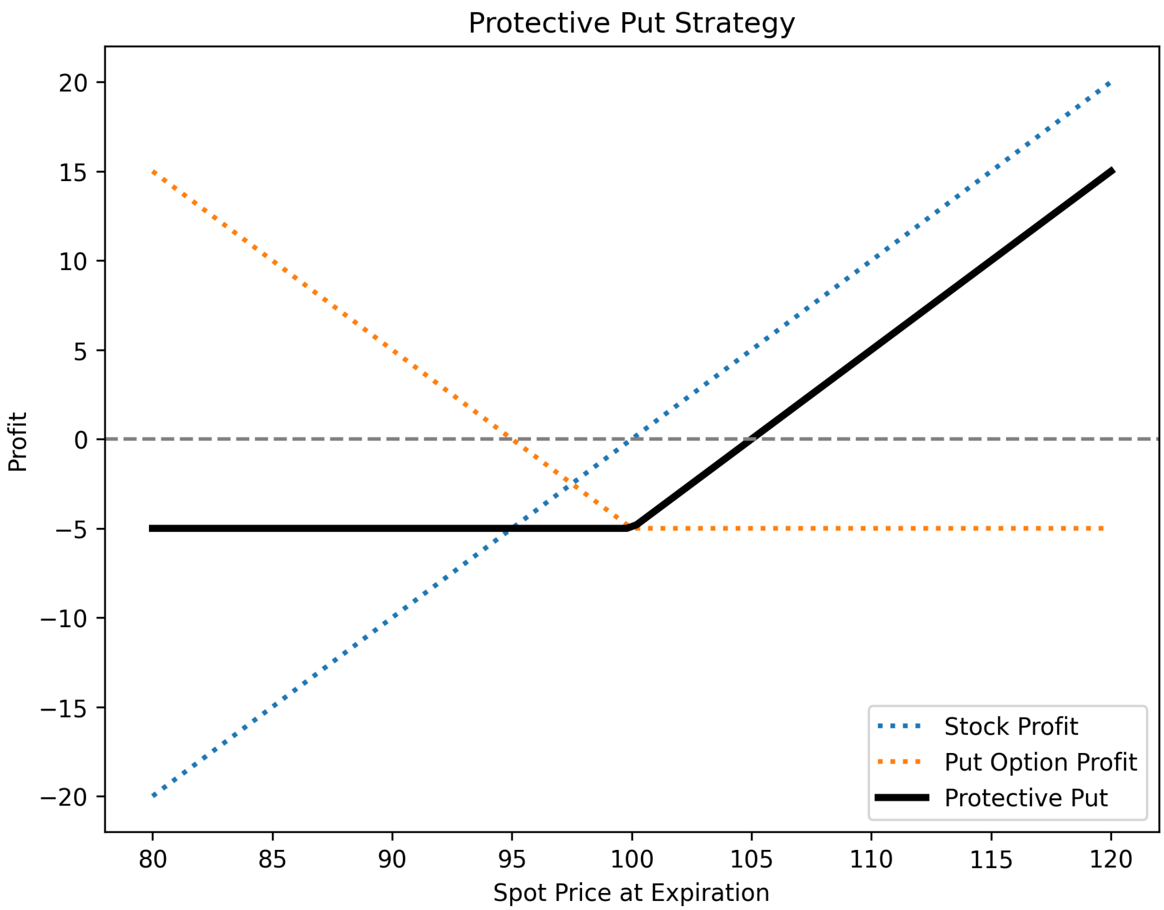
6.2 Combining Underlying and Options

Integrating stock positions with options allows for customized risk/reward profiles suited to different market views. Two key strategies illustrating this approach are the **protective put** and **covered call**.

6.2.1 Protective Put

A **protective put** involves holding a stock while purchasing a put option on the same stock. This strategy:

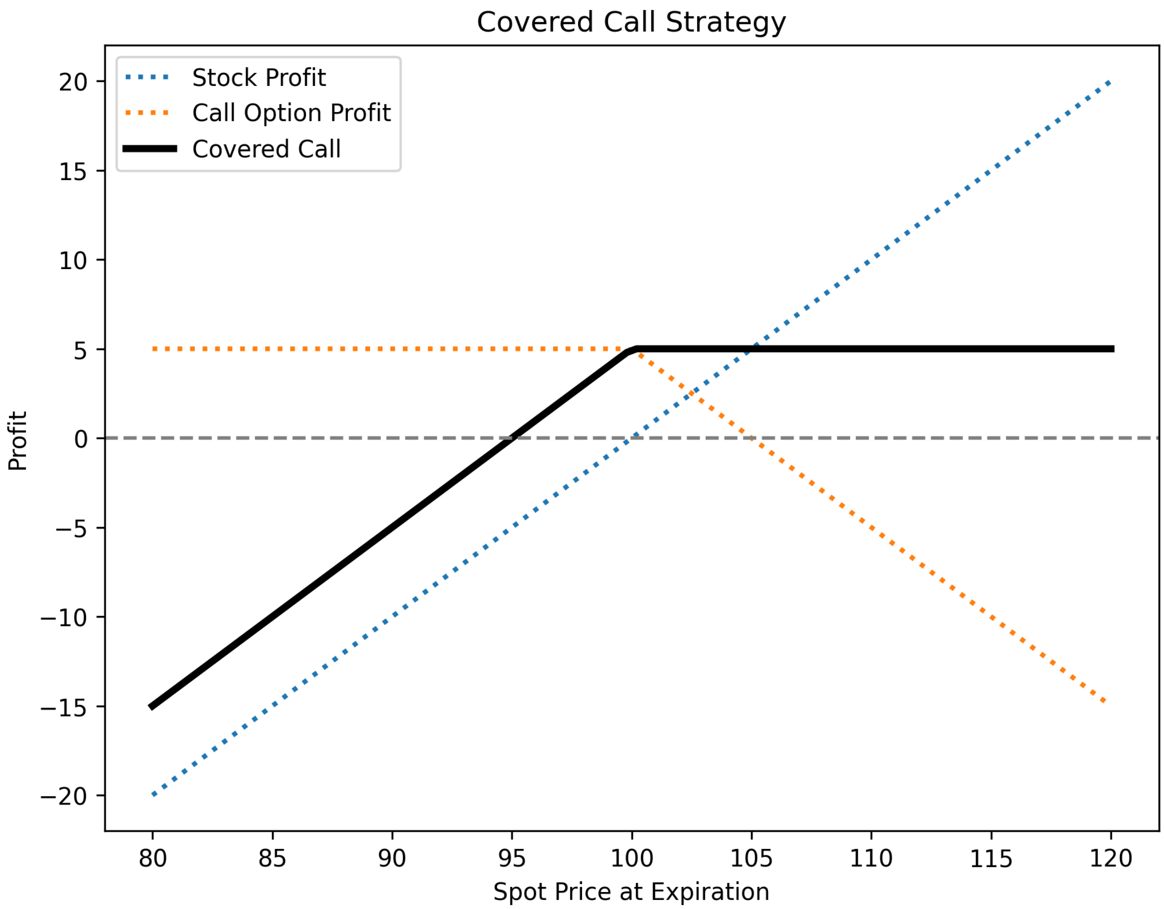
- **Protects against downside risk** by setting a minimum selling price (the put's strike price).
- **Preserves upside potential**, as gains are uncapped beyond the strike price.



6.2.2 Covered Call

A **covered call** involves owning a stock and selling a call option on it. This strategy:

- **Generates income** from the option premium, boosting returns in flat or slightly bullish markets.
- **Limits upside potential** to the call's strike price but provides partial downside protection through premium income.



6.3 Option Spreads

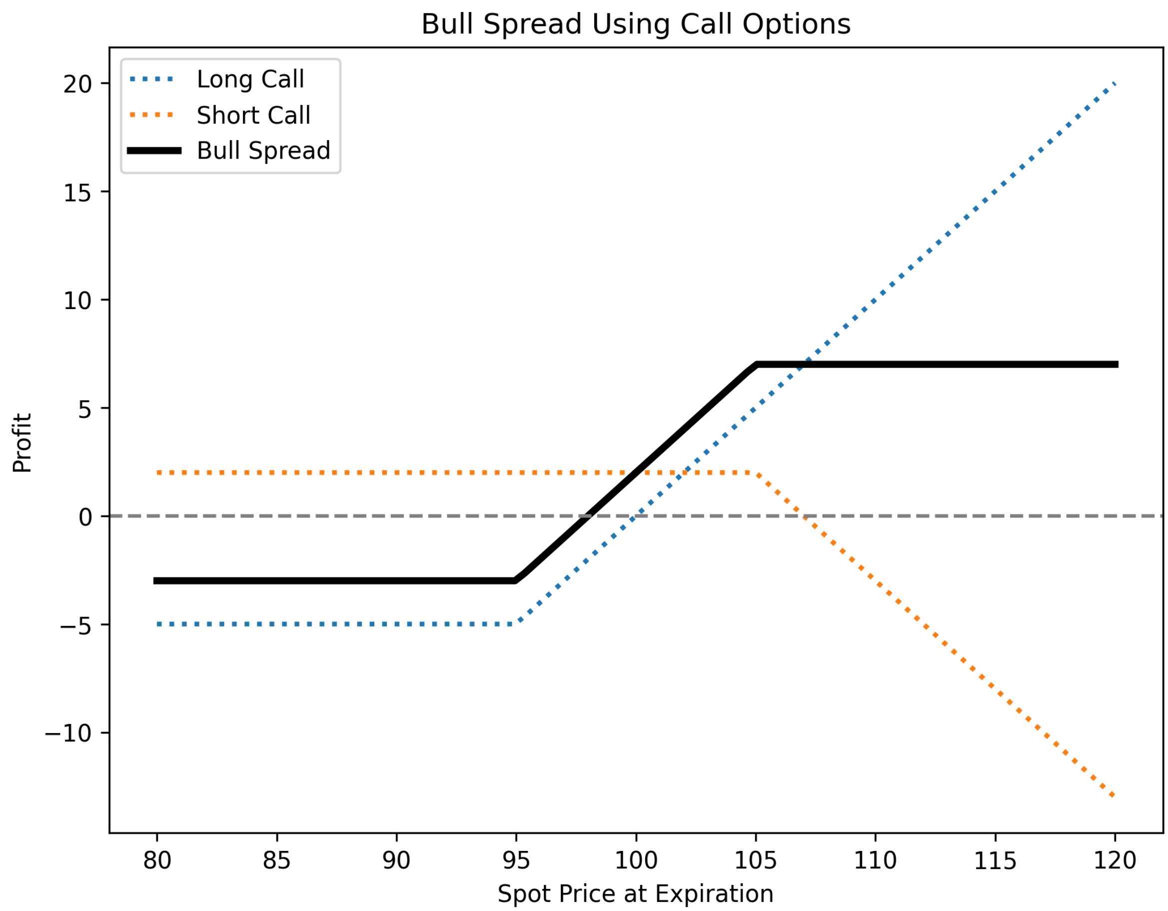
Option spreads involve holding multiple options of the same type to structure risk/reward profiles based on market expectations. These strategies range from bullish to bearish or neutral.

6.3.1 Bull Spread

A **bull spread** profits from a moderate price increase and can be constructed using calls or puts.

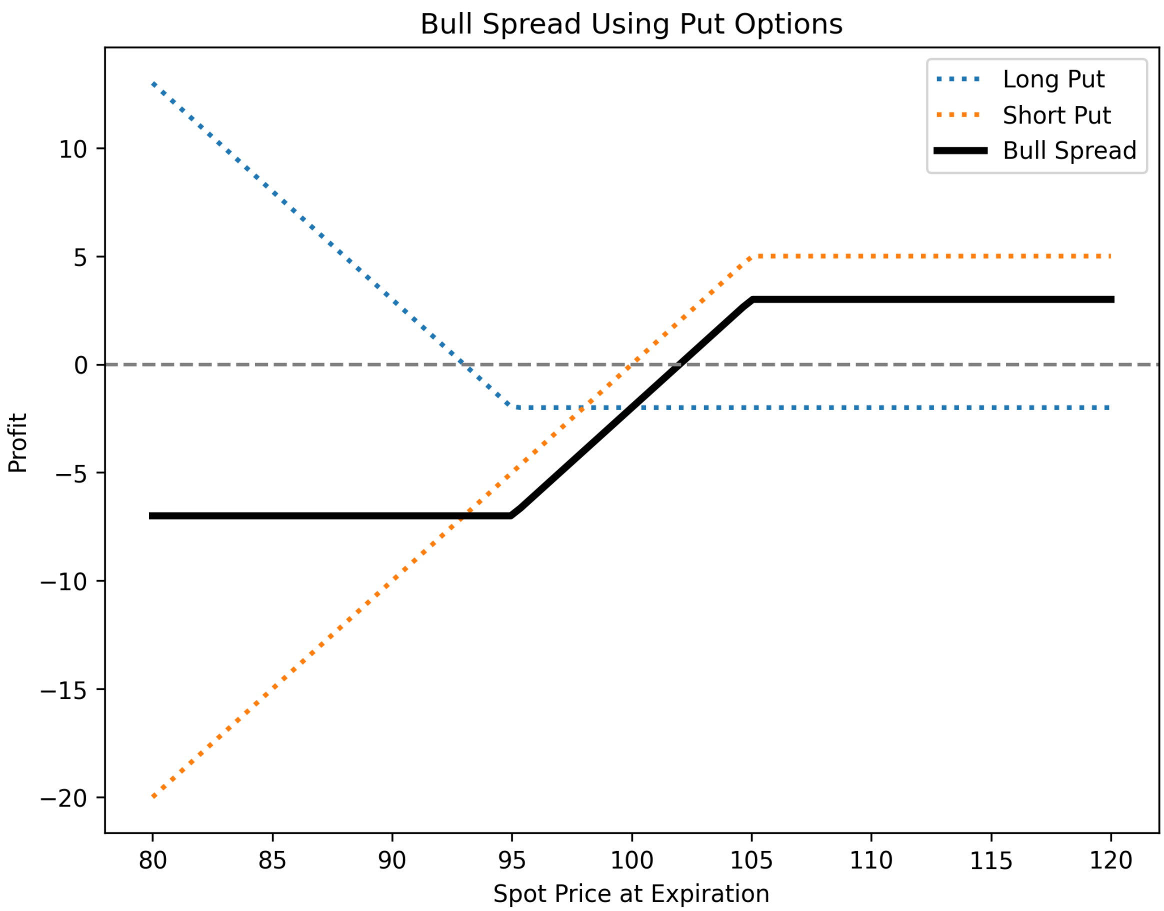
6.3.1.1 Call Bull Spread

- **Position**: Buy a call at a lower strike price, sell a call at a higher strike price (same expiration).
- **Profit**: Limited to the difference between strike prices minus the net premium paid.
- **Risk**: Limited to the net premium paid.
- **Break-even**: Lower strike price + net premium paid.



6.3.1.2 Put Bull Spread

- **Position**: Buy a put at a lower strike price, sell a put at a higher strike price (same expiration).
- **Profit**: Limited to the difference between strike prices minus the net premium paid.
- **Risk**: Limited to the net premium paid.
- **Break-even**: Higher strike price - net premium paid.

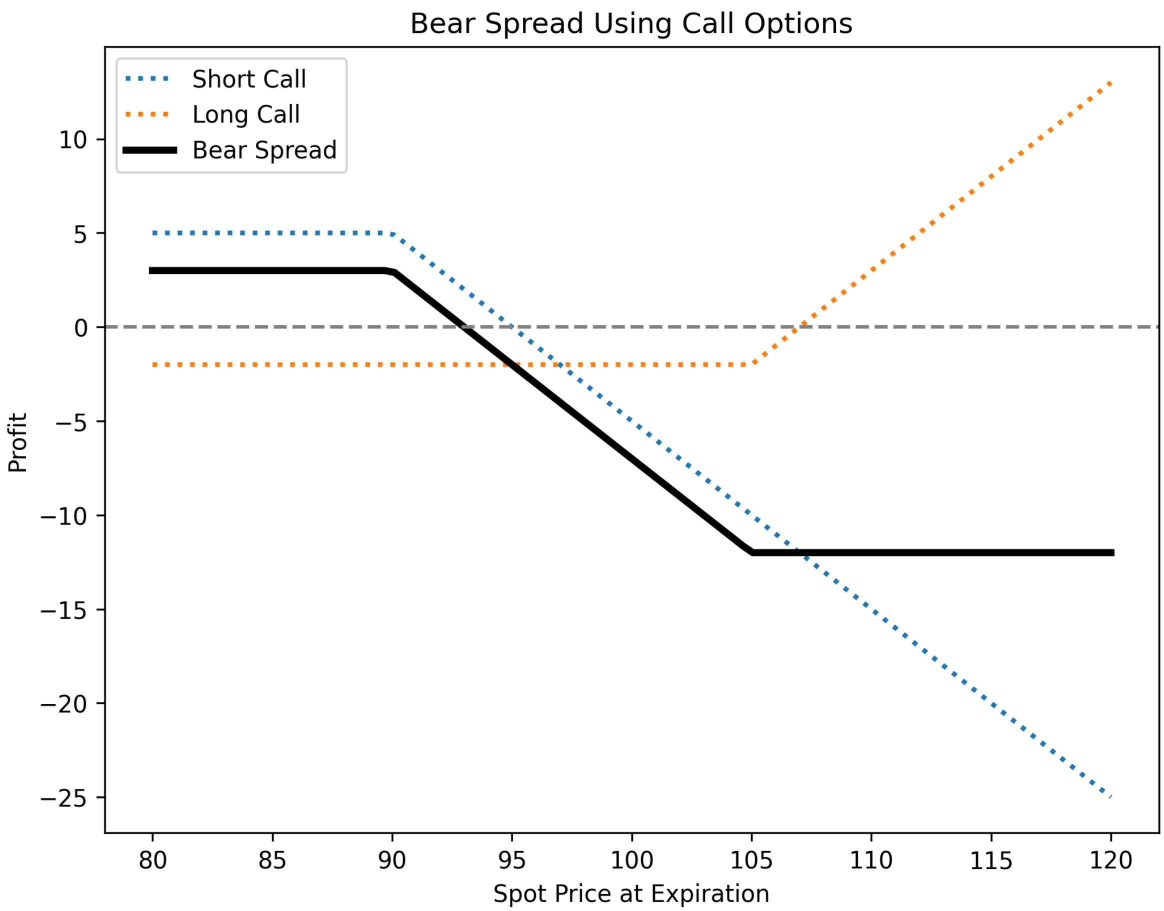


6.3.2 Bear Spread

A **bear spread** profits from a moderate price decline and can also be constructed using calls or puts.

6.3.2.1 Call Bear Spread

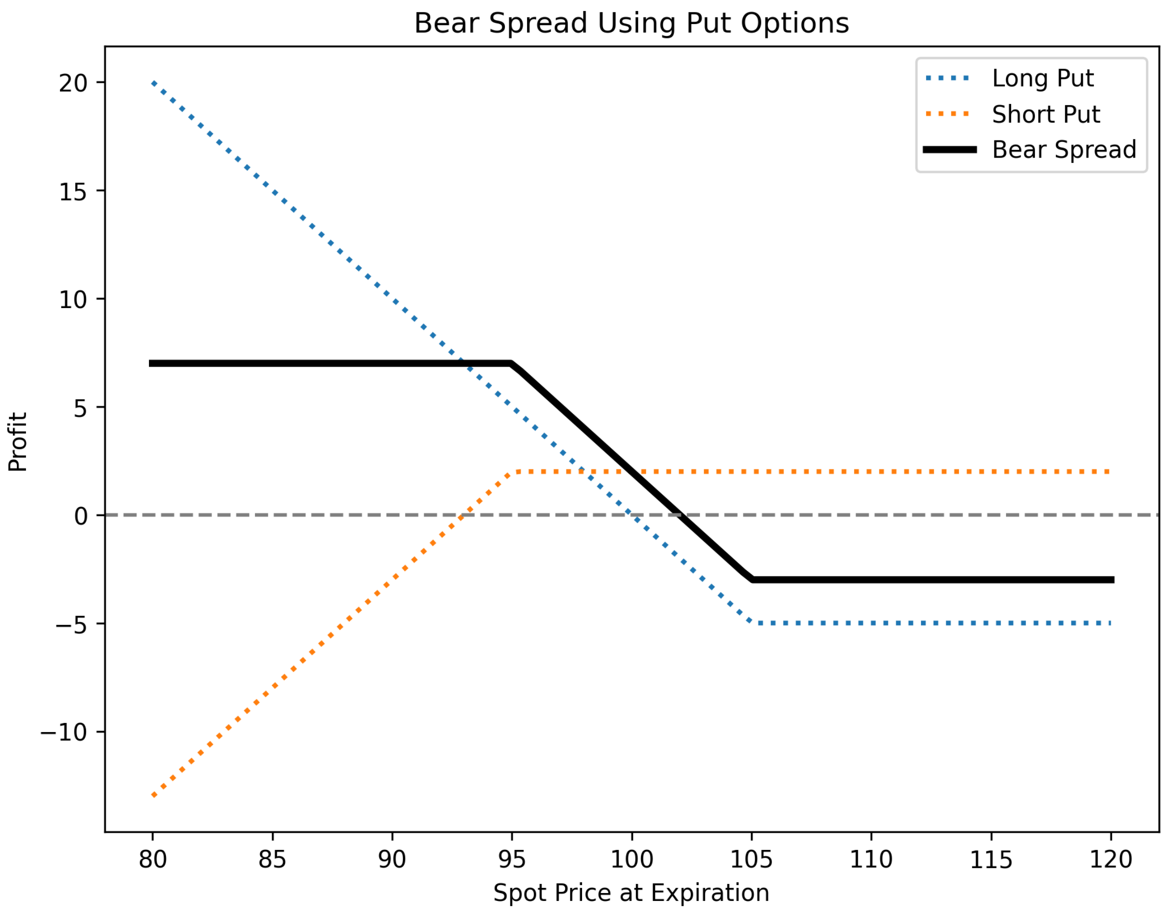
- **Position**: Buy a call at a higher strike price, sell a call at a lower strike price (same expiration).
- **Profit**: Limited to the net premium received.
- **Risk**: Limited to the difference between strike prices minus the net premium received.
- **Break-even**: Lower strike price + net premium received.



6.3.2.2 Put Bear Spread

- **Position**: Buy a put at a higher strike price, sell a put at a lower strike price (same expiration).

- **Profit**: Limited to the net premium received.
- **Risk**: Limited to the difference between strike prices minus the net premium received.
- **Break-even**: Higher strike price - net premium received.



6.3.3 Butterfly Spread

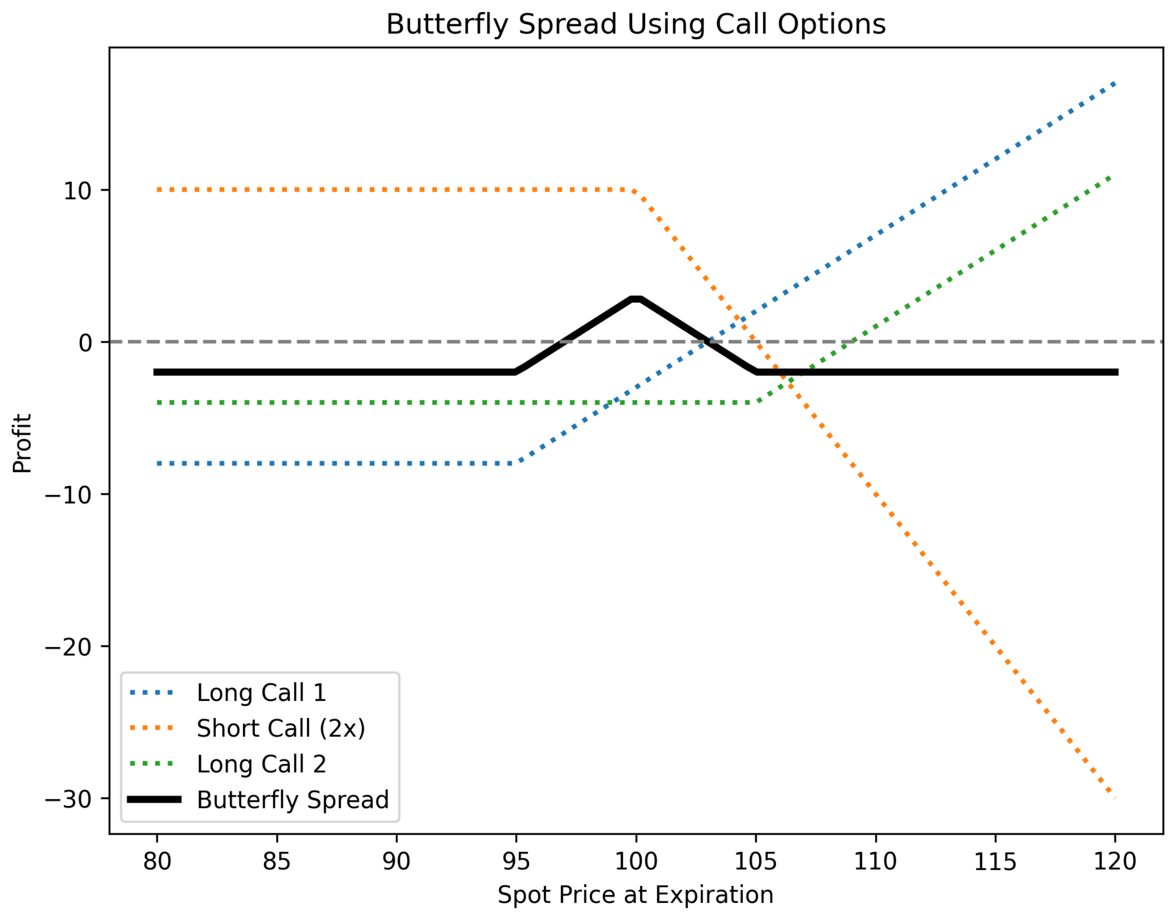
A **butterfly spread** is a neutral strategy used when minimal price movement is expected. It involves three strike prices and can be structured with calls or puts.

6.3.3.1 Call Butterfly Spread

- **Position**:

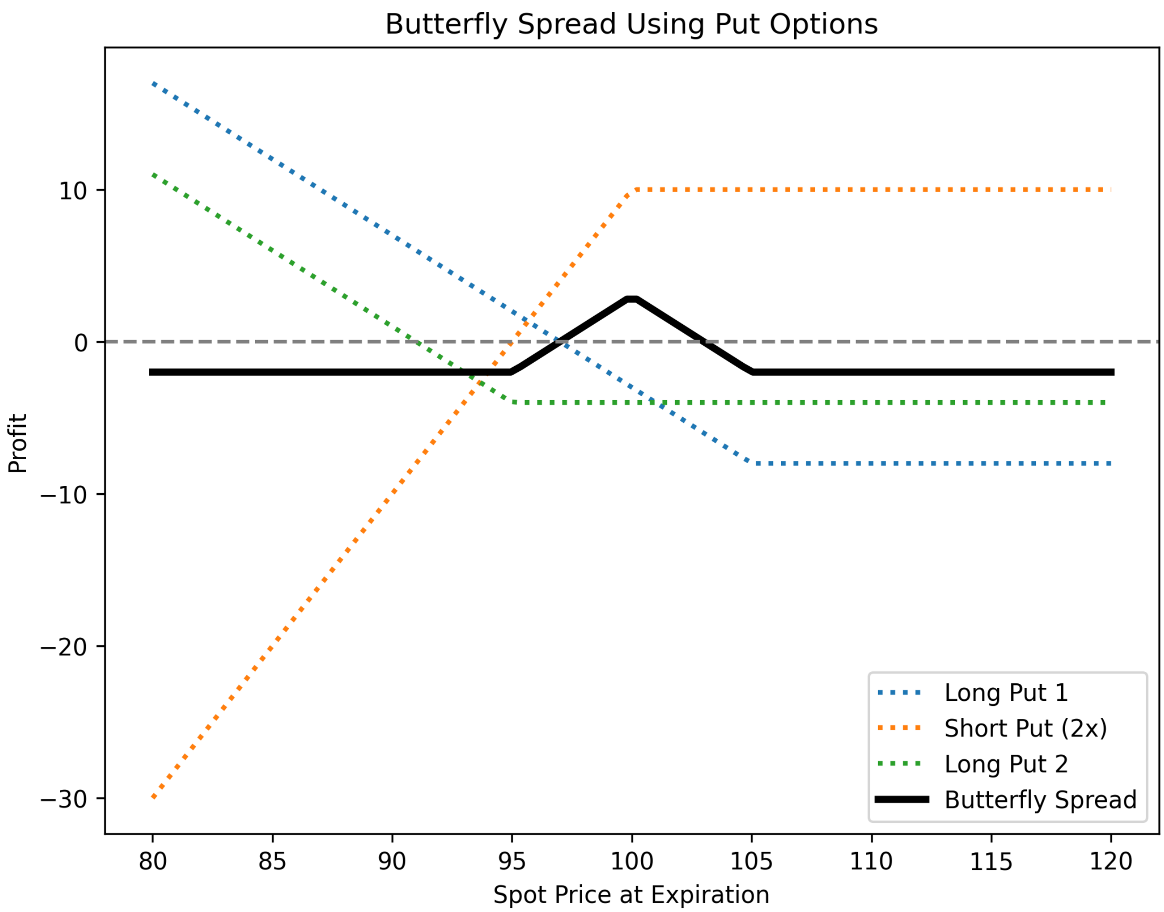
- Buy 1 call at a lower strike price (A).
    - Sell 2 calls at a middle strike price (B).
    - Buy 1 call at a higher strike price (C).
    - All options share the same expiration and have equidistant strike prices.

- **Profit**: Maximum when the underlying price equals the middle strike price at expiration.
- **Risk**: Limited to the net premium paid.
- **Break-even points**:
    - Lower: $A +$ net premium paid
    - Upper: $C -$ net premium paid



6.3.3.2 Put Butterfly Spread

- **Position**:
  - Buy 1 put at a higher strike price (A).
  - Sell 2 puts at a middle strike price (B).
  - Buy 1 put at a lower strike price (C).
  - Same expiration and equidistant strikes.
- **Profit/Risk**: Similar to the call butterfly spread.
- **Break-even points**: Identical to the call butterfly but adjusted for put strike prices.



6.3.4 Key Features of Option Spreads

- **Market Outlook**: Bull spreads suit moderate price increases, bear spreads fit moderate declines, and butterfly spreads work best in stable markets.
- **Risk & Reward**: Bull and bear spreads offer defined risk and reward, while butterfly spreads provide a high reward-to-risk ratio in low-volatility conditions.
- **Cost Efficiency**: Spreads are often more cost-effective than outright option positions.
- **Flexibility**: Investors can adjust spread width and strike prices to tailor risk exposure.

Calendar Spreads

Calendar spreads (also called time or horizontal spreads) involve options with the same underlying and strike price but **different expiration dates**. They capitalize on differences in time decay (theta). A typical calendar spread consists of selling a short-term option and buying a longer-term option of the same type (call or put).
For more details, see tastylive.

6.4 Option Combinations

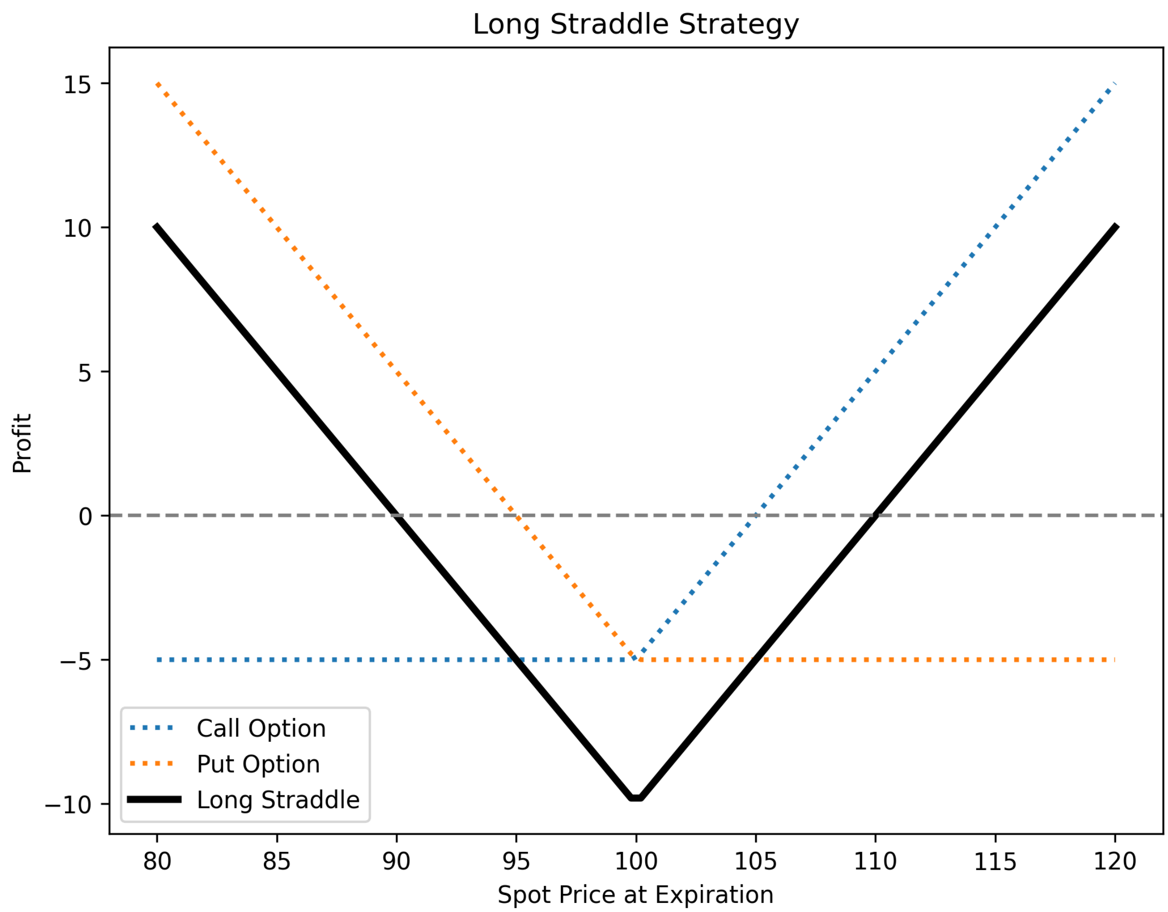
Option combinations involve two or more options of different types, allowing traders to express complex market views and hedge risks beyond single-option positions. Key strategies include **straddles, strangles, strips, and straps**.

6.4.1 Straddles

A **straddle** consists of buying or selling both a **call and a put** with the **same strike price** and expiration.

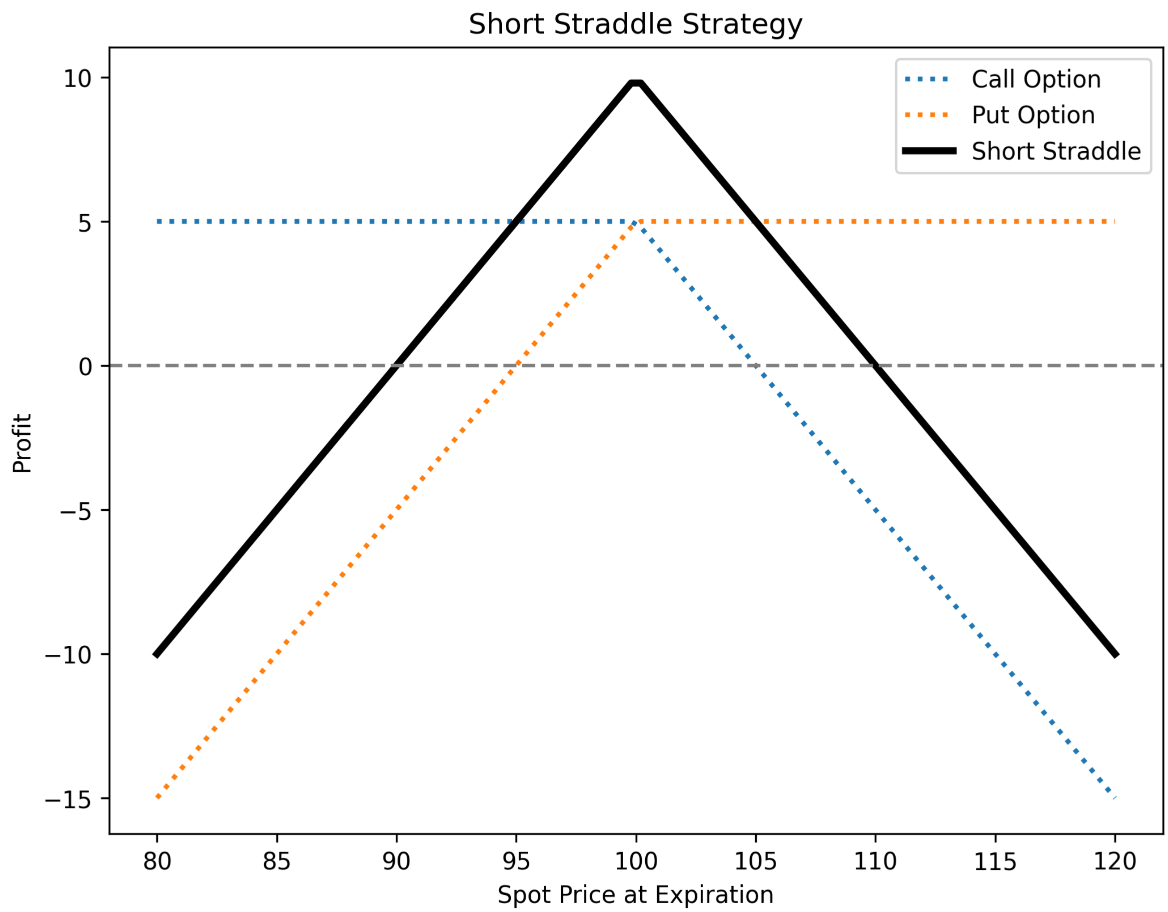
6.4.1.1 Long Straddle

- **Buy a call and a put** at the same strike price.
- Profits from **high volatility**, regardless of direction.
- **Max loss**: Total premium paid.
- **Max profit**: Unlimited if the asset moves significantly.



6.4.1.2 Short Straddle

- **Sell a call and a put** at the same strike price.
- Profits from **low volatility**, expecting price stability.
- **Max profit**: Premiums received.
- **Max risk**: Unlimited if the asset moves significantly.

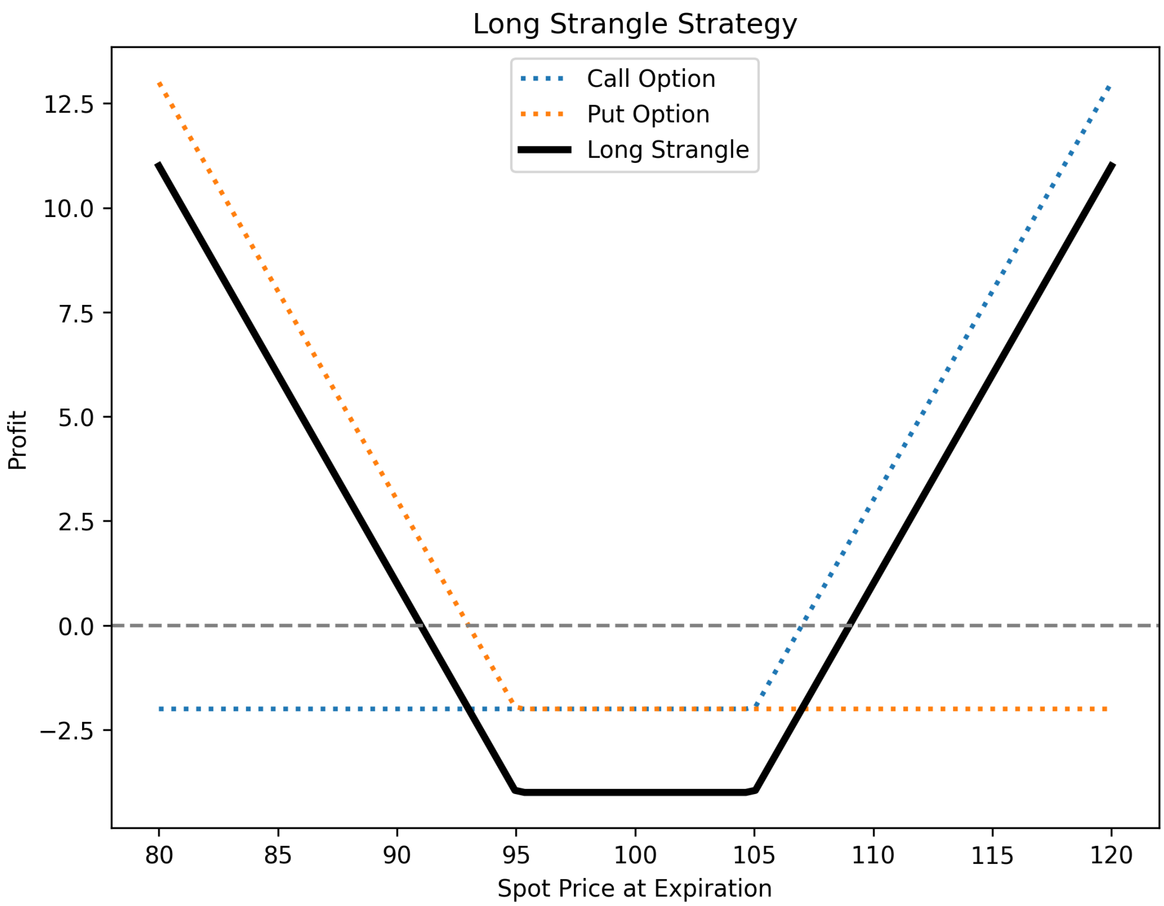


6.4.2 Strangles

A **strangle** is similar to a straddle but uses **different strike prices** (call strike > put strike).

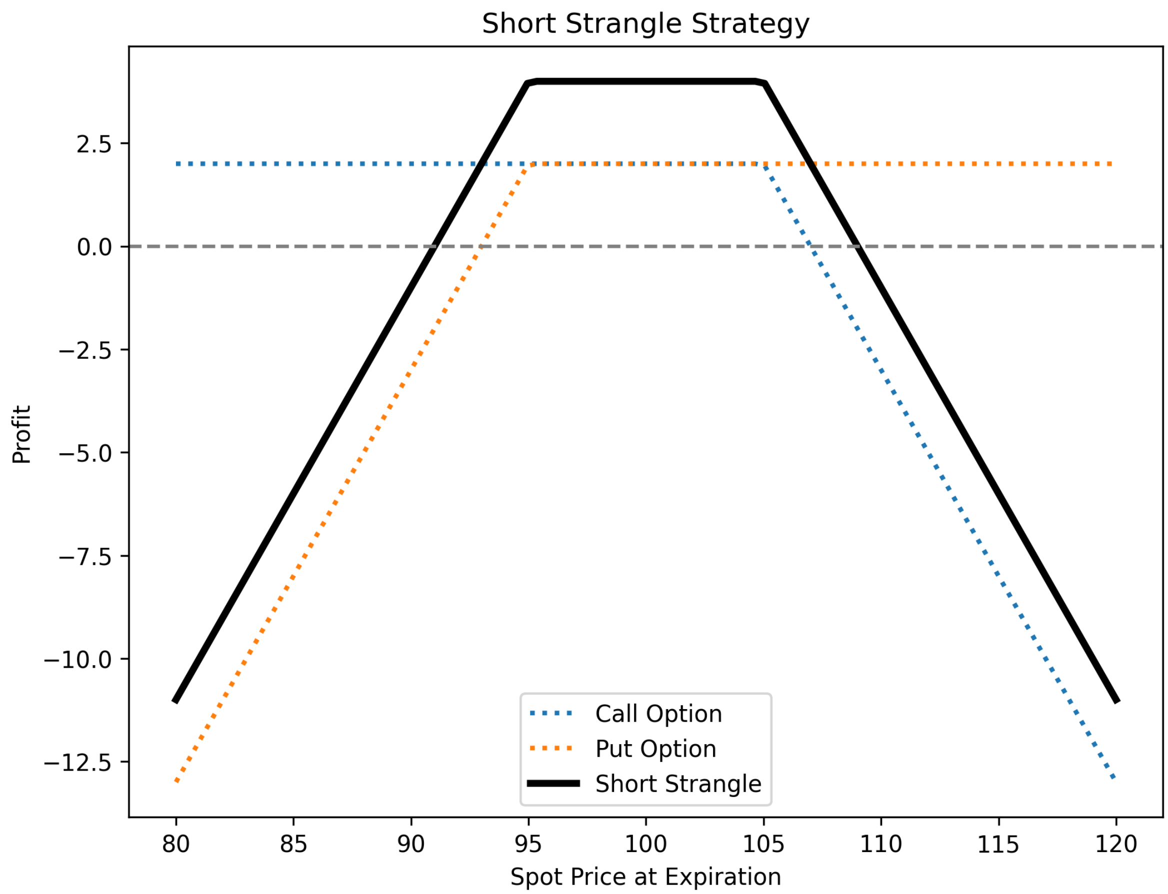
6.4.2.1 Long Strangle

- **Buy an OTM call and an OTM put**.
- Cheaper than a straddle but requires a larger price move to profit.
- **Max loss**: Total premium paid.
- **Max profit**: Unlimited on large price moves.



6.4.2.2 Short Strangle

- **Sell an OTM call and an OTM put**.
- Profits from **low volatility and time decay**.
- **Max profit**: Premiums received.
- **Max risk**: Large if the asset moves significantly.



6.4.3 Strips and Straps

These are variations of the **straddle**, introducing a directional bias while still benefiting from volatility.

- **Strip** (Bearish bias): **Buy 1 call and 2 puts** at the same strike price. More profitable if the price drops.

- **Strap** (Bullish bias): **Buy 2 calls and 1 put** at the same strike price. More profitable if the price rises.

6.4.4 Key Characteristics

- **Volatility Sensitivity**: Long positions (straddles, strangles, strips, straps) **benefit from increased volatility**, while short positions profit from **low volatility**.

- **Directional Bias**: Straddles and strangles are **neutral**, while strips and straps have a **directional tilt**.
- **Risk & Reward**:
  - **Long** positions: Limited risk (premium paid), unlimited reward.
  - **Short** positions: Limited profit, significant risk if the price moves sharply.
- **Breakeven Points**:
  - Each strategy has **two breakeven points**, requiring significant price movement for profitability in long strategies.

6.5 Practice Questions and Problems

1. What is meant by a protective put? What position in call options is equivalent to a protective put?
2. Explain two ways in which a bear spread can be created.
3. When is it appropriate for an investor to purchase a butterfly spread?
4. What trading strategy creates a reverse calendar spread?
5. What is the difference between a strangle and a straddle?
6. A call option with a strike price of \$50 costs \$2. A put option with a strike price of \$45 costs \$3. Explain how a strangle can be created from these two options. What is the pattern of profits from the strangle?
7. Explain how an aggressive bear spread can be created using put options.
8. Suppose that put options on a stock with strike prices \$30 and \$35 cost \$4 and \$7, respectively. How can the options be used to create (a) a bull spread and (b) a bear spread? Draw and explain profit/loss.
9. An investor believes that there will be a big jump in a stock price, but is uncertain as to the direction. Identify six different strategies the investor can follow and explain the differences among them.

7 Reading Week

Tip

Take some rest and revise the first six topics!

8 Option Pricing - Binomial Trees

ℹ References

- HULL, John. Options, futures, and other derivatives. Ninth edition. Harlow: Pearson, 2018. ISBN 978-1-292-21289-0.
  - Chapter 13 - Binomial Trees
- PIRIE, Wendy L. Derivatives. Hoboken: Wiley, 2017. CFA institute investment series. ISBN 978-1-119-38181-5.
  - Chapter 4 - Valuation of Contingent Claims
- Cox, J. C., S. A. Ross, and M. Rubinstein. "Option Pricing: A Simplified Approach," *Journal of Financial Economics* 7 (October 1979): 229–64.

Learning Outcomes:

- Understand the one-step binomial model for option pricing, including how to construct a riskless portfolio and compute option value.
- Grasp risk-neutral valuation and how it simplifies derivative pricing by discounting expected payoffs at the risk-free rate.
- Learn how to extend the binomial model to two steps and calculate option values at each node.
- Understand how to choose the up (u) and down (d) factors based on volatility and time step length, following the Cox-Ross-Rubinstein (CRR) model.
- Explore how the binomial model adapts to various assets, including non-dividend-paying stocks, indices, currencies, and futures.

8.1 A One-Step Binomial Model

This section introduces a fundamental method for pricing derivatives using a simple binomial model. It provides an intuitive approach to option pricing based on constructing a riskless portfolio and applying no-arbitrage principles.

The binomial model forms the foundation for understanding discrete-time derivative pricing and risk-neutral valuation, paving the way for more advanced continuous-time models.

8.1.1 A Simple Binomial Model for a Call Option

Option Characteristics:

- 3-month call option on a stock
- Strike price: $21
- Current stock price: $20
- In 3 months, stock can rise to $22 or fall to $18
- Risk-free rate: 4%

8.1.1.1 Option Pricing Steps

1. **Constructing a Riskless Portfolio**
   - Long Δ shares, short 1 call option.
   - Make the portfolio value identical in both scenarios:
     - If stock rises to $22: $\$22\Delta - 1$
     - If stock falls to $18: $\$18\Delta$
   - Solving $\$22\Delta - 1 = \$18\Delta$ gives $\Delta = 0.25$.
2. **Valuing the Riskless Portfolio**
   - Future value of the portfolio: $\$22 \times 0.25 - 1 = \$18 \times 0.25 = \$4.50$.
   - Present value: $4.50 \times e^{-0.04 \times 0.25} = \$4.455$
3. **Determining the Option's Value**
   - Value of 0.25 shares: $0.25 \times 20 = \$5$
   - Option price: $\$5 - \$4.455 = \$0.545$

8.1.2 General Binomial Option Pricing Framework

This generalization shows how to price options systematically using a binomial approach, grounded in arbitrage-free pricing and synthetic replication.

8.1.2.1 Core Concepts

- Portfolio value:
  - Up move: $S_0 u\Delta - f_u$
  - Down move: $S_0 d\Delta - f_d$
- **Delta to eliminate risk:**

$$\Delta = \frac{f_u - f_d}{S_0(u-d)}$$

  Delta reflects the option's sensitivity to changes in the underlying price.

- **No-Arbitrage Pricing:**
  - A riskless portfolio must grow at the risk-free rate.

8.1.2.2 Analytical Derivation

- Future value: $S_0 u\Delta - f_u$
- Present value: $(S_0 u\Delta - f_u)e^{-rT}$
- Initial cost: $S_0\Delta - f$

Set initial cost equal to discounted future value:

$$f = S_0\Delta - (S_0 u\Delta - f_u)e^{-rT}$$

Substitute Δ to get the pricing formula:

$$f = [pf_u + (1-p)f_d]e^{-rT}$$

with **risk-neutral probability** of an upward price movement:

$$p = \frac{e^{rT} - d}{u - d}$$

- p and $1-p$ represent the probabilities of up and down moves in a **risk-neutral world**.
- In this world, expected payoffs are discounted at the risk-free rate—capturing the core idea of derivative pricing in arbitrage-free markets.

8.1.3 Key Binomial Option Pricing Formulas

- **Option price:**

$$f = [pf_u + (1-p)f_d]e^{-rT}$$

- **Risk-neutral probability:**

$$p = \frac{e^{rT} - d}{u - d}$$

8.1.3.1 Variable Definitions:

- f: Option price
- p: Risk-neutral probability of up move
- $1-p$: Risk-neutral probability of down move
- f_u, f_d: Option payoffs in up/down states
- u, d: Up/down factors
- T: Time to maturity (in years)
- r: Risk-free interest rate (continuously compounded, annualized)

Note

Example: Binomial Option Pricing
Given: $u = 1.1$, $d = 0.9$, $r = 0.04$, $T = 0.25$, $f_u = 1$, $f_d = 0$

- **Compute p:**

$$p = \frac{e^{0.04\times 0.25} - 0.9}{1.1 - 0.9} = 0.5503$$

- **Compute f:**

$$f = e^{-0.04\times 0.25}(0.5503 \times 1 + 0.4497 \times 0) = 0.545$$

The option's value is \$0.545—its expected payoff discounted at the risk-free rate.

8.2 Risk-Neutral Valuation Framework

Risk-neutral valuation is a core concept in derivative pricing. It simplifies valuation by assuming the underlying asset grows at the **risk-free rate**, allowing us to price derivatives using **expected payoffs discounted at that same rate**.

8.2.1 Core Concept

- **Expected Stock Price under Risk Neutrality:**
  In the binomial model, with up/down probabilities p and $1-p$, the expected stock price at time T, discounted at the risk-free rate, equals the current stock price:

  $$\mathbb{E}[S_T] = S_0 e^{rT}$$

  This reflects the assumption that, in a risk-neutral world, the stock grows at rate r.

- **Binomial Trees in Derivative Pricing:**
  Binomial trees illustrate that, for pricing purposes, we can:

  - Assume the stock grows at the risk-free rate,
  - Discount future payoffs at the same rate.

  This removes the need to estimate real-world returns, focusing purely on arbitrage-free pricing in a **risk-neutral world**.

Example Revisited

Given:
$u = 1.1,\ d = 0.9,\ r = 0.04,\ T = 0.25,\ f_u = 1,\ f_d = 0$

1. **Solve for p using stock price consistency**:

$$22p + 18(1-p) = 20e^{0.04\times0.25} \Rightarrow p = 0.5503$$

2. **Expected option payoff under risk-neutral probability**:

$$0.5503 \times 1 + 0.4497 \times 0 = 0.5503$$

3. **Discounting to present value**:

$$0.5503 \times e^{-0.04\times0.25} = 0.545$$

This matches the price obtained from the no-arbitrage approach, confirming that **risk-neutral valuation and no-arbitrage yield the same result**.

8.2.2 Why the Real-World Expected Return Doesn't Matter

- **Real-world probabilities are already embedded in today's stock price.**
- When pricing derivatives, we don't need to model the asset's real-world expected return.
- All pricing is done under the **risk-neutral measure**, where every asset earns the risk-free rate.

This insight is central to modern finance: **derivative prices depend on the structure of payoffs and arbitrage conditions, not the actual return expectations of the underlying asset.**

8.3 Two-Step Binomial Trees

The two-step binomial tree extends the one-period model, allowing for more realistic option pricing over multiple time steps. It captures more possible paths for the underlying asset and enables valuation of both call and put options by discounting expected payoffs at each node.

8.3.1 Valuing a Call Option

Consider a call option with:

- Strike price: $K = \$21$
- Time step: 3 months
- Annual risk-free rate: $r = 4\%$
- Up/down factors: $u = 1.1$, $d = 0.9$
- Risk-neutral probability: $p = 0.5503$

- At final nodes (D, E, F), option values equal intrinsic payoffs.
- At nodes B and C, compute expected payoff, discounted to present:

$$\text{Node B} = e^{-0.04\times0.25}(0.5503 \times 3.2 + 0.4497 \times 0) = \$1.7433$$

$$\text{Node C} = 0 + 0 = \$0$$

$$\text{Node A} = e^{-0.04\times0.25}(0.5503 \times 1.7433 + 0.4497 \times 0) = \$0.9497$$

8.3.2 Valuing a Put Option

For a put option with:

- Strike price: $K = \$52$
- Time step: 1 year

- $r = 5\%$, $u = 1.2$, $d = 0.8$, $p = 0.6282$

Working backward from the final nodes:

$$\text{Node C} = e^{-0.05 \times 1}(0.6282 \times 4 + 0.3718 \times 20) = \$9.4636$$
$$\text{Node B} = e^{-0.05 \times 1}(0.6282 \times 0 + 0.3718 \times 4) = \$1.4147$$
$$\text{Node A} = e^{-0.05 \times 1}(0.6282 \times 1.4147 + 0.3718 \times 9.4636) = \$4.1923$$

8.3.3 Impact of the American Feature on a Put Option

American options allow early exercise, which can increase their value.

- At node C, early exercise yields \$12, replacing the original value of \$9.4636.
- Updating node A with this new value:

$$\text{Node A} = e^{-0.05 \times 1}(0.6282 \times 1.4147 + 0.3718 \times 12.0000) = \$5.0894$$

- The American feature increases the option's value from \$4.1923 to \$5.0894 due to early exercise flexibility.
- The same logic applies to American call options, though early exercise is often less advantageous due to the time value of money.

8.4 Choosing u and d for Binomial Models

In binomial models, choosing appropriate up (u) and down (d) factors is key to accurately reflecting the asset's volatility—its price fluctuation over time. A common method ties these factors directly to volatility, ensuring the model captures realistic risk and return dynamics.

8.4.1 Cox-Ross-Rubinstein (CRR) Approach

Cox, Ross, and Rubinstein (1979) proposed a widely used method that links u and d to the asset's volatility (σ) and time step length (Δt):

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = \frac{1}{u} = e^{-\sigma\sqrt{\Delta t}}$$

This structure:

- Reflects the log-normal nature of stock prices,
- Aligns the model with empirical asset behavior,
- Ensures symmetry: one step up followed by one down returns the price to its original level.

> Girsanov's Theorem and Volatility Use
>
> Girsanov's Theorem underpins the use of **real-world volatility** in risk-neutral pricing:
>
> - **Volatility is unchanged** between the real and risk-neutral worlds.
> - This allows us to observe volatility in the real world and apply it directly in binomial models under risk-neutral valuation.
>
> Though expected returns differ across these worlds, volatility does not—enabling a consistent and simplified approach to option pricing focused on discounting expected payoffs at the risk-free rate.

8.5 The Binomial Tree Model Summary

8.5.1 Core Formulas

8.5.1.1 1. Price Movement Factors

To reflect asset volatility (σ) over a time step (Δt), define:

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = \frac{1}{u} = e^{-\sigma\sqrt{\Delta t}}$$

This ensures asset prices follow a log-normal distribution, consistent with empirical behavior.

8.5.1.2 2. Risk-Neutral Probability

The risk-neutral probability of an up move is:

$$p = \frac{a - d}{u - d}, \quad \text{where } a = e^{r\Delta t}$$

Here, a represents the expected growth of the asset under the **risk-free rate** r, possibly adjusted for dividends or interest differentials.

8.5.1.3 3. Option Valuation

For a one-step binomial model, the option value is:

$$f = [pf_u + (1 - p)f_d]e^{-rT}$$

Where:

- f_u, f_d: payoffs in the up/down states
- T: time to maturity

8.5.2 Adjusting for Different Asset Types

The binomial framework is flexible—only the expression for a changes based on the underlying asset:

- **Nondividend-Paying Stocks:**

$$a = e^{r\Delta t}$$

- **Stock Indices** (with continuous dividend yield q):

$$a = e^{(r-q)\Delta t}$$

- **Currencies** (domestic rate r, foreign rate r_f):

$$a = e^{(r-r_f)\Delta t}$$

- **Futures Contracts** (cost-of-carry already included in futures price):

$$a = 1$$

This flexibility allows the binomial model to adapt to a wide range of derivative pricing scenarios.

8.6 Practice Questions and Problems

1. A stock price is currently \$40. It is known that at the end of one month it will be either \$42 or \$38. The risk-free interest rate is 8% per annum with continuous compounding. What is the value of a one-month European call option with a strike price of \$39?

> **Solution**
>
> Option price = 1.69

2. A stock price is currently \$50. It is known that at the end of six months it will be either \$45 or \$55. The risk-free interest rate is 10% per annum with continuous compounding. What is the value of a six-month European put option with a strike price of \$50?

> **Solution**
>
> Option price = 1.16

3. Explain the no-arbitrage and risk-neutral valuation approaches to valuing a European option using a one-step binomial tree.
4. A stock price is currently \$100. Over each of the next two six-month periods it is expected to go up by 10% or down by 10%. The risk-free interest rate is 8% per annum with continuous compounding. What is the value of a one-year European call option with a strike price of \$100?

> **Solution**
>
> Option price = 9.6104

5. For the situation considered in Problem 4, what is the value of a one-year European put option with a strike price of \$100? Verify that the European call and European put prices satisfy put-call parity.

> **Solution**
>
> Option price = 1.9203

6. Calculate u, d, and p when a binomial tree is constructed to value an option on a foreign currency. The tree step size is one month, the domestic interest rate is 5% per annum, the foreign interest rate is 8% per annum, and the volatility is 12% per annum.

> **Solution**
>
> u = 1.0352, d = 0.966, p = 0.4553

7. A stock index is currently 1,500. Its volatilityis 18%. The risk-free rate is 4% per annum (continuously compounded) for all maturities and the dividend yield on the index is 2.5%. Calculate values for u, d, and p when a six-month time step is used. What is the value a 12-month American put option with a strike price of 1,480 given by a two-step binomial tree.

> **Solution**
>
> Option price = 78.41

9 Option Pricing - Black-Scholes Model

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- Merton, Robert (1973). "Theory of Rational Option Pricing". *Bell Journal of Economics and Management Science*. 4 (1): 141–183.

Learning Outcomes:

- Understand the core principles of the Black-Scholes-Merton model and its role in option pricing.
- Interpret key parameters, assumptions, and limitations of the model in practical contexts.
- Grasp the role of risk-neutral valuation in derivative pricing and risk management.
- Analyze how dividends affect option pricing within the Black-Scholes-Merton framework.
- Explore how volatility impacts option values and shapes investor strategies.

9.1 Black-Scholes-Merton Model

Developed by Fischer Black and Myron Scholes (1973), the Black-Scholes-Merton model provides a **closed-form solution** for pricing European call and put options. It assumes a lognormal stock price distribution and continuous trading, and does **not** account for dividends.

The model prices options based on key variables: stock volatility, risk-free rate, strike price, and time to maturity.

The **binomial model** (Cox, Ross, and Rubinstein, 1979) offers a discrete-time alternative that accommodates **American options** and **dividends**. As the number of time steps increases, binomial results converge to the Black-Scholes formula—demonstrating their close theoretical link.

9.1.1 Model Assumptions

To derive its closed-form solution, the Black-Scholes-Merton model relies on these assumptions:

1. **European options only** (exercise at maturity)
2. **No dividends** paid during the option's life
3. **No arbitrage** opportunities
4. **Unlimited short selling** allowed
5. **No transaction costs** or taxes
6. **Constant risk-free rate** over the life of the option
7. **Constant volatility** of stock returns
8. **Geometric Brownian motion** governs asset price behavior (no jumps, continuous prices)
9. **Continuous trading and perfect liquidity**

9.1.2 Black-Scholes-Merton Pricing Formulas

Let:

- c = price of a European **call** option
- p = price of a European **put** option
- S_0 = current stock price
- K = strike price
- T = time to expiration (in years)

- r = risk-free interest rate (continuously compounded)
- σ = volatility of stock returns
- $N(x)$ = standard normal cumulative distribution function

The option prices are:

$$c = S_0 N(d_1) - Ke^{-rT} N(d_2)$$
$$p = Ke^{-rT} N(-d_2) - S_0 N(-d_1)$$

Where:

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

9.1.3 The $N(x)$ Function

$N(x)$ gives the probability that a standard normal variable is less than x. It's used to weight the expected payoffs under the risk-neutral measure. Values can be obtained from statistical tables or Excel's `NORM.DIST` function.

> Example: Black-Scholes-Merton Option Pricing
>
> **Given:**
>
> - $S_0 = 42$, $K = 40$, $r = 10\%$, $\sigma = 20\%$, $T = 0.5$
>
> **Step 1: Compute d_1 and d_2**
>
> $$d_1 = \frac{\ln(42/40) + (0.1 + 0.2^2/2) \times 0.5}{0.2\sqrt{0.5}} = 0.7693$$
> $$d_2 = 0.7693 - 0.2\sqrt{0.5} = 0.6278$$
>
> **Step 2: Find $N(d_1)$ and $N(d_2)$**
> From Excel or tables:
>
> - $N(0.7693) = 0.7791$, $N(0.6278) = 0.7349$
> - $N(-0.7693) = 0.2209$, $N(-0.6278) = 0.2651$
>
> **Step 3: Discounted strike price**
>
> $$Ke^{-rT} = 40e^{-0.1\times 0.5} = 38.049$$

Step 4: Compute call and put prices

$$c = 42 \times 0.7791 - 38.049 \times 0.7349 = 4.76$$
$$p = 38.049 \times 0.2651 - 42 \times 0.2209 = 0.81$$

9.2 Interpretation of $N(d_1)$ and $N(d_2)$

Understanding $N(d_1)$ and $N(d_2)$ helps uncover the probabilistic foundation of the Black-Scholes formula.

9.2.1 Interpretation 1: Probabilistic Meaning

- $N(d_2)$:
  Interpreted as the **risk-neutral probability** that the call option will be exercised—i.e., the stock price ends up above the strike at maturity.
- $S_0 e^{rT} N(d_1)$:
  Represents the **expected stock price in a risk-neutral world**, conditional on the option being in the money. It adjusts for scenarios where the option would expire worthless.
- **Expected Payoff and Present Value**:
  The expected payoff is:

  $$S_0 e^{rT} N(d_1) - K N(d_2)$$

  Discounting back gives the call price:

  $$c = S_0 N(d_1) - K e^{-rT} N(d_2)$$

9.2.2 Interpretation 2: Weighted Expectation Form

Rewriting the call price:

$$c = e^{-rT} N(d_2) \left(\frac{S_0 e^{rT} N(d_1)}{N(d_2)} - K \right)$$

- e^{-rT}: Present value factor

- $N(d_2)$: Probability of exercise
- $\frac{S_0 e^{rT} N(d_1)}{N(d_2)}$: Expected stock price *if* the option is exercised
- K: Exercise cost

This interpretation views the option price as the **discounted expected gain**, weighted by the probability of exercise.

9.3 Properties of the Black-Scholes Formula

Understanding how option prices respond to changes in the underlying price (S_0) provides key insights:

9.3.1 When $S_0 \to \infty$

- **Call option** (c):
  - $d_1, d_2 \to \infty \to N(d_1), N(d_2) \to 1$
  - $c \to S_0 - Ke^{-rT}$
    The call's value approaches its intrinsic value.
- **Put option** (p):
  - $N(-d_1), N(-d_2) \to 0$
  - $p \to 0$
    Exercise becomes unlikely.

9.3.2 When $S_0 \to 0$

- **Call option** (c):
  - $d_1, d_2 \to -\infty \to N(d_1), N(d_2) \to 0$
  - $c \to 0$
    The option is unlikely to be exercised.
- **Put option** (p):
  - $N(-d_1), N(-d_2) \to 1$
  - $p \to Ke^{-rT} - S_0$
    The put gains intrinsic value as the likelihood of exercise increases.

9.4 Risk-Neutral Valuation

Risk-neutral valuation is a foundational concept in derivative pricing. It simplifies option valuation by assuming all investors are **indifferent to risk**, so all assets earn the **risk-free rate**, regardless of their risk.

In the Black-Scholes-Merton model, a key insight is that the **expected return** of the underlying asset (μ) **does not appear** in the model's differential equation. This implies that the option price is unaffected by investor risk preferences. Instead, pricing can be carried out in a **risk-neutral world**—a hypothetical setting where all assets grow at the risk-free rate.

This approach abstracts from real-world risk attitudes while preserving arbitrage-free pricing, making it both elegant and widely applicable.

9.4.1 Key Principles

- **Risk-neutral assumption**: All assets are priced as if they earn the risk-free rate.
- **Exclusion of μ**: The model's differential equation omits the asset's real-world expected return, reinforcing the irrelevance of risk preferences.
- **Pricing by expected payoff**: Derivatives are priced as the **present value of expected payoffs** under risk-neutral probabilities.

9.4.2 How to Apply Risk-Neutral Valuation

A three-step process:

1. **Assume the stock earns the risk-free rate**
   Under the risk-neutral measure, expected return $= r$.
2. **Compute expected payoff**
   Calculate $E[\max(S_T - K, 0)]$ (for a call) using risk-neutral probabilities.
3. **Discount at the risk-free rate**
   Present value = expected payoff $\times\ e^{-rT}$

Example: Pricing a Call Using Risk-Neutral Valuation

Let K be the strike price, and S_T the stock price at expiration. Under risk-neutral valuation:

- Expected payoff: $E[\max(S_T - K, 0)]$

- Present value of the call option:

$$c = e^{-rT} \cdot E[\max(S_T - K, 0)]$$

This principle is the foundation of both Black-Scholes and binomial option pricing models.

9.5 The Effect of Dividends

Dividends can significantly impact option pricing. When a stock pays dividends, its price typically drops on the ex-dividend date—affecting both **European** and **American** options. The Black-Scholes model must be adjusted to account for this.

9.5.1 Valuing European Options on Dividend-Paying Stocks

To price European options on dividend-paying stocks, adjust the stock price by subtracting the **present value of expected dividends** paid during the option's life:

$$S_{\text{adj}} = S_0 - PV(\text{dividends})$$

Only dividends paid **before expiration** are included. This adjustment reflects the expected stock price drop due to dividend payouts and ensures accurate option valuation.

9.5.2 American Calls and Early Exercise with Dividends

In general, **American call options should not be exercised early**. However, when the stock pays dividends, early exercise **may be optimal**—especially just before an ex-dividend date.

9.5.2.1 When Early Exercise Makes Sense

Early exercise is more likely when the dividend exceeds the time value lost by exercising:

$$\text{Exercise if: Dividend} > K\left(1 - e^{-r(t_{i+1} - t_i)}\right)$$

Where: - K: strike price
- r: risk-free rate
- t_i, t_{i+1}: timing of dividend intervals

This condition compares the immediate dividend gain with the option's remaining time value.

9.5.3 Black's Approximation for American Calls with Dividends

Black's Approximation offers a simple way to estimate the value of American calls on *dividend-paying stocks:

1. **Price a European call** using the adjusted stock price (subtracting dividends).
2. **Price a second European call** that expires just before the final ex-dividend date.
3. **Take the maximum** of the two as an estimate for the American call value.

This method avoids complex early exercise modeling while still accounting for dividend impact. It strikes a balance between **practicality and precision**, and is widely used in industry.

9.6 Volatility in Financial Markets

Volatility measures the degree of variation in an asset's price over time. In option pricing, it represents the level of uncertainty or risk about future price movements and plays a **central role** in determining option value.

- **Definition**: Volatility (σ) quantifies the expected variability in asset returns.
- **Typical Range**: Annual volatility for stocks often falls between **15% and 60%**.
- **Impact on Options**: Higher volatility increases the value of both calls and puts—more uncertainty means a greater chance of large payoff.
- **Not Directly Observable**: Volatility must be estimated—either from historical returns or implied from option prices.

9.6.1 Historical Volatility

- **Calculation**: Standard deviation of past continuously compounded returns, usually annualized.
- **Daily Example**: For a $30 stock with 25% annual volatility:

$$\text{Daily Volatility} = 25\% \times \sqrt{\frac{1}{252}} = 1.57\%$$

- **Annualization**: Multiply daily standard deviation by $\sqrt{252}$ to express it on a yearly basis.

- **During Market Hours**: Volatility is typically higher due to real-time reactions to news and events.
- **Trading Days Convention**: Option time to maturity is measured in **trading days**, not calendar days—reflecting actual exposure to market activity.

9.6.2 Implied Volatility (IV)

- **Definition**: The market's expectation of future volatility, **implied from option prices** via models like Black-Scholes.
- **VIX Index**: A benchmark measure of implied volatility for the S&P 500.
  VIX Website

9.6.2.1 Volatility Smile and Surface

- **Volatility Smile**: Shows how implied volatility varies across strike prices.
- **Volatility Surface**: Extends the smile across different maturities.
- **Market Anomalies**: Deviations from a flat surface (assumed in BSM) highlight real-world frictions, changing risk preferences, or event risk.

9.6.3 Role of Implied Volatility in Option Trading

- **Market Expectations**: Higher IV signals higher expected future price movement $\rightarrow$ more expensive options.
- **Quote Standard**: Options are often quoted by implied volatility instead of price.
- **Relative Valuation**: IV allows comparison of options beyond intrinsic value or time decay.
- **Repricing Over Time**: As market sentiment shifts, IV adjusts—impacting option values and trading strategies.
- **Sentiment Indicator**: Rising IV often signals increased fear or uncertainty.
- **Communication Tool**: IV serves as a universal metric for discussing risk and valuation—used by traders, regulators, and risk managers.

9.7 Practice Questions and Problems

1. Name some assumptions of the Black-Scholes-Merton model. Compare with the real world and explain potential issues.
2. Explain the principle of risk-neutral valuation.

3. Calculate the price of a three-month European put option on a non-dividend-paying stock with a strike price of \$50 when the current stock price is \$50, the risk-free interest rate is 10% per annum, and the volatility is 30% per annum. (Tip: Use Excel function NORMDIST to calculate $N(x)$)

> **Solution**
>
> Put option price = 2.37

4. What difference does it make to your calculations in the previous Problem if a dividend of \$1.50 is expected in two months? (Tip: Use Excel function NORMDIST to calculate $N(x)$)

> **Solution**
>
> Put option price = 3.03

5. What is the price of a European call option on a non-dividend-paying stock when the stock price is \$52, the strike price is \$50, the risk-free interest rate is 12% per annum, the volatility is 30% per annum, and the time to maturity is three months?

> **Solution**
>
> Call option price = 5.06

6. What is the price of a European put option on a non-dividend-paying stock when the stock price is \$69, the strike price is \$70, the risk-free interest rate is 5% per annum, the volatility is 35% per annum, and the time to maturity is six months?

> **Solution**
>
> Put option price = 6.40

7. Consider an option on a non-dividend-paying stock when the stock price is \$30, the exercise price is \$29, the risk-free interest rate is 5% per annum, the volatility is 25% per annum, and the time to maturity is four months.

   a. What is the price of the option if it is a European call?
   b. What is the price of the option if it is an American call?
   c. What is the price of the option if it is a European put?
   d. Verify that put–call parity holds.

Solution

European call = 2.52 European put = 1.05

8. What is implied volatility? How can it be calculated?
9. The volatility of a stock price is 30% per annum. What is the standard deviation of the percentage price change in one trading day?

Solution

1-day std. = 1.9%

10. What is the actual implied volatility of the S&P 500? Try to interpret the value.
11. A call option on a non-dividend-paying stock has a market price of $2.50. The stock price is $15, the exercise price is $13, the time to maturity is three months, and the risk-free interest rate is 5% perannum. What is the implied volatility?

Solution

Implied volatility = 39.6%

10 Options on Indices, Currencies, and Futures

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Learning Outcomes:

- Understand how index options are used for hedging and portfolio protection, and their advantages over stock-based strategies.
- Apply the Black-Scholes model to value index options.
- Use currency options to hedge FX risk, and value them using spot and forward prices.
- Evaluate European futures options using Black's model, and differentiate options on futures from those on physical assets.

10.1 Options on Stock Indices

Index options allow investors to speculate on or hedge against movements in a broad market index, without trading individual stocks. Major U.S. index options include:

- **S&P 100 Index (OEX and XEO):** OEX options are American-style; XEO options are European-style. Both provide exposure to the top 100 S&P stocks.
- **S&P 500 Index (SPX):** European-style options tracking 500 large-cap U.S. stocks. Widely used by institutional investors for hedging and tactical positioning.

- **Dow Jones Index (DJX):** Priced at 1/100th of the Dow Jones Industrial Average, offering a low-cost hedge or exposure to 30 major U.S. stocks.
- **Nasdaq 100 Index (NDX):** European-style options on the 100 largest non-financial Nasdaq stocks, heavily weighted toward technology.

ll are exchange-traded, settled in cash, and each contract represents 100 times the index level, avoiding physical delivery.

10.1.1 Using Index Options for Portfolio Insurance

Portfolio insurance uses index put options to protect against market declines.

Notation:

- S_0: current index level
- K: strike price
- β: portfolio beta relative to the index

Strategy:

- For $\beta = 1.0$, buy one put per $\$100 \times S_0$ of portfolio value.
- For $\beta \neq 1.0$, buy β puts per $\$100 \times S_0$.
- Choose K based on desired insurance threshold.

Example: $\beta = 1.0$

Portfolio:

- Value = \$500,000
- $\beta = 1.0$
- Index = 1,000
- Target insured value = \$450,000

Implementation:

- Required: $500{,}000/(100 \times 1{,}000) = 5$ put options
- Strike price: $K = 900$ (10% below current level)

If index falls to 880:

- Portfolio drops to approx. $440,000
- Option payoff:

$$\text{Payoff} = 5 \times (900 - 880) \times 100 = \$10{,}000$$

- Net portfolio value: $440,000 + $10,000 = $450,000

Example: $\beta = 2.0$

Portfolio:

- Value = $500,000
- $\beta = 2.0$
- Index = 1,000
- Target insured value = $450,000
- Risk-free rate = 12% p.a., dividend yield = 4%

Contracts Required:

$$\frac{500{,}000}{100 \times 1{,}000} \times 2 = 10$$

put options

10.1.1.1 Determining Strike Price via CAPM

- Index rises to 1,040 in 3 months: 4% return
- Total return: 5% (4% price + 1% dividends)
- Excess return: $5\% - 3\% = 2\%$ (quarterly risk-free rate)
- Portfolio excess return: $2\% \times 2 = 4\%$

- Net return: $4\% + 3\% - 1\% = 6\%$
- Projected portfolio value:

$$500{,}000 \times 1.06 = 530{,}000$$

- **Result:** Similar calculations can be carried out for other values of the index at the end of the three months. Appropriate strike price for the 10 put option contracts that are purchased is 960 (or 955 when we include dividends).

Index in 3 Months	Portfolio Value in 3 Months ($)
1,080	570,000
1,040	530,000
1,000	490,000
960	450,000
920	410,000
880	370,000

10.2 Valuation of Stock Index Options

Valuing European index options relies on understanding asset price dynamics. The probability distribution of the index price is the same in two cases:

1. When the index starts at S_0 and pays a continuous dividend yield q.
2. When the index starts at $S_0 e^{-qT}$ with no dividends.

This equivalence allows us to adjust the starting price to $S_0 e^{-qT}$ and apply the standard no-dividend valuation framework, simplifying the analysis.

10.2.1 Lower Bounds and Put-Call Parity

The intrinsic value sets a lower bound on option prices, preventing arbitrage.

- **Call Lower Bound:**
$$c \geq \max(S_0 e^{-qT} - K e^{-rT}, 0)$$
- **Put Lower Bound:**
$$p \geq \max(K e^{-rT} - S_0 e^{-qT}, 0)$$

- **Put-Call Parity:**
$$c + Ke^{-rT} = p + S_0 e^{-qT}$$

This relationship links European calls and puts with the same K and T.

10.2.2 Black-Scholes with Dividend Yield

The Black-Scholes model is extended to incorporate continuous dividend yields q:

- **Call:**
$$c = S_0 e^{-qT} N(d_1) - Ke^{-rT} N(d_2)$$
- **Put:**
$$p = Ke^{-rT} N(-d_2) - S_0 e^{-qT} N(-d_1)$$

Where:
$$d_1 = \frac{\ln(S_0/K) + (r - q + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

10.2.3 Valuation Using Forward Prices

European options can also be valued using the forward price $F_0 = S_0 e^{(r-q)T}$:

- **Call:**
$$c = e^{-rT}[F_0 N(d_1) - KN(d_2)]$$
- **Put:**
$$p = e^{-rT}[KN(-d_2) - F_0 N(-d_1)]$$

Where:
$$d_1 = \frac{\ln(F_0/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

This method is especially effective for dividend-paying assets, using forward or futures prices directly.

10.2.4 Implied Forward Prices and Dividend Yields

Option prices can be used to infer forward prices and dividend yields:

- **Implied Forward Price:**
$$F_0 = K + (c - p)e^{rT}$$
- **Implied Dividend Yield:**
$$q = -\frac{1}{T} \ln \left(\frac{c - p + Ke^{-rT}}{S_0} \right)$$

These estimates help uncover market expectations and are key inputs in OTC markets and early exercise decisions for American options.

10.3 Currency Options

Currency options hedge against foreign exchange risk and are widely used by firms with international operations. They are traded both on exchanges (e.g., NASDAQ OMX) and over the counter (OTC), playing a key role in financial risk management.

Currencies behave like dividend-paying assets, with the foreign interest rate r_f acting as the yield. This allows the use of stock option pricing models, treating r_f analogously to a dividend yield.

10.3.1 Range Forward Contracts

Range forwards are tailored contracts that limit currency exposure within a specified band:

- To **hedge currency payments**, sell a put at K_1 and buy a call at K_2 ($K_2 > K_1$).
- To **hedge currency receipts**, buy a put at K_1 and sell a call at K_2.
- The premium received typically offsets the premium paid, keeping net cost low.

These contracts provide partial protection while allowing limited participation in favorable exchange rate movements.

10.3.2 Black-Scholes for Currency Options

Adapting Black-Scholes to currency options yields:

- **Call:**
$$c = S_0 e^{-r_f T} N(d_1) - K e^{-rT} N(d_2)$$
- **Put:**
$$p = K e^{-rT} N(-d_2) - S_0 e^{-r_f T} N(-d_1)$$

Where:
$$d_1 = \frac{\ln(S_0/K) + (r - r_f + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

10.3.3 Valuation Using Forward Prices

Alternatively, use the forward exchange rate $F_0 = S_0 e^{(r-r_f)T}$:

- **Call:**
$$c = e^{-rT}[F_0 N(d_1) - K N(d_2)]$$
- **Put:**
$$p = e^{-rT}[K N(-d_2) - F_0 N(-d_1)]$$

Where:
$$d_1 = \frac{\ln(F_0/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

This forward-based approach simplifies valuation by embedding interest rate differentials directly into the pricing.

10.4 Options on Futures and Black's Model

Futures options grant the right, but not the obligation, to enter into a futures contract at a set price before expiration. These are usually American-style and expire shortly before or on the first delivery date of the underlying futures contract.

10.4.1 Mechanics of Futures Options

10.4.1.1 Call Options

Exercising a call on futures gives the holder a **long futures position** and a cash payout equal to the difference between the most recent settlement price and the strike price.

Example

- Strike = 320¢/lb, futures price = 331¢, latest settlement = 330¢
- Each contract = 25,000 lbs
- Cash payout: $25{,}000 \times (330 - 320)¢ = \$2{,}500$
- Close futures position for: $25{,}000 \times (331 - 330)¢ = \$250$
- **Total payoff:** \$2,750

10.4.1.2 Put Options

Exercising a put gives a **short futures position** and a cash payout based on the strike minus the settlement price.

Example

- Strike = 600¢/bushel, futures = 580¢, settlement = 579¢
- Each contract = 5,000 bushels
- Cash payout: $5{,}000 \times (600 - 579)¢ = \$1{,}050$
- Close short for: $5{,}000 \times (579 - 580)¢ = -\$50$
- **Total payoff:** \$1,000

10.4.2 Futures Option Payoffs (Immediate Closeout)

The payoff for futures options, if the position in the futures contract is closed out immediately after exercising the option, is straightforward:

- **Call:**
$$\text{Payoff} = F - K$$
- **Put:**
$$\text{Payoff} = K - F$$

Where F is the futures price at exercise, and K is the strike.

10.4.3 Why Use Futures Options?

1. **Spot Option Equivalence:**
   When expiration coincides, European futures and spot options have the same value.
2. **Greater Liquidity:**
   Futures markets are often more liquid than spot markets, improving execution quality.

3. **No Physical Delivery:**
   Exercise results in a futures position, not physical delivery—useful for commodities.
4. **Integrated Markets:**
   Futures options and futures trade on the same exchange, simplifying trading.
5. **Lower Costs:**
   Centralized trading and standardized contracts often reduce transaction costs.

10.4.4 Put-Call Parity and Lower Bounds for European Futures Options

10.4.4.1 Put-Call Parity

Consider two portfolios:

1. A European call plus cash: $c + Ke^{-rT}$
2. A European put plus a long futures position and cash: $p + F_0 e^{-rT}$

To prevent arbitrage, both must have equal value at maturity:

$$c + Ke^{-rT} = p + F_0 e^{-rT}$$

This parity links European futures calls and puts with the same strike K and maturity T.

10.4.4.2 Lower Bounds

Minimum prices to prevent arbitrage:

- **Call:**
$$c \geq (F_0 - K)e^{-rT}$$
- **Put:**
$$p \geq (K - F_0)e^{-rT}$$

10.4.5 Black's Model for Futures Options

Futures require no initial investment, so their expected return under the risk-neutral measure is zero. This makes them analogous to stocks paying a dividend yield $q = r$, simplifying valuation.

Black (1976) adapted the Black-Scholes model for European options on futures:

- **Call:**
$$c = e^{-rT}[F_0 N(d_1) - K N(d_2)]$$
- **Put:**
$$p = e^{-rT}[K N(-d_2) - F_0 N(-d_1)]$$

Where:

$$d_1 = \frac{\ln(F_0/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

10.4.6 Futures vs. Spot Option Valuation

- **Contango ($F_0 > S_0$):**
American calls on futures > spot calls; American puts on futures < spot puts.
- **Backwardation ($F_0 < S_0$):**
American calls on futures < spot calls; American puts on futures > spot puts.

10.5 Practice Questions and Problems

10.5.1 Index Options

1. A stock index is currently 300, the dividend yield on the index is 3% per annum, and the risk-free interest rate is 8% per annum. What is a lower bound for the price of a six-month European call option on the index when the strike price is 290?

> **Solution**
>
> Lower bound = 16.90

2. Consider a stock index currently standing at 250. The dividend yield on the index is 4% per annum, and the risk-free rate is 6% per annum. A three-month European call option on the index with a strike price of 245 is currently worth \$10. What is the value of a three-month put option on the index with a strike price of 245?

Solution

Put option value = 3.84

3. Calculate the value of a three-month at-the-money European call option on a stock index when the index is at 250, the risk-free interest rate is 10% per annum, the volatility of the index is 18% per annum, and the dividend yield on the index is 3% per annum.

Solution

Call option value = 11.15

4. An index currently stands at 696 and has a volatility of 30% per annum. The risk-free rate of interest is 7% per annum and the index provides a dividend yield of 4% per annum. Calculate the value of a three-month European put with an exercise price of 700.

Solution

Put option value = 40.6

10.5.2 Currency Options

1. A foreign currency is currently worth \$1.50. The domestic and foreign risk-free interest rates are 5% and 9%, respectively. Calculate a lower bound for the value of a six-month call option on the currency with a strike price of \$1.40 if it is (a) European and (b) American.

Solution

Lower bound European = 0.069 Lower bound American = 0.10

2. Calculate the value of an eight-month European put option on a currency with a strike price of 0.50. The current exchange rate is 0.52, the volatility of the exchange rate is 12%, the domestic risk-free interest rate is 4% per annum, and the foreign risk-free interest rate is 8% per annum.

Solution

Put option value = 0.0162

3. A currency is currently worth \$0.80 and has a volatility of 12%. The domestic and foreign risk-free interest rates are 6% and 8%, respectively. Use a two-step binomial tree to value (a) a European four-month call option with a strike price of 0.79 and (b) an American four-month call option with the same strike price.

> **Solution**
>
> - European Option Value: \$0.0235
> - American Option Value: \$0.0250

10.5.3 Futures Options

1. Consider a two-month futures call option with a strike price of 40 when the risk-free interest rate is 10% per annum. The current futures price is 47. What is a lower bound for the value of the futures option if it is (a) European and (b) American?

> **Solution**
>
> Lower bound European = 6.88 Lower bound American = 7

2. Consider a four-month futures put option with a strike price of 50 when the risk-free interest rate is 10% per annum. The current futures price is 47. What is a lower bound for the value of the futures option if it is (a) European and (b) American?

> **Solution**
>
> Lower bound European = 2.90 Lower bound American = 3

3. Calculate the value of a five-month European futures put option when the futures price is \$19, the strike price is \$20, the risk-free interest rate is 12% per annum, and the volatility of the futures price is 20% per annum.

> **Solution**
>
> Put option value = 1.50

4. A futures price is currently 25, its volatility is 30% per annum, and the risk-free interest rate is 10% per annum. What is the value of a nine-month European call on the futures with a strike price of 26?

Solution

Call option value = 2.01

5. A futures price is currently 60 and its volatility is 30%. The risk-free interest rate is 8% per annum. Use a two-step binomial tree to calculate the value of a six-month European call option on the futures with a strike price of 60. If the call were American, would it ever be worth exercising it early?

Solution

- European Option Value: 4.3155
- American Option Value: 4.4026

6. Suppose that a one-year futures price is currently 35. A one-year European call option and a one-year European put option on the futures with a strike price of 34 are both priced at 2 in the market. The risk-free interest rate is 10% per annum. Identify an arbitrage opportunity.

Solution

Arbitrage profit = $1

10.5.4 Strategic Considerations

1. Would you expect the volatility of a stock index to be greater or less than the volatility of a typical stock? Explain your answer.

2. Does the cost of portfolio insurance increase or decrease as the beta of a portfolio increases? Explain your answer.

3. Explain how corporations can use range forward contracts to hedge their foreign exchange risk when they are due to receive a certain amount of a foreign currency in the future.

4. An index currently stands at 1,500. Six-month European call and put options with a strike price of 1,400 and time to maturity of six months have market prices of 154.00 and 34.25, respectively. The risk-free rate is 5%. What is the implied dividend yield?

Solution

Implied dividend yield = 1.99%

5. Consider an American futures call option where the futures contract and the option contract expire at the same time. Under what circumstances is the futures option worth more than the corresponding American option on the underlying asset?

11 The Greek Letters

ℹ References

- HULL, John. Options, futures, and other derivatives. Ninth edition. Harlow: Pearson, 2018. ISBN 978-1-292-21289-0.
  - Chapter 19 - The Greek Letters
- PIRIE, Wendy L. Derivatives. Hoboken: Wiley, 2017. CFA institute investment series. ISBN 978-1-119-38181-5.
  - Chapter 4 - Valuation of Contingent Claims

Learning Outcomes:

- Identify and describe the Greek letters used in options trading.
- Explain the concept of Delta and its importance in hedging strategies.
- Understand Gamma and its impact on the curvature of the options price relative to the stock price.
- Comprehend Theta and how time decay affects the value of options.
- Explore Vega and its relationship with volatility in the pricing of options.
- Recognize Rho and its sensitivity to changes in interest rates.
- Develop strategies for managing Delta, Gamma, and Vega to optimize risk and return in options portfolios.

11.1 Case Study

- A bank sells a European call option for \$300,000, covering 100,000 shares of a stock that does not pay dividends.
- **Initial Stock Price** (S_0): \$49
- **Strike Price** (K): \$50
- **Risk-Free Rate** (r): 5%
- **Volatility** (σ): 20%
- **Time to Expiration** (T): 20 weeks
- **Expected Return** (μ): 13%
- **Black-Scholes-Merton Valuation**: \$240,000

- **Objective**: Determine how the bank can hedge its risk to secure a $60,000 profit.

11.1.1 Naked Position

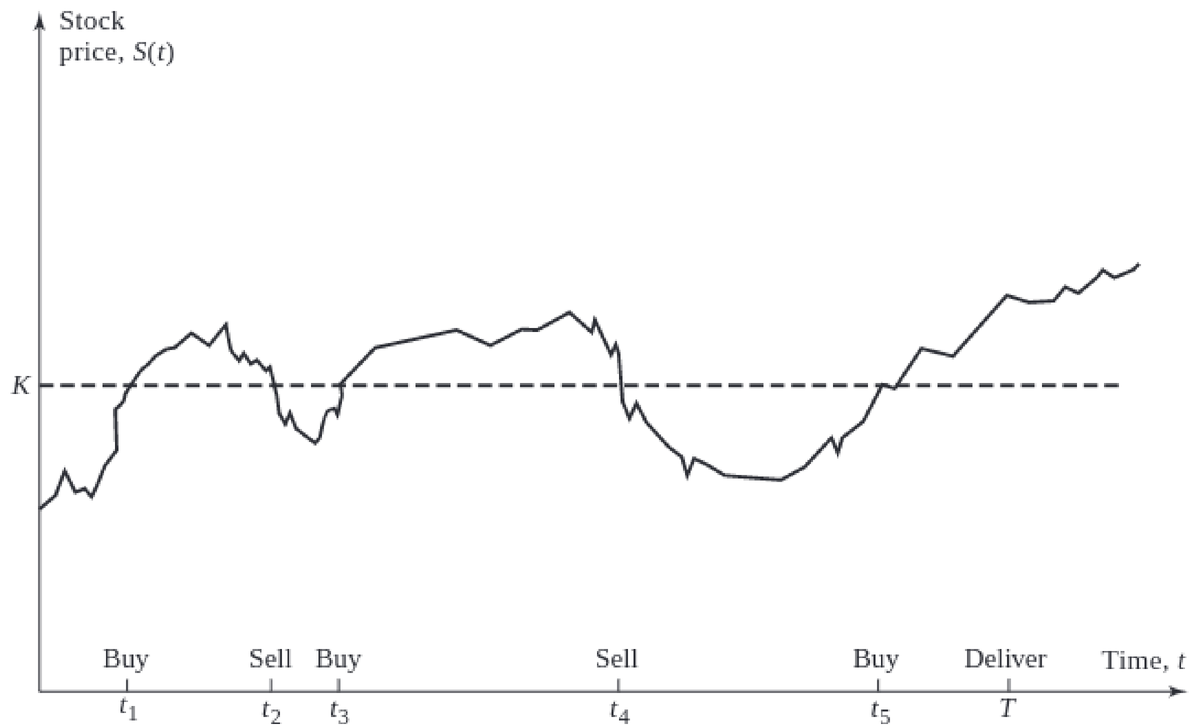
- **Description**: The bank takes no protective action against the position's exposure.
- **Risk Analysis**: This strategy leaves the position fully exposed to market fluctuations, with potential unlimited losses if the stock price rises significantly.

11.1.2 Covered Position

- **Description**: The bank purchases 100,000 shares immediately.
- **Risk Analysis**: Although this strategy can mitigate the risk if the stock's price increases above the strike price, it requires significant capital investment and is exposed to loss if the stock price declines.

11.1.3 Dynamic Hedging: Stop-Loss Strategy

- **Buy**: 100,000 shares are purchased when the stock price hits $50.
- **Sell**: 100,000 shares are sold when the stock price drops below $50.
- **Risk**: This approach attempts to limit losses by dynamically adjusting the position based on stock price movements. However, it may lead to high transaction costs and potential slippage; the actual transaction price may not always align with the trigger price due to market volatility.



The most efficient way to hedge would be to use **Greek letters.**

11.2 The Greek Letters

Greek letters are vital tools in options trading, providing a quantitative understanding of the sensitivity of an option's price to various factors. These metrics are crucial for effective risk management and strategic decision-making in trading portfolios.

- The Greeks are typically calculated using the Black-Scholes-Merton (BSM) model, a foundational tool in financial mathematics for valuing options.
- The Greek letters are the partial derivatives with respect to the model parameters that are liable to change.
- In practice, the volatility parameter in the BSM model is set to the implied volatility of the option, a modification often referred to as the "practitioner Black-Scholes" model.
- The discussion here pertains primarily to European options on non-dividend-paying stocks.

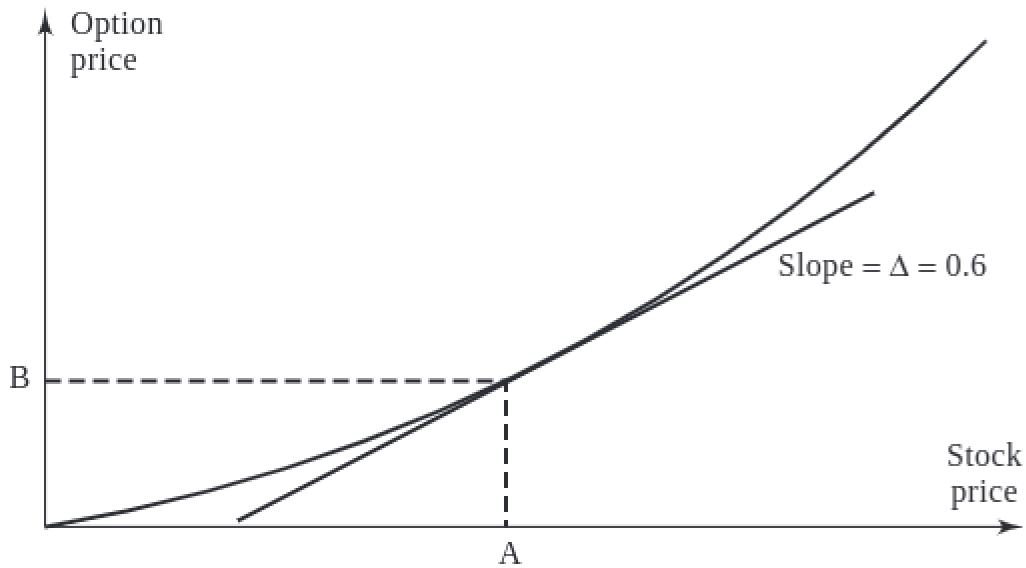
11.2.1 Description of Key Greeks

The Greeks measure the sensitivity of the option's price to one of the underlying parameters, **holding all other parameters constant**. Each Greek addresses a different risk factor:

- **Delta (Δ)**: Measures the rate of **change of the option's price** with respect to changes in the underlying asset's price. Specifically, delta is the first derivative of the option's price with respect to the stock's price. For instance, a delta of 0.6 suggests the option's price will move approximately \$0.60 for every \$1.00 movement in the underlying stock.
- **Gamma (Γ)**: Represents the rate of **change of delta** with respect to changes in the underlying asset's price. Gamma helps assess the stability of an option's delta, providing insight into how delta might change as the stock price varies. High gamma values indicate that delta is highly sensitive to changes in the stock price.
- **Theta (Θ)**: Measures the sensitivity of the option's price to **the passage of time**, known as "time decay." Theta indicates the expected rate of change in the option's price for a one-day decrease in its time to expiration. This is particularly important as the option approaches its expiry date.
- **Vega (ν)**: Quantifies the sensitivity of the option's price to **changes in the volatility** of the underlying asset. A vega of 1.5 suggests that the option's price is expected to change by \$1.50 for every 1% change in the implied volatility of the underlying stock.
- **Rho (ρ)**: Measures the sensitivity of the option's price to **changes in the risk-free interest rate**. For instance, a rho of 0.05 indicates that the option's price will change by \$0.05 for every 1% change in interest rates.

11.3 Understanding Delta (Δ)

Delta is a measure of an option's price sensitivity relative to changes in the price of the underlying asset. It's expressed as the amount an option's price is expected to move for a one-unit change in the price of the underlying asset.



The followint graph showing how delta varies with stock price for call and put options.

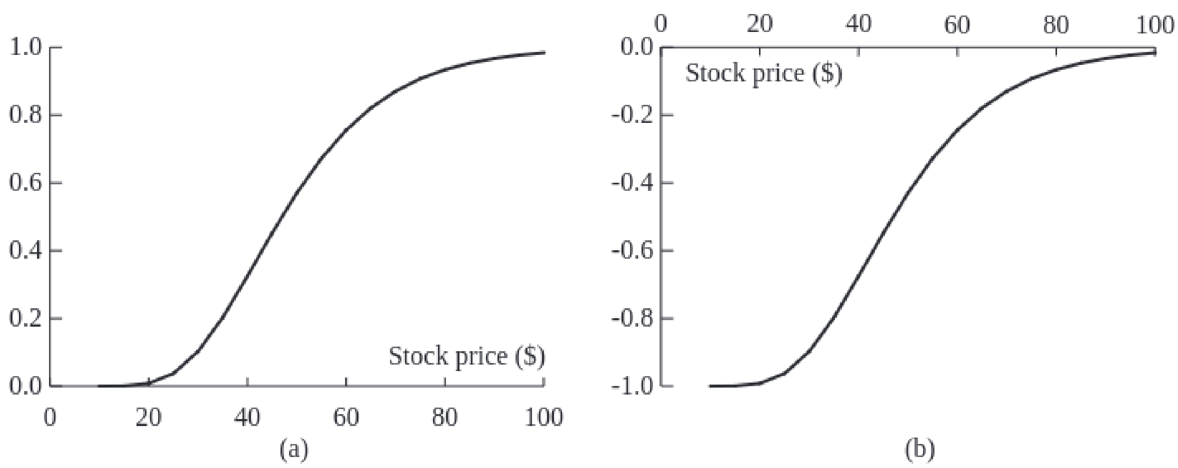


Figure 11.1: Variation of Delta With Stock Price (K=50)

- **Call Option (a)**: Delta increases as the stock price rises, peaking near 1 as the option goes deep in-the-money.
- **Put Option (b)**: Delta decreases (becomes more negative) as the stock price drops, indicating increased sensitivity as the option moves further in-the-money.

The next graph showing how delta changes as time to maturity decreases.

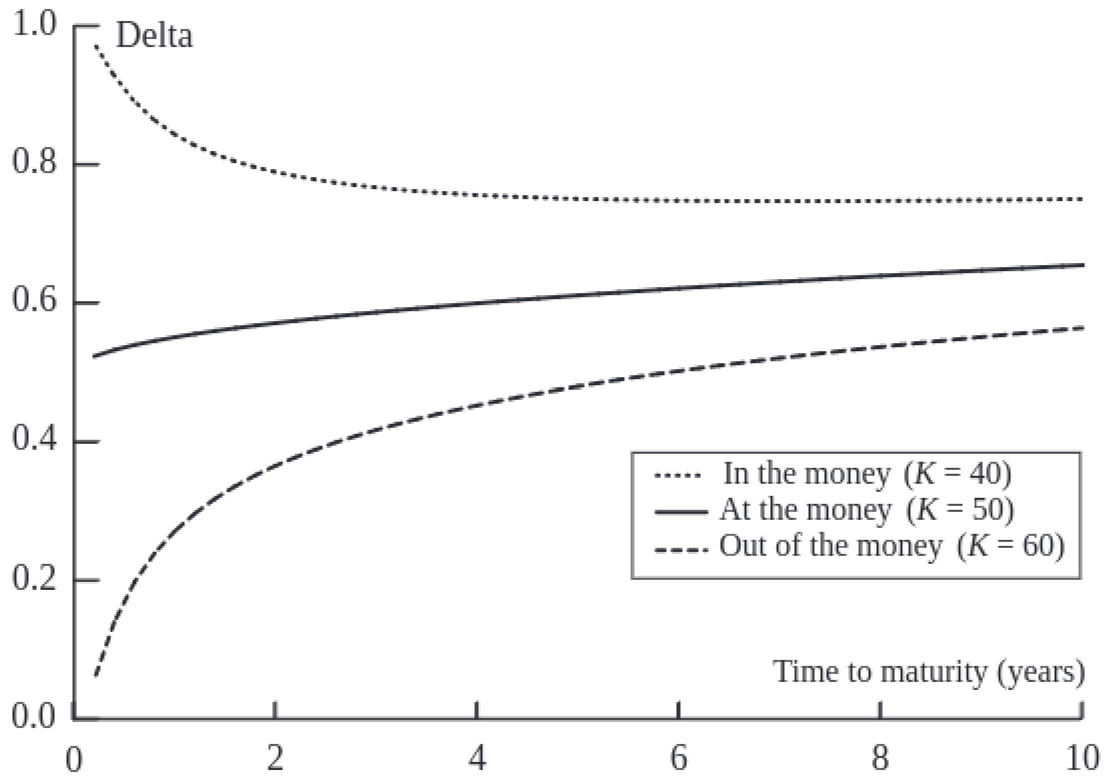


Figure 11.2: Variation of Delta with Time to Maturity

Delta's sensitivity to time decay depends on the moneyness of the option. For options that are exactly at-the-money, delta converges to 0.5. For out-of-the-money options, it approaches zero because such options are likely to expire worthless, whereas for in-the-money options, delta converges to 1.

11.3.1 Delta Hedging: Principles and Practices

Delta hedging is a strategy used to mitigate risk in option trading by setting up a position in the underlying asset. The position's size is determined by the option's delta, aiming to neutralize the effect of small price movements of the underlying asset.

To achieve delta neutrality, the required number of units to hedge, N_H, is calculated using the formula:

$$N_H = -\left(\frac{\text{Portfolio delta}}{\text{Delta of the hedging instrument}}\right)$$

A delta-neutral portfolio implies a total delta of zero, indicating no sensitivity to small price movements in the underlying stock. Note that the **delta of the stock itself is always 1**.

- **Rebalancing**: Delta values change as the stock price fluctuates and as time passes, necessitating periodic rebalancing of the hedge (buy high, sell low).
- **Delta for Calls and Puts**: The delta of a European call option on a non-dividend paying stock is represented by $N(d_1)$, whereas for a put, it's $N(d_1) - 1$.

Example: Delta Hedging I

Consider an investor who holds a short position in 1,000 call options, each with a delta of -0.6. Create a delta-neutral portfolio to shield against small price movements.

- **Current Delta**: $1,000 \times -0.6 = -600$
- **Hedging Action**: Purchase 600 shares of the stock (since each stock has a delta of 1).

This strategy ensures that any gain or loss on the options is countered by a corresponding loss or gain in the stock holdings.

Example: Delta Hedging II

Consider a portfolio comprising 1,000 shares of a non-dividend-paying stock. Each share contributes a delta of 1 to the portfolio, resulting in a total portfolio delta of 1,000.
To achieve delta neutrality using call options with a delta of 0.40, we calculate the required number of calls to sell:

$$N_H = -\left(\frac{1,000}{0.40}\right) = -2,500$$

This means selling 2,500 call options ensures that the positive delta from the stock holdings is exactly offset by the negative delta contribution from the short call positions.

Example: Delta Hedging III

Consider a scenario where you have a short position in 10,000 shares of a non-dividend-paying stock:

11.3.1.1 Using Call Options for Hedging:

- **Option Delta** (Δ_c): 0.668
- To neutralize the negative delta of -10,000 from the short stock position, you would need to purchase call options to increase delta:

$$N_H = -\left(\frac{-10,000}{0.668}\right) = 14,970$$

Purchasing 14,970 call options will add a positive delta, balancing out the negative delta from the short stock position.

11.3.1.2 Using Put Options for Hedging:

- **Option Delta** (Δ_p): -0.332
- To balance the negative delta, you can use put options which inherently have a negative delta:

$$N_H = -\left(\frac{-10,000}{-0.332}\right) = -30,120$$

Selling 30,120 put options would counteract the negative delta from the short stock position, achieving neutrality.

11.3.2 Hedging Cost Scenarios from Case Study

The following table presents a simulation of hedging dynamics when the option expires in-the-money, with total hedging costs amounting to $263,300:

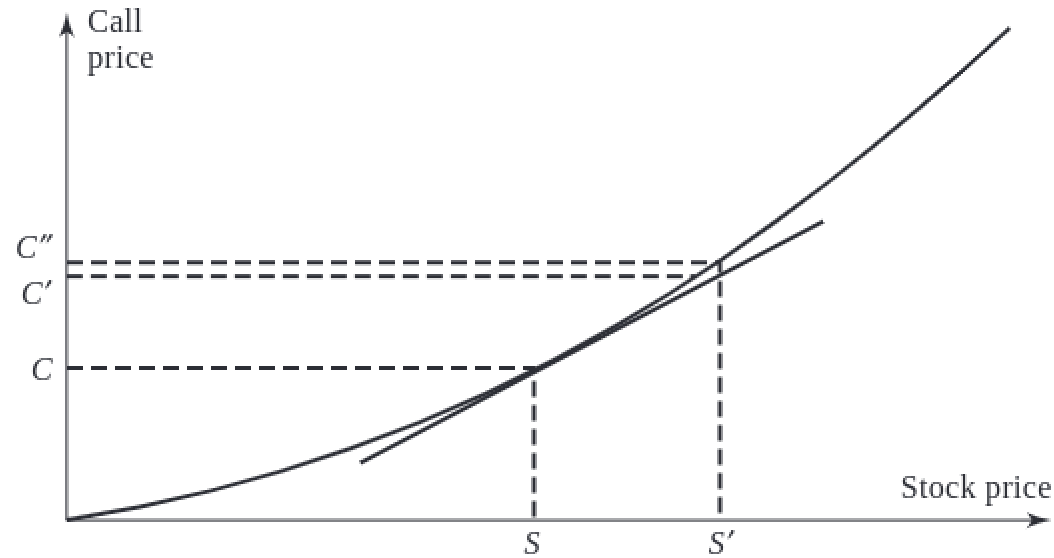
Week	*Stock price*	*Delta*	*Shares purchased*	*Cost of shares purchased ($000)*	*Cumulative cost including interest ($000)*	*Interest cost ($000)*
0	49.00	0.522	52,200	2,557.8	2,557.8	2.5
1	48.12	0.458	(6,400)	(308.0)	2,252.3	2.2
2	47.37	0.400	(5,800)	(274.7)	1,979.8	1.9
3	50.25	0.596	19,600	984.9	2,966.6	2.9
4	51.75	0.693	9,700	502.0	3,471.5	3.3
5	53.12	0.774	8,100	430.3	3,905.1	3.8
6	53.00	0.771	(300)	(15.9)	3,893.0	3.7
7	51.87	0.706	(6,500)	(337.2)	3,559.5	3.4
8	51.38	0.674	(3,200)	(164.4)	3,398.5	3.3
9	53.00	0.787	11,300	598.9	4,000.7	3.8
10	49.88	0.550	(23,700)	(1,182.2)	2,822.3	2.7
11	48.50	0.413	(13,700)	(664.4)	2,160.6	2.1
12	49.88	0.542	12,900	643.5	2,806.2	2.7
13	50.37	0.591	4,900	246.8	3,055.7	2.9
14	52.13	0.768	17,700	922.7	3,981.3	3.8
15	51.88	0.759	(900)	(46.7)	3,938.4	3.8
16	52.87	0.865	10,600	560.4	4,502.6	4.3
17	54.87	0.978	11,300	620.0	5,126.9	4.9
18	54.62	0.990	1,200	65.5	5,197.3	5.0
19	55.87	1.000	1,000	55.9	5,258.2	5.1
20	57.25	1.000	0	0.0	5,263.3	

- The cost reflects the higher intrinsic value and sensitivity (delta) as the option gains worth.
- If the option expires out-of-the-money, the costs should be slightly lower, as reduced delta correlates with a decreased likelihood of the option finishing in-the-money.

11.4 Understanding Gamma (Γ)

Gamma (Γ) is the second derivative of the option's price with respect to the underlying asset's price. It measures the rate of change of delta (Δ) and is crucial for understanding the curvature or the convexity of the option's value in relation to the stock price. It's crucial for assessing the stability of an option's delta, thus influencing how often a delta-hedged portfolio needs rebalancing. Here are some key characteristics of gamma:

- **Zero for Stocks**: Gamma for a long or short position in a single share of stock is zero because a stock's delta does not change.
- **Symmetry for Calls and Puts**: Gamma is identical for both call and put options.
- **Non-negativity**: Gamma is always non-negative. It reaches its highest value when an option is at-the-money, highlighting increased sensitivity at this point.
- **Risk Measurement**: Gamma quantifies the non-linearity risk—risk that remains in a delta-neutral portfolio due to price movements of the underlying asset.



11.4.1 Gamma and Hedging Dynamics

Gamma plays a critical role in addressing potential errors in delta hedging, particularly when stock prices exhibit significant movement:

- **Small Changes**: For minor fluctuations in stock price, delta hedging generally performs well.
- **Larger Shifts**: For more substantial stock price movements, delta-plus-gamma hedging provides greater accuracy, accommodating the curvature in the relationship between an option's price and the stock price.
- **Gamma Risk**: This term refers to the risk arising from sudden, significant movements in stock prices—often referred to as jumps—which can leave a previously well-hedged position exposed. Gamma risk is particularly relevant in markets characterized by high volatility or discontinuous price movements.
- **Theta and Gamma Interaction**: Typically, when theta is large and negative, indicating substantial time decay benefits, gamma tends to be large and positive. This inverse relationship highlights the trade-offs between potential profitability from time decay and the risk from larger price movements.

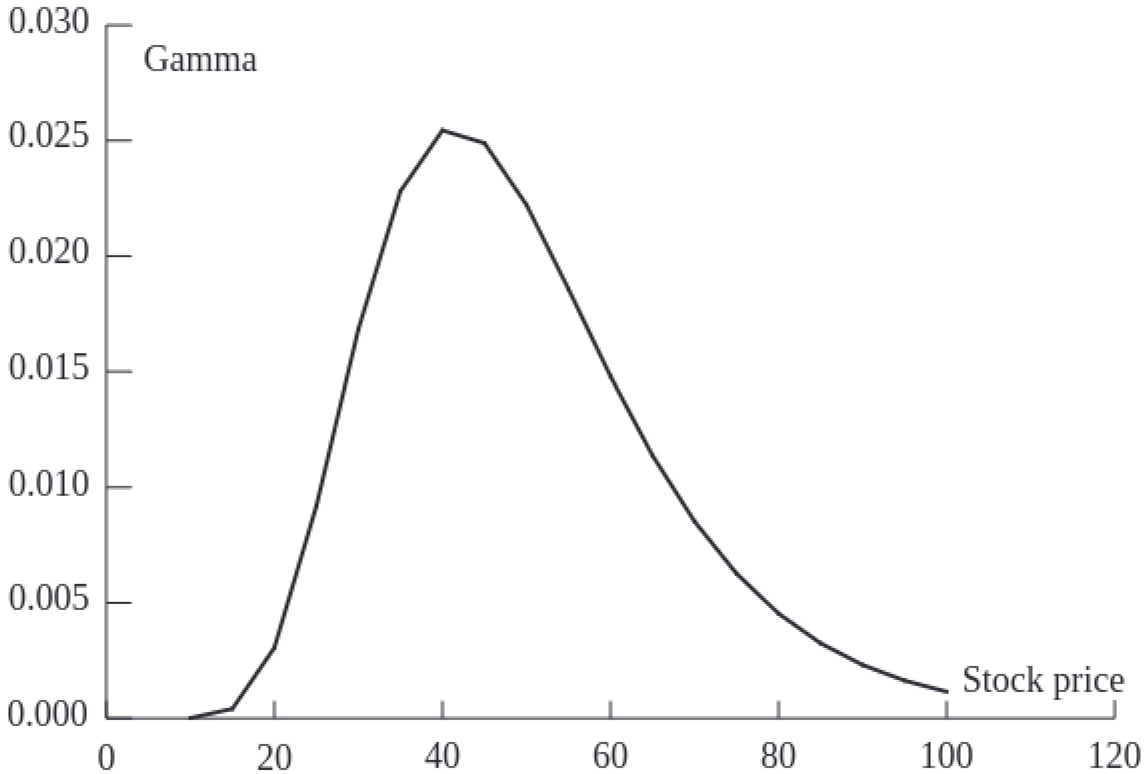


Figure 11.3: Gamma variation with stock price

This chart demonstrates how gamma peaks when an option is at-the-money and diminishes as the option moves deeper into or out of the money. This pattern underscores the heightened sensitivity and potential risk/reward near the at-the-money point.

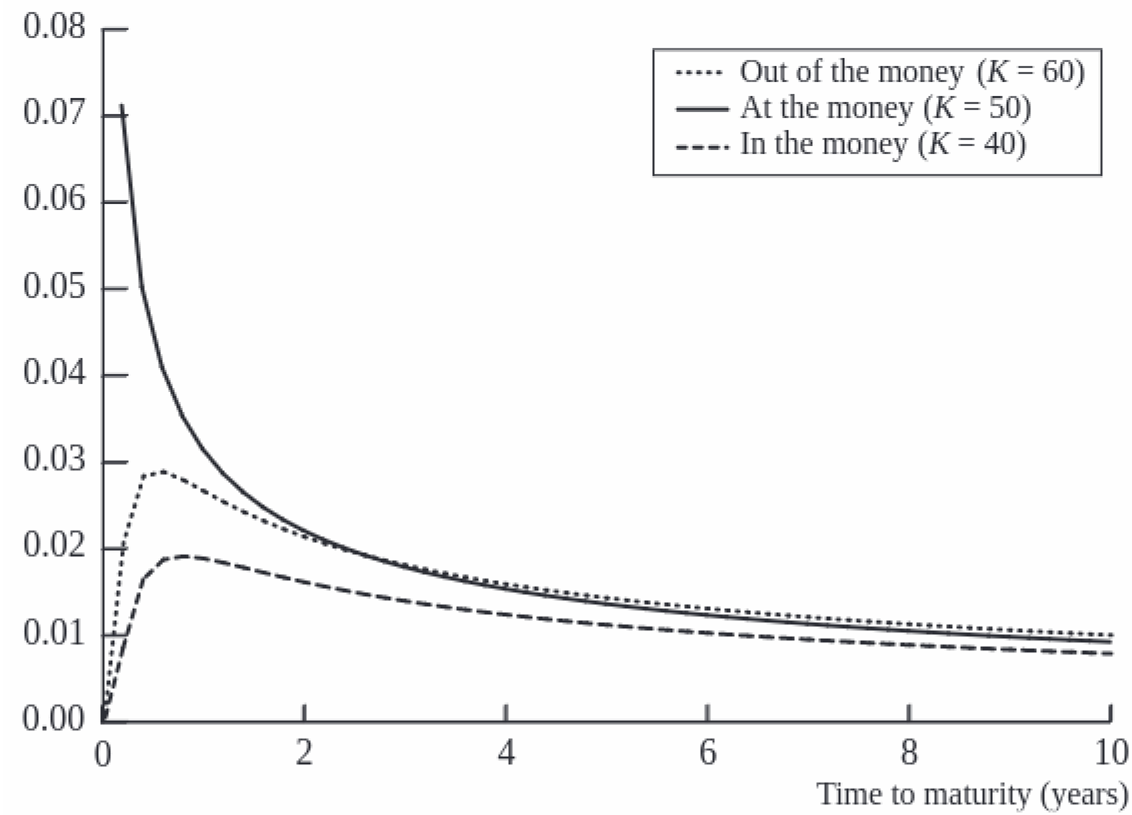


Figure 11.4: Gamma variation over time

Gamma increases as the option approaches expiration, particularly for at-the-money options. This increase reflects growing sensitivity to stock price movements as time to exercise decreases, emphasizing the importance of precise hedging strategies in the final days before an option's expiration.

11.5 Understanding Theta (Θ)

Theta (Θ) represents the sensitivity of the price of an option or a portfolio of derivatives to the passage of time, assuming all other factors remain constant. It is a critical component in options pricing, reflecting the temporal decay of an option's value.

- **Directionality**: Theta is typically negative for both calls and puts. This reflects the loss in time value as options approach their expiration date.
- **Non-applicability to Stocks**: Since stocks do not have an expiration date, they inherently have a theta of zero.

- **Impact on Options**: Each day closer to expiration decreases the option's time value, reducing its price if other conditions remain unchanged.

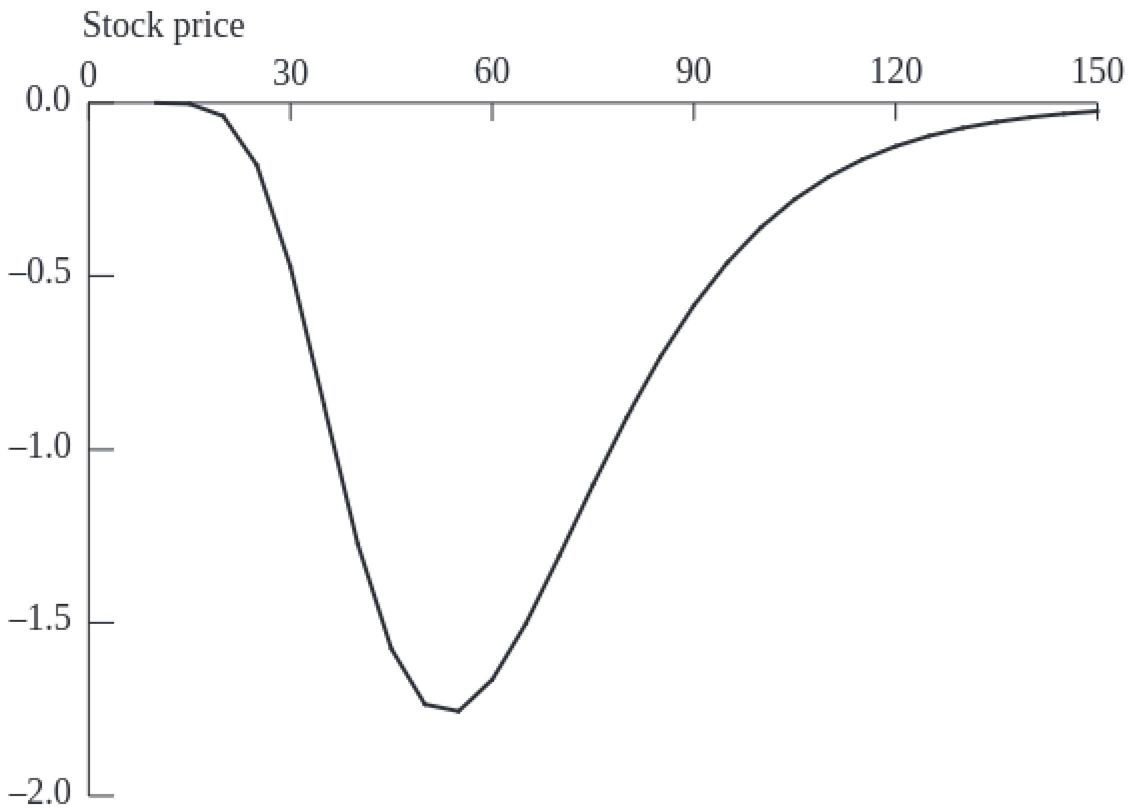


Figure 11.5: Theta variation with stock price

The graph illustrates how theta behaves relative to the stock price for a call option. Notably, theta becomes increasingly negative as the option moves around at-the-money.

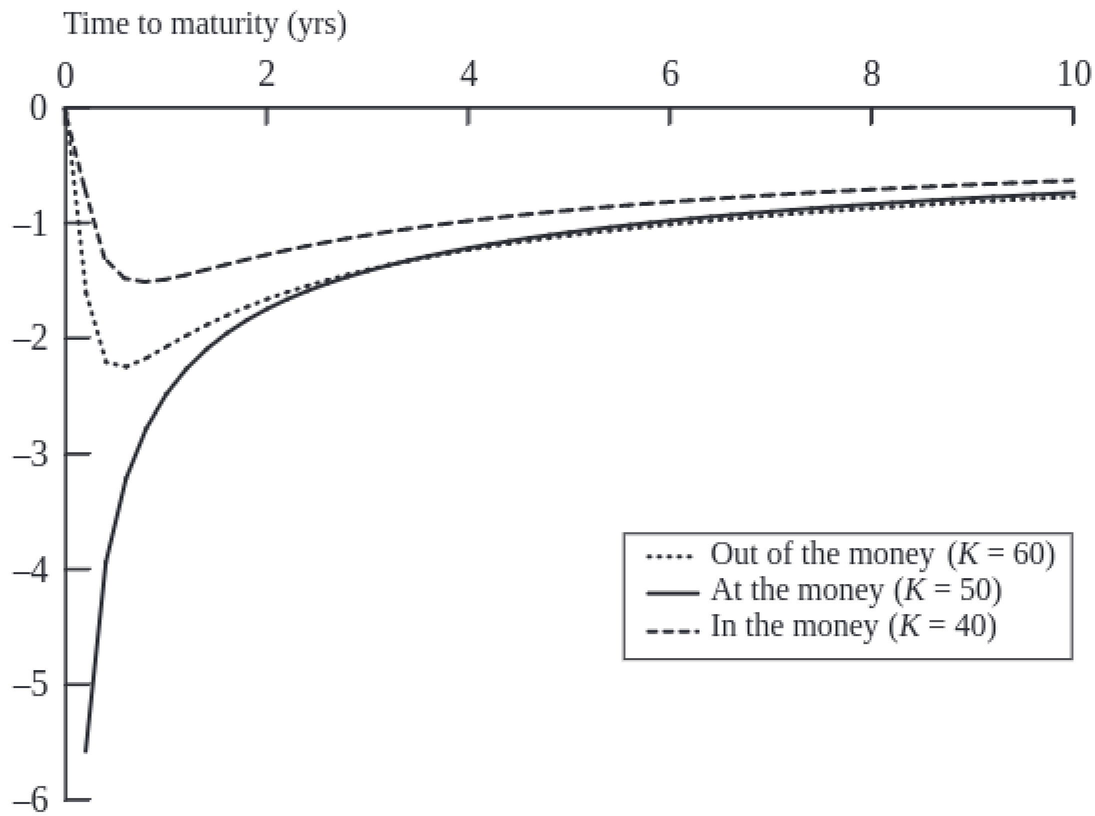


Figure 11.6: Theta variation over time

This visualization shows theta's behavior as the option nears its expiry. Theta typically becomes more negative as expiration approaches, indicating an acceleration in the rate of time value decay. This is most pronounced for at-the-money options, where the uncertainty about finishing in-the-money is highest.

- **Theta vs. Hedge Parameters**: Unlike delta, which can be hedged, theta represents a guaranteed decline in value over time and is not a hedgeable risk. However, its predictability and inevitability make it a significant factor in options strategy.
- **Descriptive Utility**: Despite its non-hedgeable nature, theta is considered a useful descriptive statistic for portfolios, particularly in strategies aimed at earning through time decay, such as selling options.
- In a delta-neutral setting, where the portfolio is insulated against small price movements in the underlying asset, theta can serve as an indirect measure of gamma exposure. This relationship arises because high gamma values, which indicate greater sensitivity to price changes, usually accompany high rates of time decay (theta). Thus, monitoring theta provides insights into potential gamma risks in the portfolio.

11.6 Understanding Vega (ν)

Vega represents the rate of change in the value of an option or a derivatives portfolio with respect to changes in volatility. This Greek is crucial in options trading, reflecting how the price of options reacts to fluctuations in the underlying asset's volatility.

Vega highlights the impact of volatility, an essential but unobservable market factor that significantly influences option pricing. Future volatility, being an estimation, adds a layer of complexity and risk in predicting option values.

- **Directionality**: Vega is always positive. An increase in the implied volatility of the underlying asset generally leads to an increase in the value of both call and put options.
- **Symmetry**: The vega of a call option is equal to the vega of a put option with the same terms.
- **Calculation**: For a portfolio, vega can be understood as a weighted average of the vegas of the individual positions, reflecting the aggregate sensitivity to a uniform shift in implied volatilities across the portfolio.
- **Volatility Sensitivity**: Of all the Black-Scholes-Merton (BSM) model variables (stock price, strike price, time to expiration, risk-free rate, and volatility), an option's price is most sensitive to changes in volatility. This sensitivity makes vega a key focus in risk management and trading strategies.

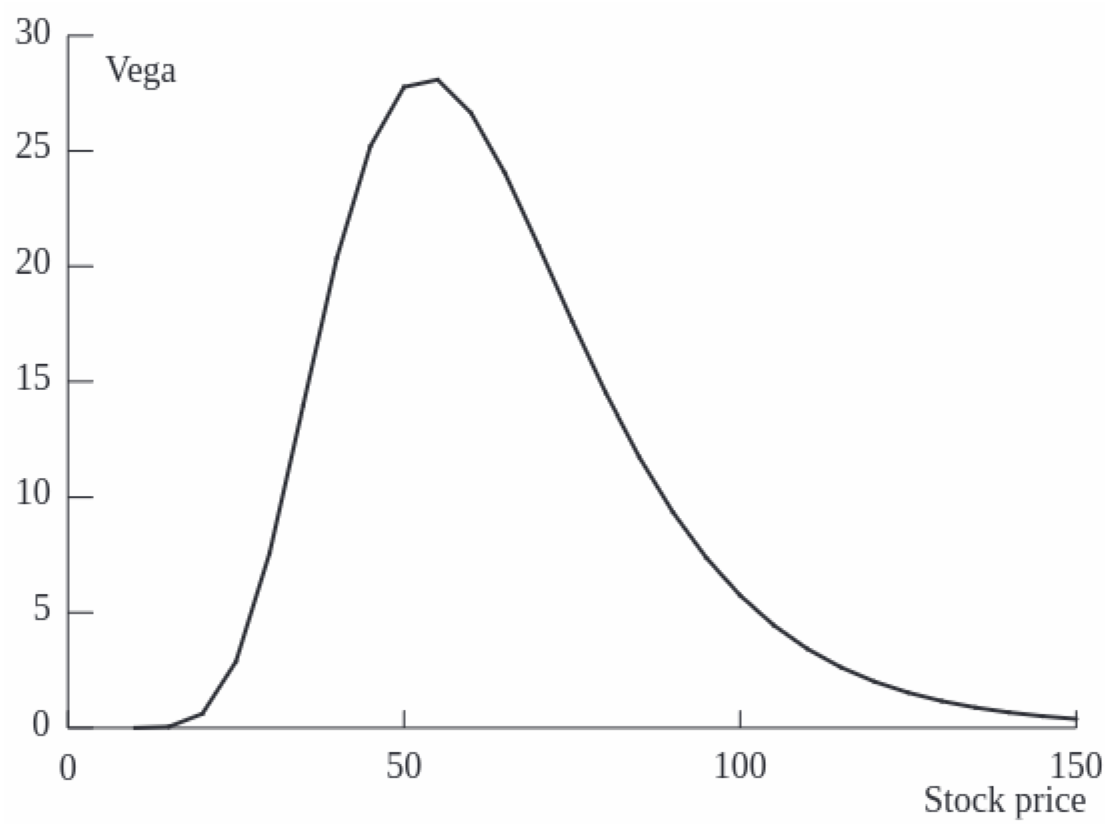


Figure 11.7: Vega variation with stock price

The chart illustrates how vega changes in relation to the stock price. Vega tends to be higher when an option is at-the-money and decreases as the option moves deeper into or out of the money. This pattern reflects the increased sensitivity to volatility changes when the option's strike price is near the current stock price, where the uncertainty about the outcome at expiration is greatest.

11.7 Understanding Rho (ρ)

Rho represents the sensitivity of an option or a derivatives portfolio's value to changes in the risk-free interest rate. It is one of the lesser-focused Greeks but plays a crucial role in environments where interest rate fluctuations are significant.

- **Directionality**: Rho is typically positive for call options and negative for put options. This reflects the different financial implications of options:

  - **Call Options**: A positive Rho for call options indicates that their value increases as interest rates rise. This is because higher rates reduce the present value of the exercise price paid at expiration, making it cheaper in present terms to buy the stock at a future date.
  - **Put Options**: Conversely, a negative Rho for put options means their value decreases as interest rates rise. Higher rates increase the present value of the proceeds received from selling the stock in the future, thus decreasing the attractiveness of holding a put.

The sensitivity to interest rates through Rho might seem minor compared to other Greeks like Delta or Vega, but it can become significant in certain financial environments:

- **High Interest Rate Volatility**: In periods of significant interest rate changes, Rho becomes a more critical factor for traders, especially those dealing with long-dated options where the cumulative impact of rate changes can be substantial.
- **Portfolio Management**: For portfolios that include a mix of long-dated options and bonds, understanding and managing Rho is vital to hedge interest rate risks effectively.

11.8 Strategic Management of Delta, Gamma, and Vega

Proper management of the Greek values—Delta, Gamma, and Vega—is essential for maintaining balanced options portfolios that align with specific risk management objectives. Here, we outline techniques to achieve neutrality in these Greeks through strategic positioning in options and the underlying asset.

- **Delta**: This Greek can be adjusted directly by taking positions in the underlying asset. For instance, buying or selling stock can increase or decrease the portfolio's delta.
- **Gamma and Vega**: Adjustments to gamma and vega generally require engaging in positions in options or other derivatives, as these Greeks measure sensitivity to changes in the underlying asset's price and volatility, respectively.

The general approach involves **first hedging vega and/or gamma** using options, and then as a **final step, using underlying assets** to reduce delta to zero. This sequence is important because hedging vega and gamma can alter delta, whereas hedging using underlying assets does not affect vega or gamma.

Example: Achieving Greek Neutrality

	Delta	Gamma	Vega
Portfolio	0	-5000	-8000

Option 1	0.6	0.5	2.0
Option 2	0.5	0.8	1.2

1. **Achieving Delta and Gamma Neutrality**:
   - **Required Position**: Long 10,000 units of Option 1 and short 6,000 units of the underlying asset.
   - **Rationale**: This positioning uses the delta and gamma from Option 1 to offset the existing negative gamma in the portfolio while also balancing delta.

2. **Achieving Delta and Vega Neutrality**:
   - **Required Position**: Long 4,000 units of Option 1 and short 2,400 units of the underlying asset.
   - **Rationale**: This configuration uses Option 1's vega to counteract the portfolio's negative vega while adjusting the delta with a corresponding position in the underlying asset.

3. **Achieving Delta, Gamma, and Vega Neutrality Across Multiple Options**:
   - **System of Equations**:
     - For Gamma:
     $$-5000 + 0.5w_1 + 0.8w_2 = 0$$
     - For Vega:
     $$-8000 + 2.0w_1 + 1.2w_2 = 0$$
   - **Solution**: Long 400 units of Option 1 and 6,000 units of Option 2, with a short position of 3,240 units in the underlying asset to achieve complete neutrality in delta, gamma, and vega.

11.8.1 Hedging in Practice

- **Daily Delta Neutrality**: Traders typically adjust their portfolios to be delta-neutral at least once per day to manage risk effectively against price movements in the underlying asset.
- **Opportunistic Gamma and Vega Adjustments**: While daily adjustments for delta are common, adjustments for gamma and vega are made as opportunities arise, allowing traders to better manage the curvature and volatility risks associated with their positions.
- **Economies of Scale**: As portfolios increase in size, the relative cost of hedging per option decreases, providing cost efficiencies in larger portfolios.
- **Scenario Analysis**: This involves testing different market scenarios and their effects on the portfolio, considering various assumptions about asset prices and volatilities. This

analysis helps in understanding potential impacts and preparing for various market conditions.

11.9 Practice Questions and Problems

1. Explain how a stop-loss trading rule can be implemented for the writer of an out-of-the money call option. Why does it provide a relatively poor hedge?
2. What does it mean to assert that the delta of a call option is 0.7? How can a short position in 1,000 options be made delta neutral when the delta of each option is 0.7?
3. What does it mean to assert that the theta of an option position is -0.1 when time is measured in years? If a trader feels that neither a stock price nor its implied volatility will change, what type of option position is appropriate?
4. What is meant by the gamma of an option position? What are the risks in the situation where the gamma of a position is large and negative and the delta is zero?
5. A bank's position in options on the dollar–euro exchange rate has a delta of 30,000 and a gamma of 80,000. Explain how these numbers can be interpreted. The exchange rate (dollars per euro) is 0.90. What position would you take to make the position delta neutral? After a short period of time, the exchange rate moves to 0.93. Estimate the new delta. What additional trade is necessary to keep the position delta neutral? Assuming the bank did set up a delta-neutral position originally, has it gained or lost money from the exchange-rate movement?
6. A financial institution has the following portfolio of stock options:

Type	Position	Delta	Gamma	Vega
Call	-1,000	0.5	2.2	1.8
Call	-500	0.8	0.6	0.2
Put	-2,000	-0.4	1.3	0.7
Call	-500	0.7	1.8	1.4

   A traded option is available with a delta of 0.6, a gamma of 1.5, and a vega of 0.8.

   1. What position would make the portfolio delta neutral?
   2. What position in the traded option and in stocks would make the portfolio both gamma neutral and delta neutral?
   3. What position in the traded option and in stocks would make the portfolio both vega neutral and delta neutral?

4. Suppose that a second traded option with a delta of 0.1, a gamma of 0.5, and a vega of 0.6 is available. How could the portfolio be made delta, gamma, and vega neutral?

Solution

1. Long 450 stocks.
2. Long 4,000 options and short 1,950 stocks.
3. Long 5,000 options and short 2,550 stocks.
4. Long 3,200 options 1, long 2,400 options 2, and short 1,710 stocks.

12 Structured Products I

> **References**
>
> - BLÜMKE, Andreas. How to invest in structured products: a guide for investors and investment advisors. Chichester: Wiley, 2009. xvi, 374. ISBN 9780470746790.

Learning Outcomes:

- Identify the key characteristics and benefits of structured products as investment vehicles.
- Describe the role and impact of the European Structured Investment Products Association (EUSIPA) in the structured products market.
- Differentiate between various types of structured products, including investment certificates and barrier options.
- Evaluate the mechanisms and behavior of capital protection and participation investment products through theoretical and case study approaches.

12.1 Introduction to Structured Products

12.1.1 Advantages of Investing in Structured Products

Structured products enjoy significant popularity across **European states**, primarily due to their capacity to balance risk and reward. These financial instruments offer several compelling advantages:

1. **Capital Protection and Market Participation**: Investors are drawn to structured products because they provide a mechanism to protect all or part of the invested capital while still allowing participation in the gains of the capital markets.

2. **Accessibility to Regional Markets**: These products facilitate access to regional markets and asset classes that may otherwise be inaccessible through direct investments.

3. **Diverse Return Profiles**: Structured products can be designed with a wide array of return profiles, adapting to various investor needs and market conditions. They remain effective across different market movements—providing potential returns in rising, sideways, or falling markets.

4. **Liquidity**: Market makers enhance the liquidity of structured products by continuously buying and selling them, which ensures a stable market presence and availability.
5. **Sophisticated Investment Strategies**: Investors can implement complex investment strategies typically available to advanced traders through the acquisition of a single structured product.
6. **Regulatory Oversight and Transparency**: These products are often listed on official, regulated markets, adding a layer of security and transparency for investors.
7. **Tax Advantages**: In some jurisdictions, structured products offer favorable tax conditions, enhancing their attractiveness as investment options.

12.1.2 European Structured Investment Products Association (EUSIPA)

- Official Website: EUSIPA
- Market Insights: EUSIPA Market Reports

Founded in 2009, EUSIPA represents a collective of national issuer associations from multiple European countries including Austria, France, Germany, Italy, Sweden, Belgium, the UK, Switzerland, and The Netherlands. As an international non-profit association governed under Belgian law, EUSIPA also maintains a presence in the EU transparency register.

EUSIPA aims to foster transparency and establish uniform market standards across Europe. It serves as a pivotal platform for its members to engage in meaningful dialogue with European policymakers, ensuring that the voices of issuers are heard in the regulatory landscape.

12.2 Definition and Nature of Structured Products (Investment Certificates)

> Definition by Andreas Blümke
>
> Structured products are financial assets, which consist of various elemental components, combined to generate a specific risk-return profile (not replicable with stocks and bonds) adapted to an investor's needs.

Structured products, often referred to as investment certificates, are essentially securitized derivatives. These are complex financial contracts encapsulated within a single security that trades on exchanges similar to stocks. These instruments are crafted and issued by financial institutions and are utilized by both retail and institutional investors. They can be traded on stock exchanges or dealt directly between parties in over-the-counter (OTC) transactions.

12.2.1 Credit Risk Associated with Structured Products

Structured products carry inherent credit risks as they **are issued in the form of bearer bonds**. This means the issuer's entire assets back the liability on these products. The quality and safety of structured products are intrinsically linked to the creditworthiness of the issuer. Like traditional bonds, these products are **exposed to issuer risk**, which implies that in the event of issuer bankruptcy, both bonds and structured products are treated as part of the bankruptcy estate.

To mitigate such risks, investors are advised to:

- Opt for products issued by financially robust institutions.
- Diversify their investments across various issuers.
- Continuously monitor the financial health of the issuers over time.

12.2.2 Categorization of Structured Products

Structured products do not follow a universal standard for categorization. However, common classifications are often derived from industry associations such as:

- **EUSIPA**: European Structured Investment Products Association provides a framework for categorizing structured products within Europe.
  - View the EUSIPA Derivatives Map for details.
- **SVSP**: Swiss Structured Products Association offers an alternative categorization scheme.
  - Explore the SVSP Derivative Map for comparison.

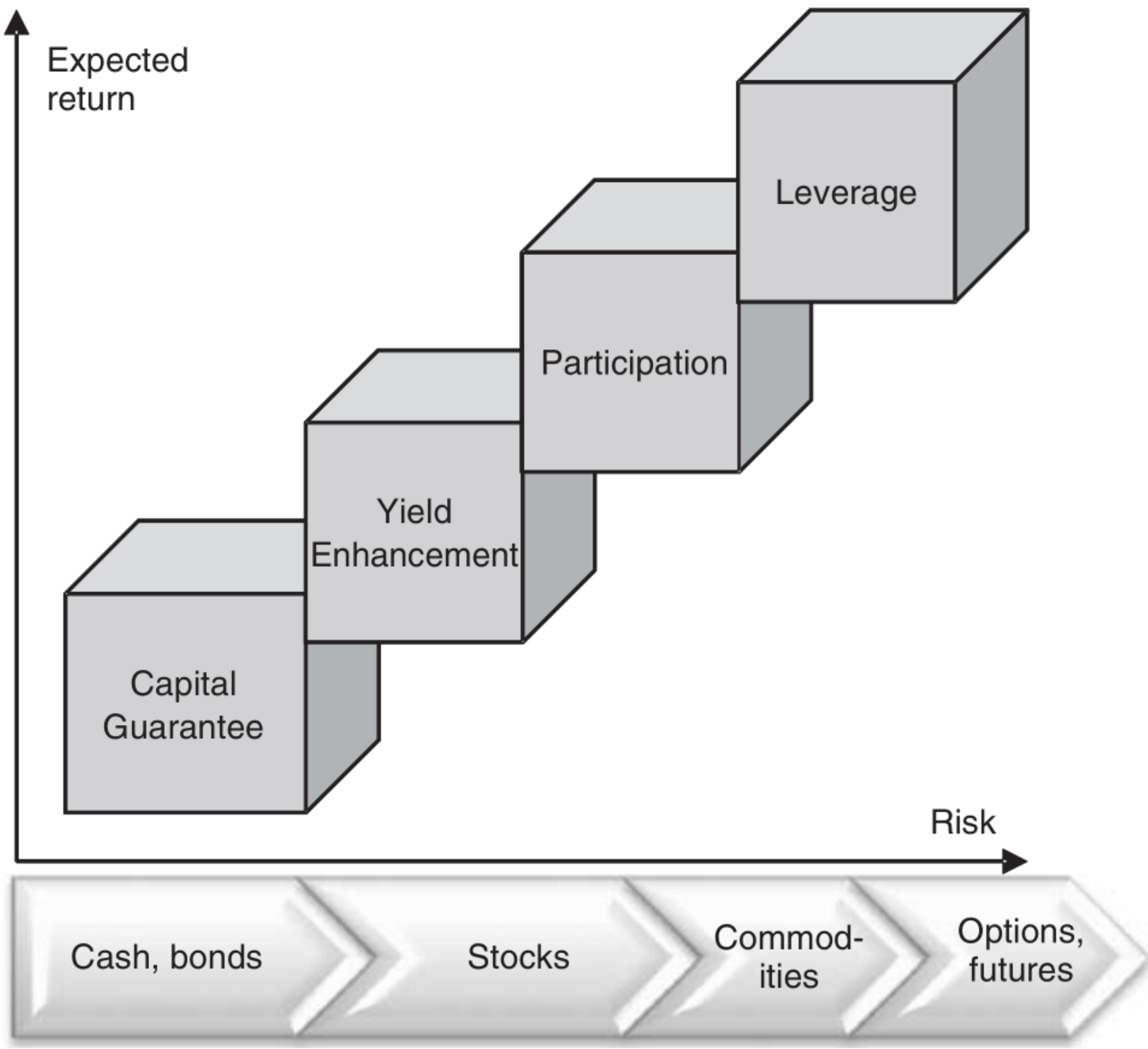


Figure 12.1: Classical structured products representation

12.2.3 Investors and Sellers of Structured Products

12.2.3.1 Investors: Private and Institutional Buyers

Structured products attract both private and institutional investors when traditional investment avenues do not meet their specific needs. These needs might include the desire for returns higher than the risk-free rate while still benefiting from capital protection. Structured products are particularly appealing in scenarios where conventional investments either do not provide sufficient returns or fail to address **specific financial goals and risk profiles**.

12.2.3.2 Sellers: Financial Institutions and Their Motivations

Sellers of structured products, typically financial institutions, are driven by profit. They utilize sophisticated mathematical models to determine a 'fair value' of the product at issuance. This fair value is then increased by a spread which covers various costs associated with the product over its lifetime. These costs include but are not limited to:

- Secondary market activities
- Listing fees
- Production of term-sheets
- Settlement processes

Among these, the most **significant cost factor is hedging**. Hedging expenses are challenging to predict in advance as they depend on market dynamics over the product's life. The profit from structured products, however, is only realized at their expiry, and having a large and diverse portfolio helps in more effectively hedging as some products can offset the risks of others.

12.2.4 Platforms to Find Structured Products

Structured products can be accessed and traded through various platforms, which serve as marketplaces for these financial instruments:

- **Deutsche Bank X-markets**: Offers a wide range of structured products for different investment strategies.
  - Website: Deutsche Bank X-markets
- **Börse Stuttgart**: Known for its user-friendly approach to trading structured products among other securities.
  - Website: Börse Stuttgart
- **Börse Frankfurt**: One of Europe's largest trading centers for securities, including derivatives and structured products.
  - Website: Börse Frankfurt

12.2.5 Additional References

For further research and detailed insights into the market for structured products, the following resources are invaluable:

- European Structured Investment Products Association (EUSIPA): Website
- German Derivatives Association: Website

- Swiss Structured Products Association (SSPA): Website
- UK Structured Products Association: Website
- Italian Association of Certificates and Investment Products (ACEPI): Website

12.3 Barrier Options

> Definition
>
> Barrier options are exotic call or put options that include a barrier condition placed above or below the strike that, when crossed, either transfroms the exotic option into a plain vanilla option ('in' barriers) or cancels it altogether ('out' barriers).

Barrier options are integrated into various structured financial products such as barrier reverse convertibles and bonus certificates. Recognized for their complexity, these options introduce a conditional component to the standard option mechanism, making the final payoff uncertain until the option's maturity.

12.3.1 Types of Barrier Options

Barrier options can be classified into four main types based on the direction of the barrier and its effect:

- **Up & Out**: The option becomes void if the underlying asset's price goes above the barrier.
- **Up & In**: The option comes into existence when the underlying asset's price goes above the barrier.
- **Down & Out**: The option becomes void if the underlying asset's price falls below the barrier.
- **Down & In**: The option comes into existence when the underlying asset's price falls below the barrier.

Additionally, these options can feature a rebate, a predefined amount paid to the option holder if the barrier is breached before maturity.

12.3.2 Pricing Dynamics

Barrier options are generally more cost-effective than their plain vanilla counterparts due to the added condition of the barrier. The pricing dynamics vary significantly between 'in' and 'out' options:

- **For 'In' Options**: As the maturity increases, the price approaches that of a plain vanilla option, especially as the price of the underlying asset approaches the barrier.
- **For 'Out' Options**: Longer maturities reduce the price, potentially approaching zero as the asset's price nears the barrier.

12.3.3 Barrier Styles

Structured products typically incorporate barrier options with specific monitoring styles:

- **American-style Barriers**: These allow the barrier condition to be triggered at any point during the option's life, including intraday events, and are known for continuous monitoring.
- **European-style Barriers**: These restrict the barrier condition to only be checked at the maturity of the option.

Although most structured products utilize American-style barriers due to cost-effectiveness, the investor focus generally lies elsewhere rather than on the barrier type itself.

Example	American Barrier (Original)	European Barrier
Shark Note	Barrier 131.5%, Rebate 7.5%	Barrier 123%, Rebate 7.5%
Barrier Reverse Convertible	Coupon 10.4%, Barrier 75%	Coupon 8.6%, Barrier 75%
Bonus Certificate	Bonus 9%, Barrier 65%	Bonus 2.5%, Barrier 65%

- **Window-style Barriers**: Situated between American and European styles, window barrier options are designed to activate or deactivate only during specific periods within the product's life. For example, the barrier in some reverse convertibles might only be relevant during the final three months of a one-year term. Although not common, window barriers provide an opportunity for investors to avoid premature knockouts, with the value difference between American and window options typically being minimal.

12.4 Capital Protection Investment Products

Definition

Capital guaranteed products ensure the redemption of the initial capital invested at maturity, while also allowing participation to varying degrees in the performance of an underlying risky asset.

These products are distinguished by three primary features:

1. **Limited Loss Potential**: The potential loss is confined to the level of the capital guarantee, not accounting for the issuer's credit risk.
2. **Participation in Underlying Assets**: Investors gain exposure to the performance of selected assets.
3. **Minimal Guaranteed Income**: Typically, these products offer low or no guaranteed income, focusing instead on capital preservation and growth through asset performance.

It's crucial to consider the **opportunity costs**, such as foregone dividends or the risk-free rate. While attractive at first glance, the actual benefit depends significantly on the performance of the underlying asset at maturity:

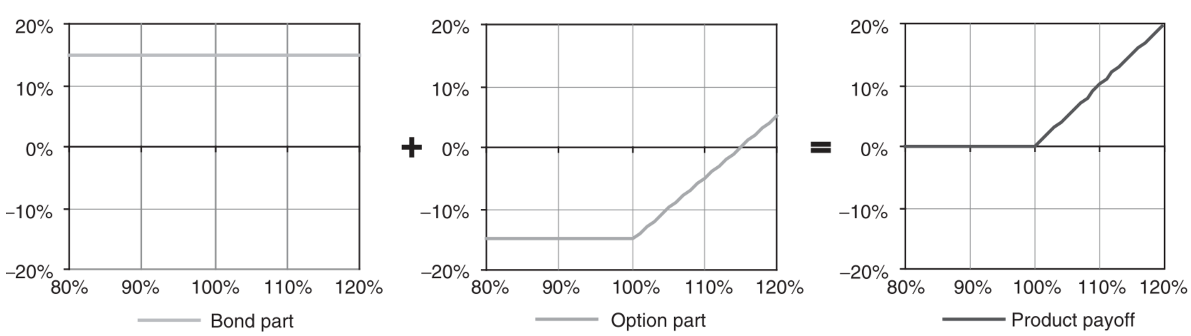
- If the asset performs well, the capital guarantee becomes redundant.
- If the asset performs poorly, it might have been better not to invest.

Capital guaranteed products serve as a prudent option for investors who prefer not to remain in cash but are also uncertain about future market directions.

12.4.1 Uncapped Capital Protection

Classical construction of capital guaranteed products involves:

1. Issuing a zero-coupon bond matching the maturity of the product to ensure capital return.
2. Purchasing a call option on the underlying risky asset to allow for profit participation.



12.4.1.1 Discount and Participation Formula

If we consider an interest rate of 4% with a 5-year maturity:

$$\text{Zero-bond Price} = \frac{100\%}{(1+4\%)^5} = 82.19\%$$

The available funds to purchase options (the discount) is:

$$\text{Discount} = 100\% - 82.19\% = 17.81\%$$

Participation rate is calculated as:

$$\text{Participation} = \frac{\text{Discount}}{\text{Option cost}}$$

12.4.1.2 Factors Influencing Product Viability

Two critical factors affect the desirability and effectiveness of capital guaranteed products:

- **Interest Rates**: Higher rates increase the discount, thereby enhancing the capacity to purchase options.
- **Volatility of the Underlying**: Lower volatility reduces option costs, improving participation rates.

Options for enhancing attractiveness include reducing the capital guarantee below 100%, adding caps or exotic features like knock-out barriers, and utilizing out-of-the-money options.

12.4.2 Capital Protection Products Modifications

12.4.2.1 Exchangeable Certificates

Figure 12.2: Exchangeable Certificates

- **Market Expectations:** Rising volatility, sharply rising or falling underlying.
- Minimum redemption at maturity equals the capital protection (e.g., 100% of nominal).
- Value may fall below capital protection during the product's life.
- Unlimited upside above the strike price, with possible coupon payments.

12.4.2.2 Capped Capital Protection

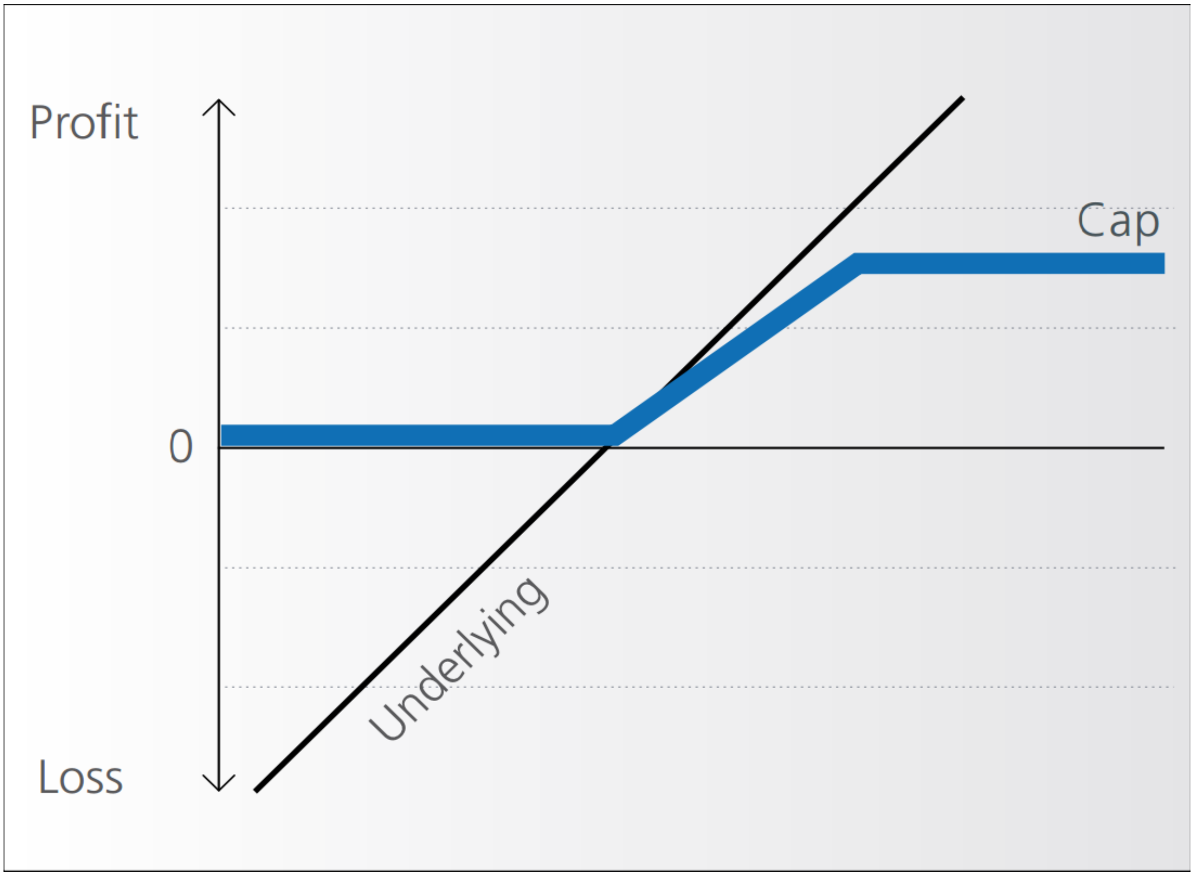


Figure 12.3: Capped Capital Protection

- **Market Expectations:** Rising underlying, potential for sharp falls.
- Guaranteed minimum redemption at expiry.
- Participation in positive performance up to a specified cap.
- Limited profit potential due to the cap.

12.4.2.3 Capital Protection with Knock-Out (Shark Note)

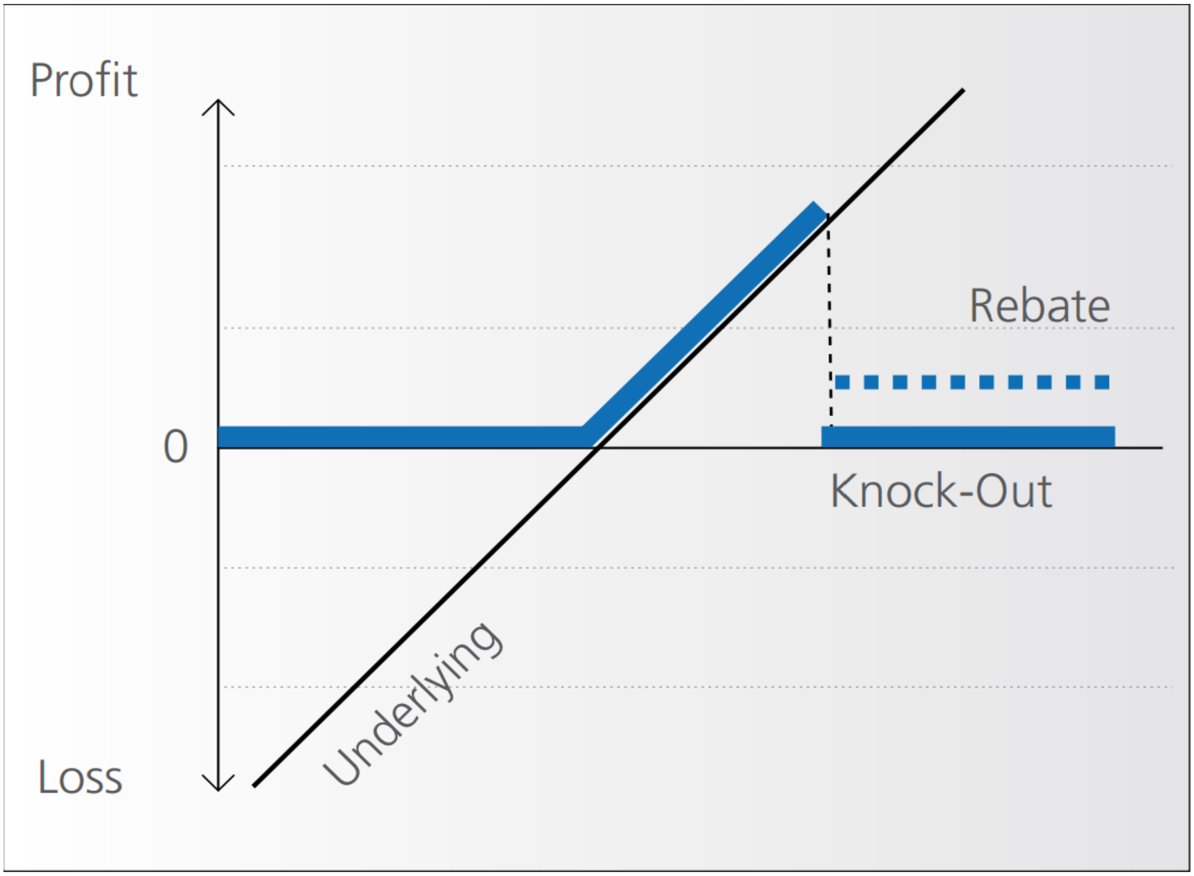


Figure 12.4: Shark Note - Knock-Out

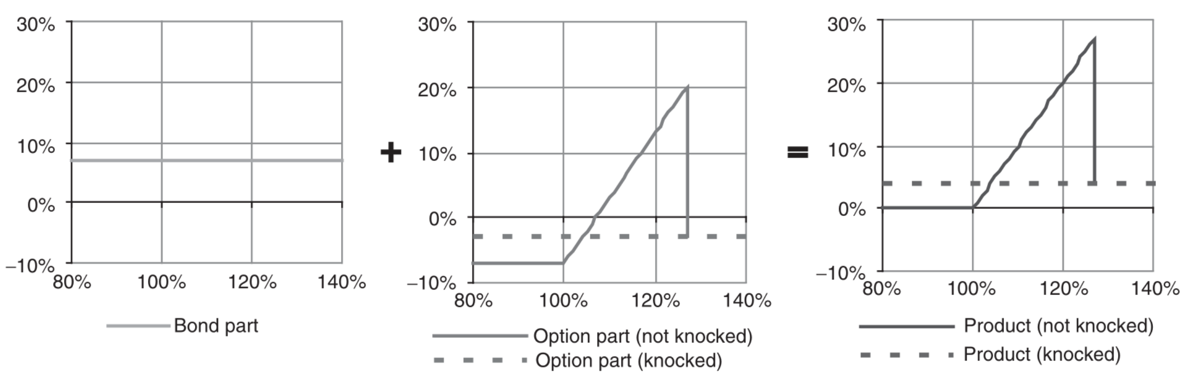


Figure 12.5: Shark Note Construction

- **Market Expectations:** Rising underlying, unlikely to breach a set barrier.
- Full capital protection with a short- to medium-term horizon.
- Uses an up-and-out call option; if the barrier is touched, a rebate may be paid.
- Participation in performance until a barrier is hit; if breached, participation ends and a rebate might be paid.

Redemption Scenarios:

- If the underlying is below 100% of its initial value at maturity: 100% capital guarantee.
- If above 100% without touching the barrier: 100% + participation.
- If the barrier is touched: 100% + rebate.

Strategic Use:

- Set a high barrier to minimize knock-out risk or a high rebate to enhance returns if knock-out occurs.
- Autocall feature allows early redemption if the barrier is breached, suitable for reinvestment without waiting for product maturity.

12.4.2.4 Capital Protection with Coupon

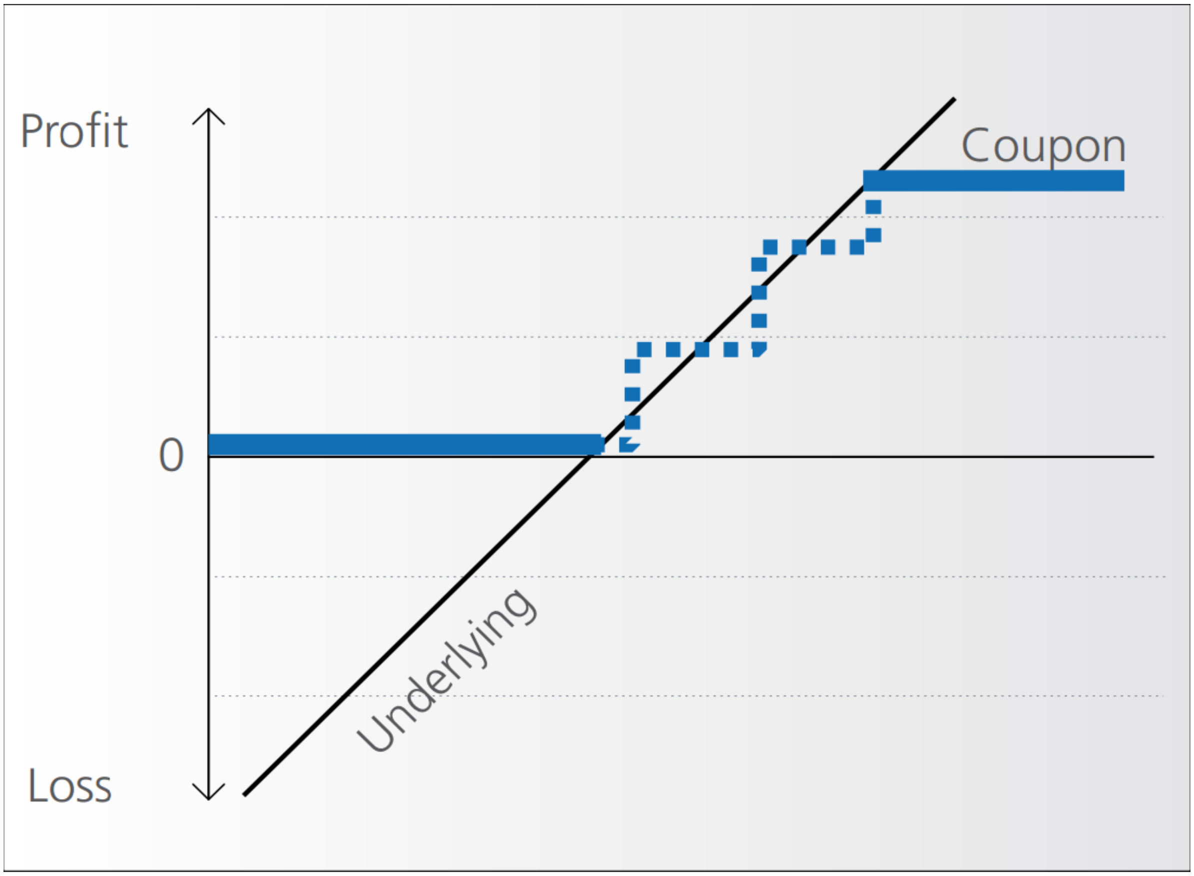


Figure 12.6: Capital Protection with Coupon

- **Market Expectations:** Rising underlying, potential sharp falls.
- Guaranteed capital protection at maturity.
- Periodic coupon payments linked to the performance of the underlying.
- Limited upside potential.

12.4.3 General Recommendations

- Match the product with your investment horizon; the capital guarantee is effective only at maturity.
- Avoid products with lower capital guarantee than 90% (e.g., 90% guarantee means that underlying must perform by more than 10% to be break even, not considering any opportunity cost).
- Ensure the participation rate is at least 80%.

- Verify the issuer's credit rating.
- Prefer shorter maturity periods for products like Shark Notes to reduce risk (no more than two or three years).

12.5 Behavior of Capital Guarantee Products - Case Study

Characteristic	Details
Underlying Risky Asset	Eurostoxx50 Index
Maturity	4 years
Implied Volatility	23%
Asset's Dividend Yield (p.a.)	4%
Interest Rate Level (4-year swap rate, p.a.)	4.5%
Capital Guarantee Level	100%
Initial Participation	100%

Key Variables:

- **Interest Rate:** A primary determinant of the price of the zero-coupon bond component of the product.
- **Volatility:** Crucial for the valuation of the call option embedded in the product. High volatility increases the potential upside, impacting the option's price more significantly when the asset is at-the-money.

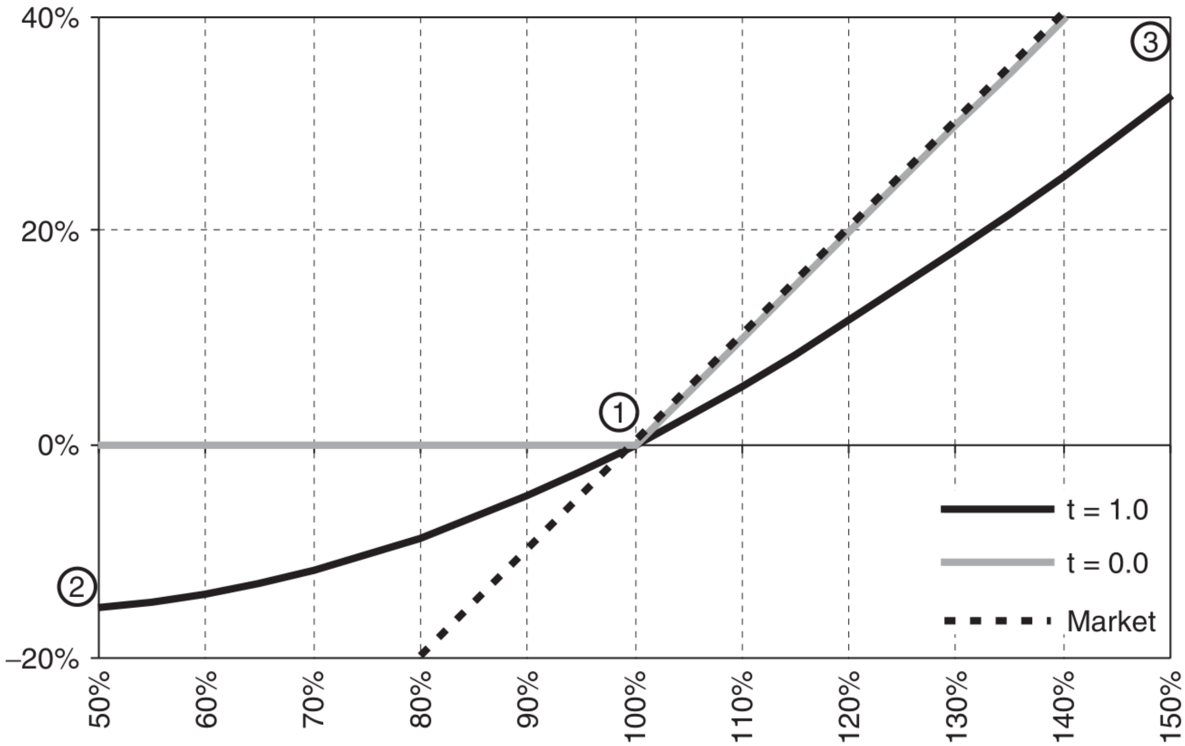


Figure 12.7: Spot price variations for classical capital guaranteed products

This graph demonstrates the relationship between the spot price of the underlying asset and the price of the capital guaranteed product. The slope of the line, particularly when it approaches a 45-degree angle, indicates a delta of 100%, where the product's price moves one-to-one with the underlying asset.

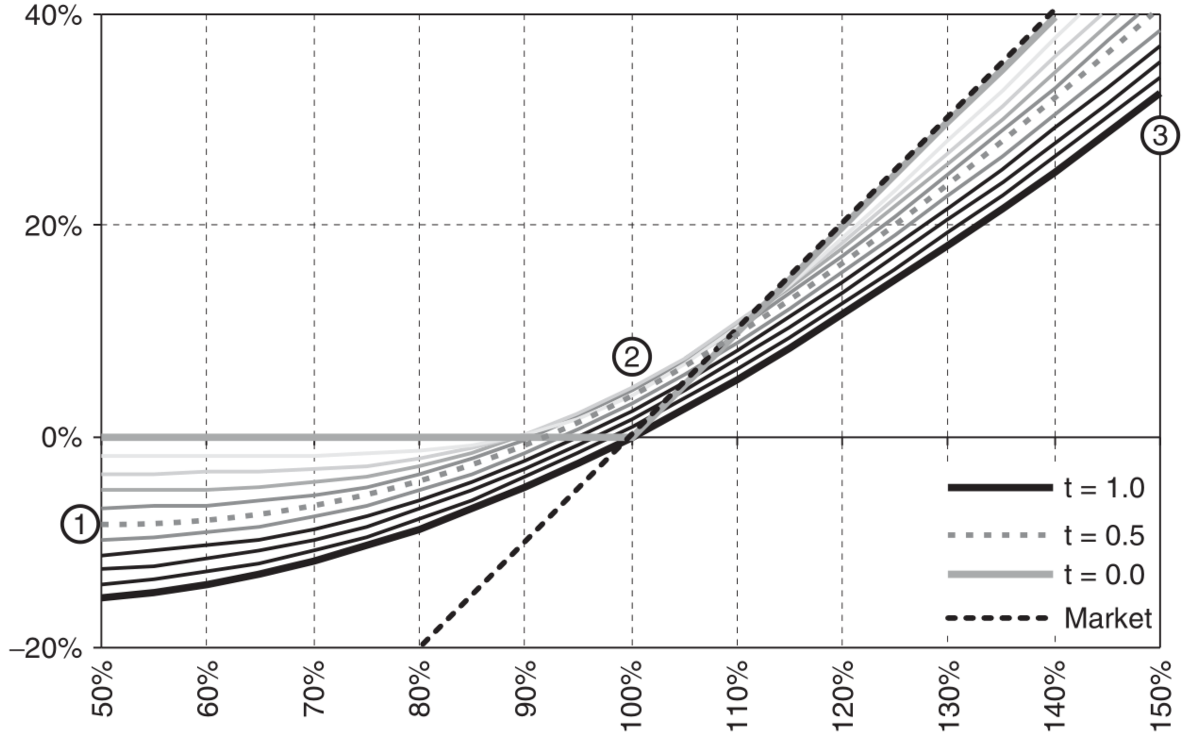


Figure 12.8: Price as a function of maturity

The slope of this graph illustrates how the delta of the product changes with time. Typically, the product's participation at inception ranges from 40%-60% of its maturity level, reflecting the initial risk profile and pricing.

Note: Capital guarantee is contingent on purchasing the product at par and is valid *only at maturity, not before. The guarantee level remains constant throughout the* product's life.

12.5.1 Impact of Volatility and Interest Rate Variations

- **Implied Volatility Fluctuations:**
  - An increase in implied volatility (e.g., from 23% to 33%) can enhance the value of the call option component, as shown in the corresponding graph.
  - A decrease (e.g., from 23% to 13%) typically lowers the call option's value due to reduced potential for high returns.

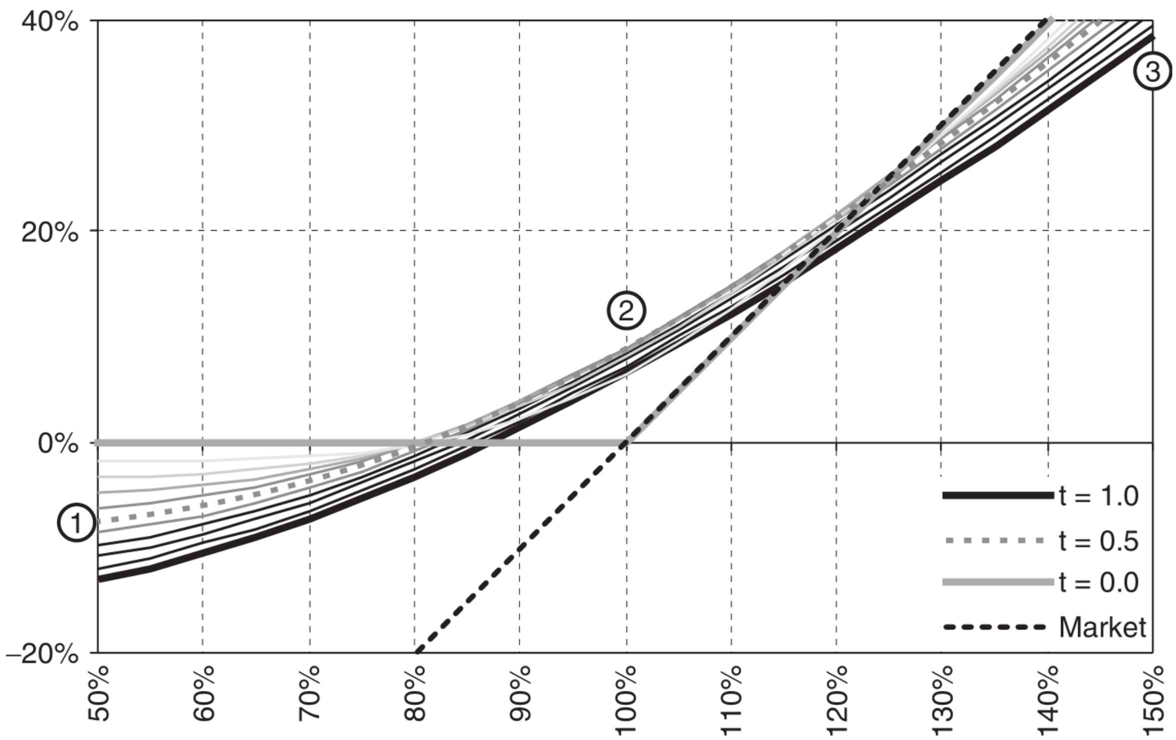


Figure 12.9: Implied volatility increase (from 23% to 33%)

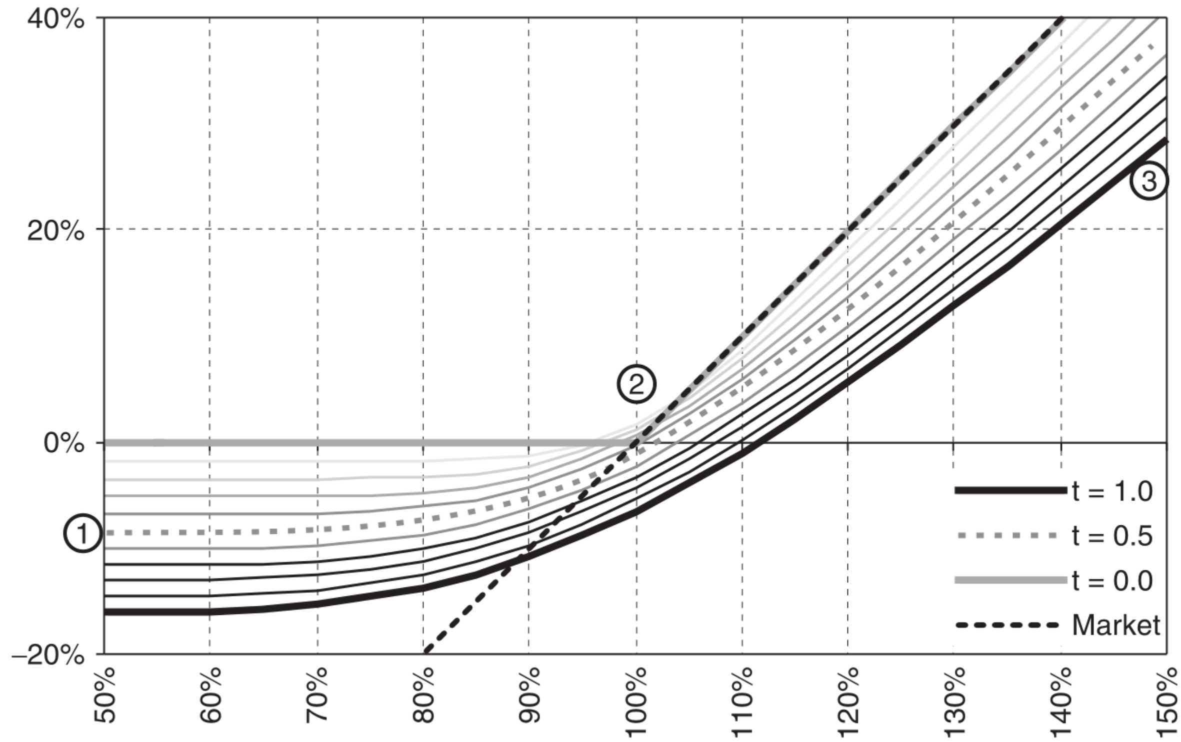


Figure 12.10: Implied volatility decrease (from 23% to 13%)

Volatility is more influential when the product is at-the-money and can mitigate some losses through increased option premiums during downturns in the asset's price.

- **Interest Rate Impact:**
  - Rising interest rates lead to lower prices for the zero-coupon bond component, influencing the overall valuation negatively, especially if the underlying asset's price falls simultaneously.
  - Conversely, falling interest rates increase the bond's value, cushioning any adverse effects from a drop in the underlying asset's price.

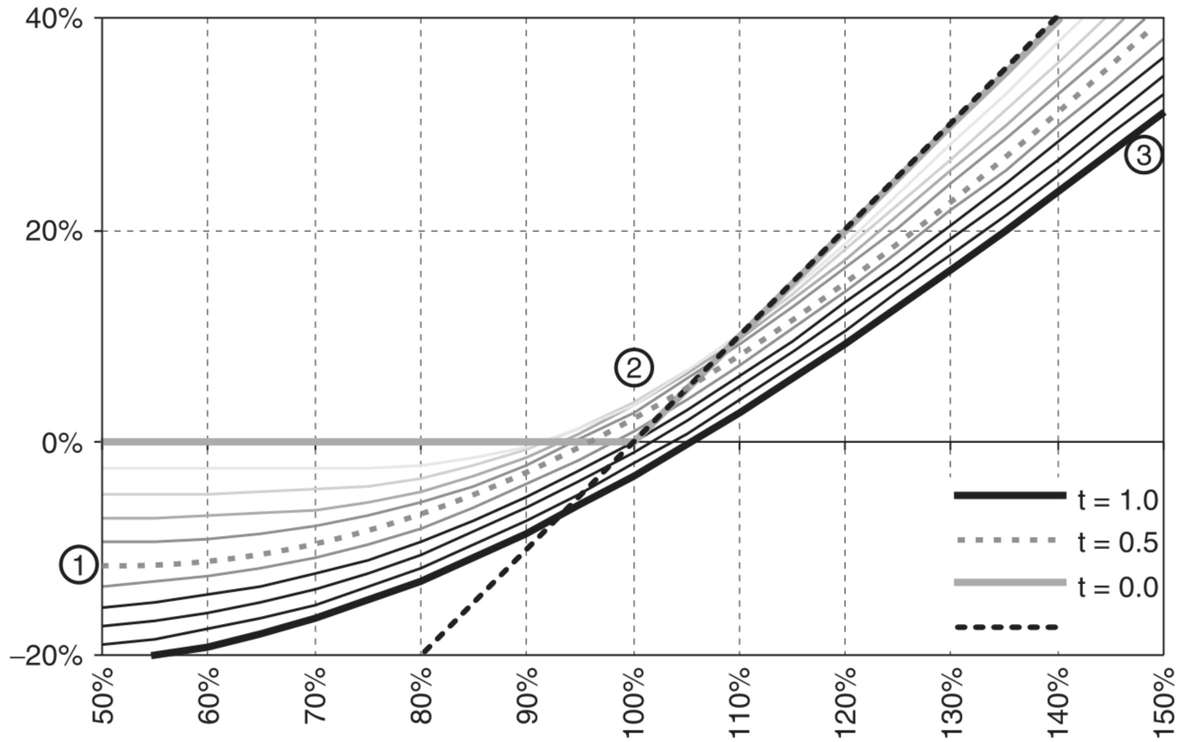


Figure 12.11: Interest rate increase (from 4.5% to 6.5%)

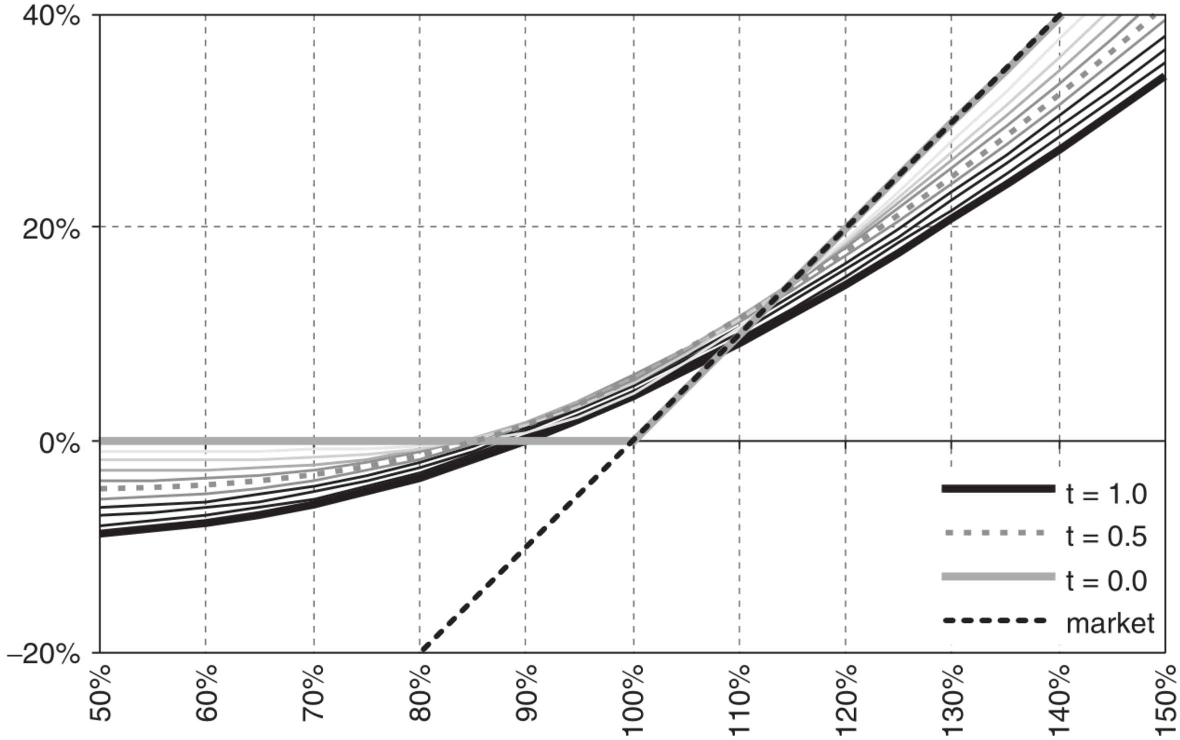


Figure 12.12: Interest rate decrease (from 4.5% to 2.5%)

12.5.2 Conclusive Insights

- Shifts in implied volatility influence the capital guaranteed product most when ***at-the-money***.
- An ***increase in volatility*** tends to ***increase the value*** of the product.
- A positive shift in ***interest rates lowers the value*** of the product most when the embedded call is ***out-of-the-money***.
- When the call is deep in-the-money, an interest rate shift has less impact.
- Everything else held equal, ***the passing of time*** (without movement on the spot) ***is positive for the value*** of the product over time, since the time value lost on the call is more than offset by the gain in the zero bond value.

12.6 Participation Investment Products

Definition

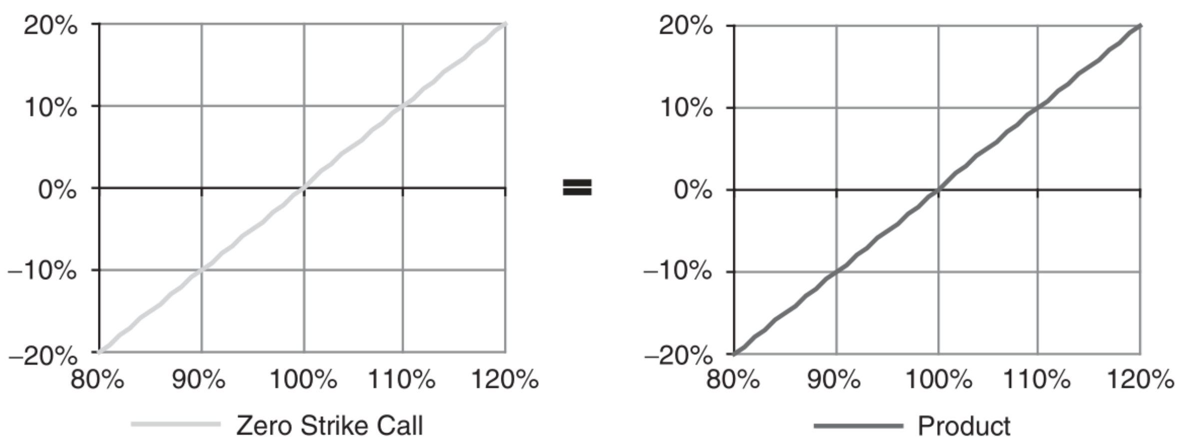
Participation products are investment vehicles that link returns directly to the performance of their underlying assets, sometimes featuring conditional downside protection or a leveraged upside.

Key Characteristics:

- **Risk Profile:** Generally higher risk compared to capital protection and yield enhancement products due to the absence of capital guarantees.
- **Underlying Assets:** Typically stocks or stock indices, but can also include commodities, real estate, and more exotic assets.
- **Liquidity and Efficiency:** Often very liquid, these products compete directly with ETFs in providing exposure to specific markets, themes, or regions.

12.6.1 Tracker Certificate

- **Function:** Mirrors the performance of one or more underlying assets. Commonly tracks excess returns (excluding dividends/yields).
- **Structure:** Comprised of a zero-strike call option, which values the asset minus any discounted dividends or yields.



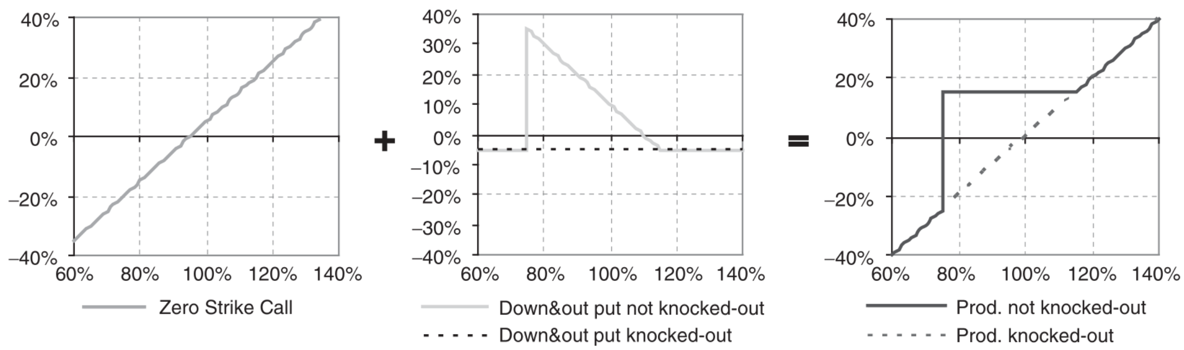
Advantages:

- **Access to Difficult Markets:** Allows investment in markets or assets that are otherwise inaccessible via traditional instruments, such as private equity or certain commodities.

- **Tax Efficiency:** May offer favorable tax treatments compared to direct investments in the underlying assets.

12.6.2 Bonus Certificate

- **Function:** Combines the features of a tracker certificate with conditional downside protection.
- **Structure:** Includes a zero-strike call option and a long down-and-out put option.

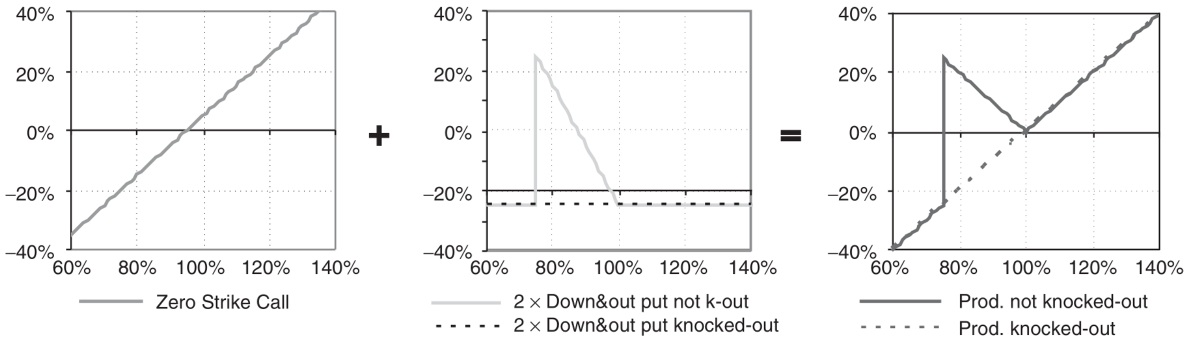


Key Parameters:

- **Participation Level:** Degree to which the investor gains from positive performance of the underlying.
- **Bonus Level:** Additional return offered if the underlying performs above a certain threshold without breaching a downside barrier.
- **Barrier Level:** The price level below which the downside protection is activated.
- **Maturity:** Typically short, recommended no longer than two to three years.

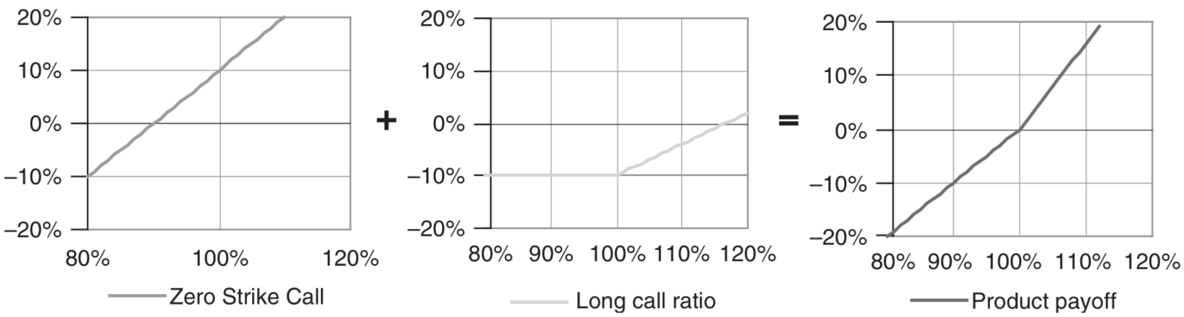
12.6.3 Twin-Win Certificate

- **Function:** Provides positive participation in both the upside and downside movements of the underlying asset.
- **Structure:** Consists of a long zero-strike call and double down-and-out put options.



12.6.4 Outperformance Certificate

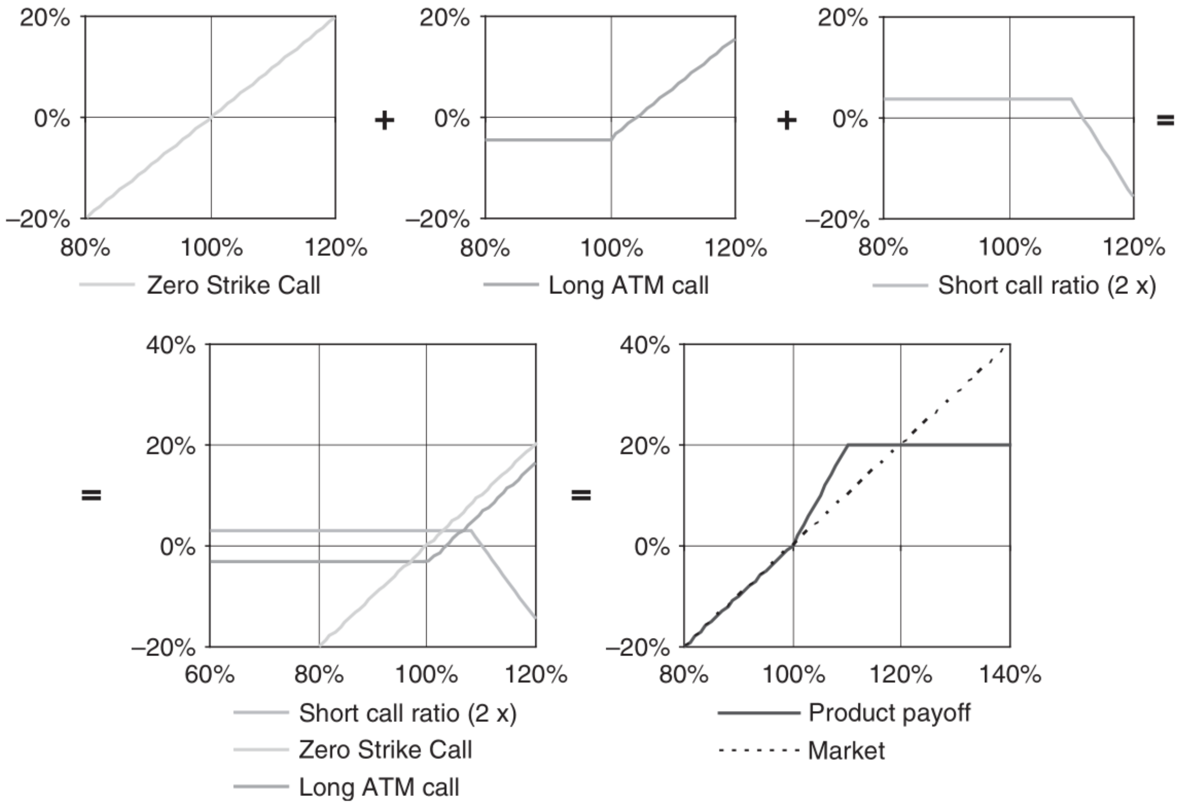
- **Function:** Designed for aggressive investment strategies, providing enhanced returns if the underlying outperforms expected dividends.
- **Structure:** Combines a zero-strike call with multiple long at-the-money calls, funded by dividends.



- **Volatility and Dividends:** Optimal conditions include low volatility (for cheaper options) and high dividend yields.

12.6.5 Capped Outperformance Certificate

- **Function:** Provides leveraged exposure to the underlying asset up to a capped level.
- **Structure:** A zero-strike call combined with a long at-the-money call and two short calls at higher strikes.



- **Scenario Planning:** Best utilized in a moderately bullish scenario with falling volatility.
- **Timing and Maturity:** Critical due to the short duration of the product, typically 3-9 months.

12.7 Behavior of Participation Products

Participation products such as bonus, turbo, airbag, and outperformance certificates are heavily influenced by factors like implied volatility, dividend yields, and interest rates. These factors shape the product's performance and its strategic suitability for different market conditions.

12.7.1 Bonus Certificate - Case Study

Factor (Increase)	Impact on Product's Price	Effect Level
Spot Price	Positive	Maximum
Implied Volatility	Variable	High
Implied Correlation	Positive	Medium-Low

Factor (Increase)	Impact on Product's Price	Effect Level
Interest Rates	Negative	Low
Dividends	Negative	Medium

- **Initial Sensitivity:** At issuance, the delta of a bonus certificate is approximately 1, meaning its price moves almost one-for-one with the underlying asset. However, this sensitivity decreases if the spot price approaches the barrier.
- **Price Stability:** Bonus certificates tend to underperform during market downturns due to the drop in market value, despite their protective features.

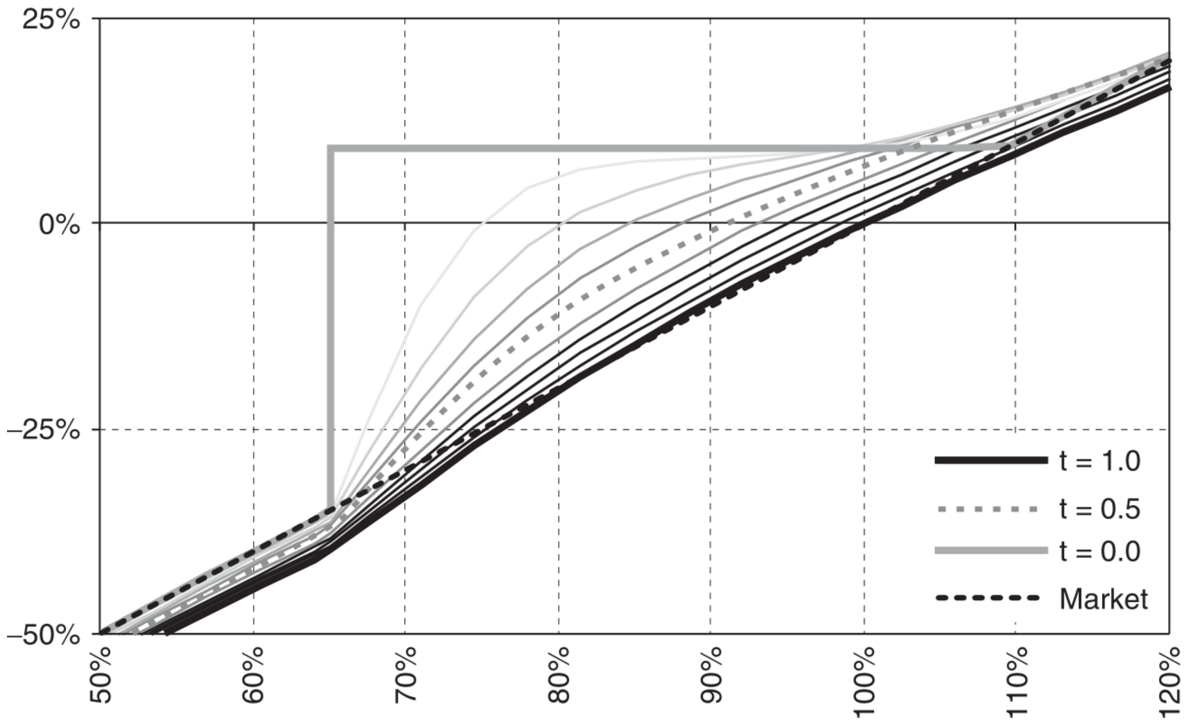


Figure 12.13: Price as a function of maturity

- **Maturity:** Shorter maturities are preferable to reduce exposure to prolonged market volatility.
- **Barrier Level:** The barrier should be set considering the worst-case market scenario to ensure effective downside protection.
- **Leverage and Sensitivity:** As the spot price nears the barrier, the certificate's delta becomes highly volatile, which can lead to significant price swings.

12.7.1.1 Impact of Implied Volatility

- **Initial Conditions:** High implied volatility at issuance allows for better pricing of protective options.
- **Post-Issuance Volatility:** A decrease in implied volatility post-issuance generally benefits the mark-to-market value of the certificate.

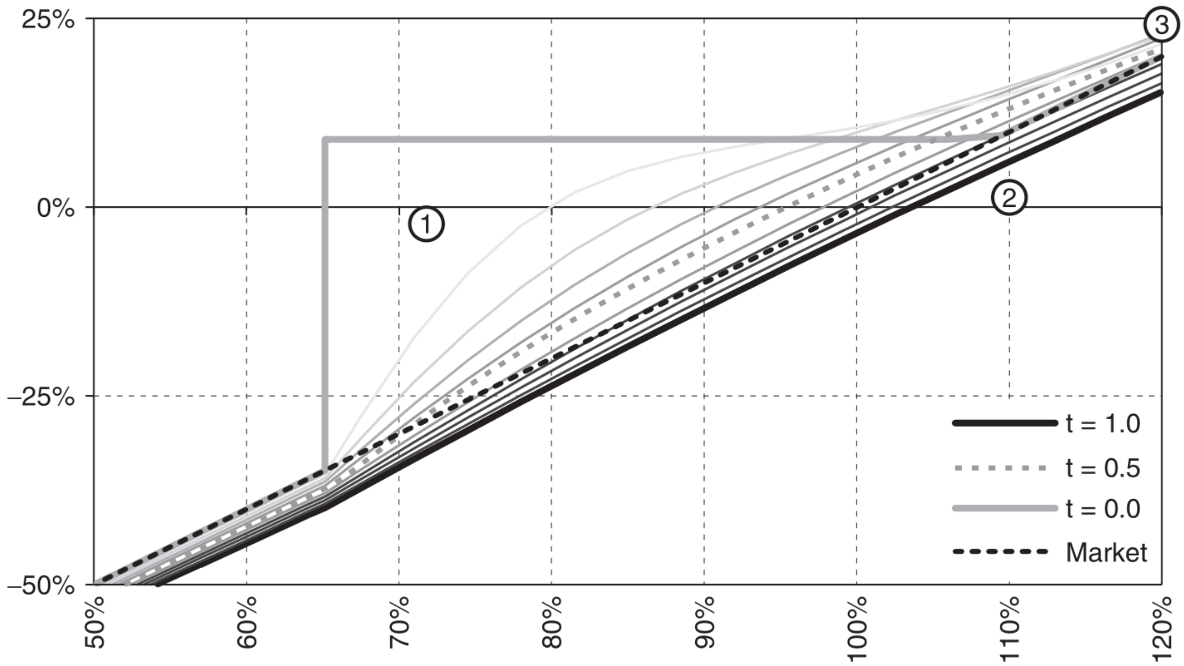


Figure 12.14: Increase in implied volatility (from 23% to 33%)

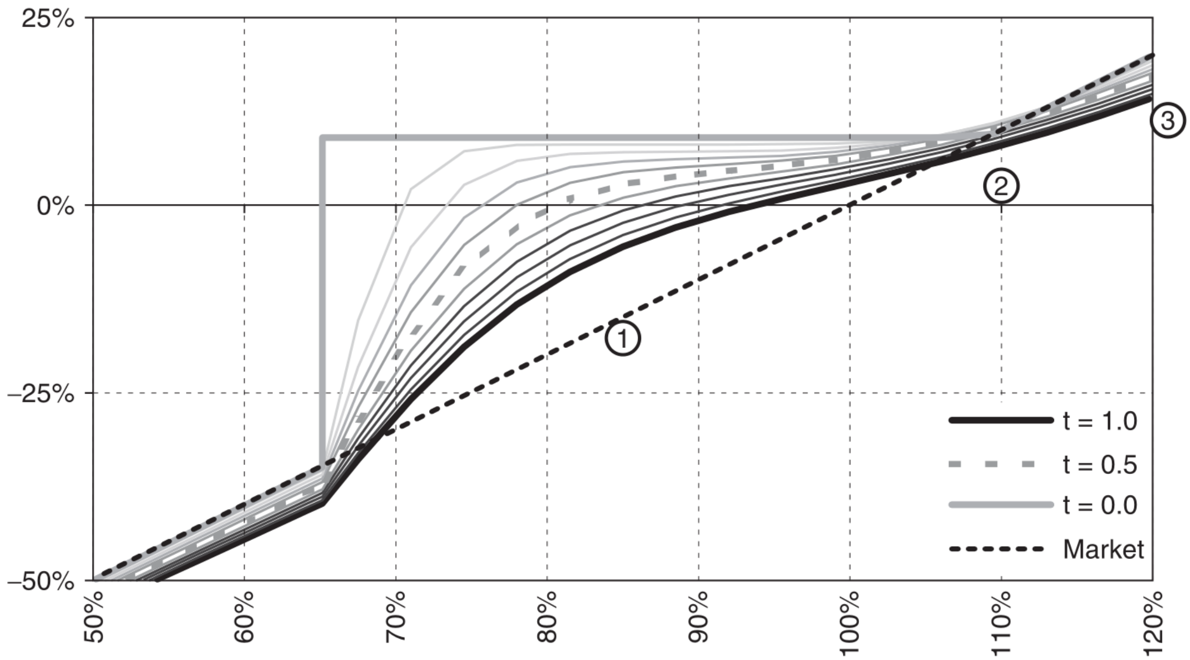


Figure 12.15: Decrease in implied volatility (from 23% to 13%)

12.8 Practice Questions and Problems

1. What is the value of the capital guarantee certificate at maturity if the value of the underlying asset ends at 120, the strike price of the certificate is 100, level of guarantee 100%, and participation 80%?

> **Solution**
>
> 116

2. What is the value of the capital guarantee certificate at maturity if the value of the underlying asset ends at 90, the strike price of the certificate is 100, level of guarantee 100%, and participation 80%?

> **Solution**
>
> 100

3. What is the value of the capital guarantee certificate at maturity if the value of the underlying asset ends at 63, the strike price of the certificate is 40, the cap is 50, level of guarantee 100%, and participation 130%?

Solution

53

4. What is the value of the capital guarantee certificate with knock-out at maturity if the value of the underlying asset ends at 50, the strike price of the certificate is 55, the knock-out barrier is 67, level of guarantee 100%, and participation 110%? The highest value of the underlying asset during maturity was 70.

Solution

55

5. What is the value of the capital guarantee certificate with knock-out at maturity if the value of the underlying asset ends at 63, the strike price of the certificate is 55, the knock-out barrier is 67, level of guarantee 100%, and participation 110%? The highest value of the underlying asset during maturity was 65.

Solution

63.8

6. What is the value of the outperformance certificate at maturity if the value of the underlying asset ends at 56, the strike price of the certificate is 40, and the participation is 170%?

Solution

67.2

7. What is the value of the outperformance certificate at maturity if the value of the underlying asset ends at 77, the strike price of the certificate is 40, the cap is 70, and the participation is 130%?

Solution

79

8. What is the value of the bonus certificate at maturity if the value of the underlying asset ends at 35, the strike price of the certificate is 40, the knock-out barrier is 30? The lowest value of the underlying asset during maturity was 35.

> **Solution**
>
> 40

9. What is the value of the bonus certificate at maturity if the value of the underlying asset ends at 35, the strike price of the certificate is 40, the knock-out barrier is 30? The lowest value of the underlying asset during maturity was 25.

> **Solution**
>
> 35

10. What is the value of the twin-win certificate at maturity if the value of the underlying asset ends at 44, the strike price of the certificate is 50, the knock-out barrier is 40? The lowest value of the underlying asset during maturity was 35.

> **Solution**
>
> 44

11. What is the value of the twin-win certificate at maturity if the value of the underlying asset ends at 44, the strike price of the certificate is 50, the knock-out barrier is 40? The lowest value of the underlying asset during maturity was 41.

> **Solution**
>
> 56

13 Structured Products II

References

- BLÜMKE, Andreas. How to invest in structured products: a guide for investors and investment advisors. Chichester: Wiley, 2009. xvi, 374. ISBN 9780470746790.

Learning Outcomes:

- Identify and explain the mechanisms and purposes of different yield enhancement structures such as reverse convertibles and discount certificates.
- Analyze strategic variations in yield enhancement products, including ways to enhance the coupon and general investment strategies.
- Evaluate the behavioral dynamics of yield enhancement products under varying market conditions.
- Utilize case studies, such as knock-in reverse convertibles, to understand practical applications and implications of product behaviors.
- Describe common special features in structured products, including autocall and callable options, and their impact on product performance and investor options.

13.1 Yield Enhancement Investment Products

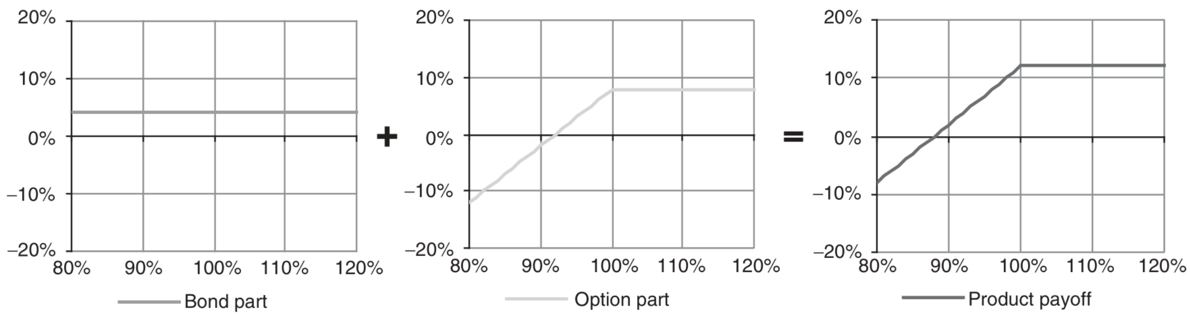
Definition

Yield enhancement products are financial constructs with capped upside potential and without capital guarantee, aiming to generate high returns relative to bond yields. The risk level may approach that of the underlying assets in unfavorable market scenarios.

- **Category Significance:** This is the largest category of structured products in terms of both the variety of structures available and the volume of investments.
- **Capped Upside:** Historically, all yield enhancement products feature a capped upside, contrasting with participation products that offer unlimited upside potential.
- **Popular Types:** The most prevalent forms are reverse convertibles and discount certificates, which share similar payout profiles but differ in construction.

13.1.1 Reverse Convertibles

- **Components:**
  1. **Zero-Coupon Bond:** Purchased to cover the product's lifetime, providing the base return.
  2. **Short Plain Vanilla Put Option:** Sold on the underlying risky asset to generate additional premium.

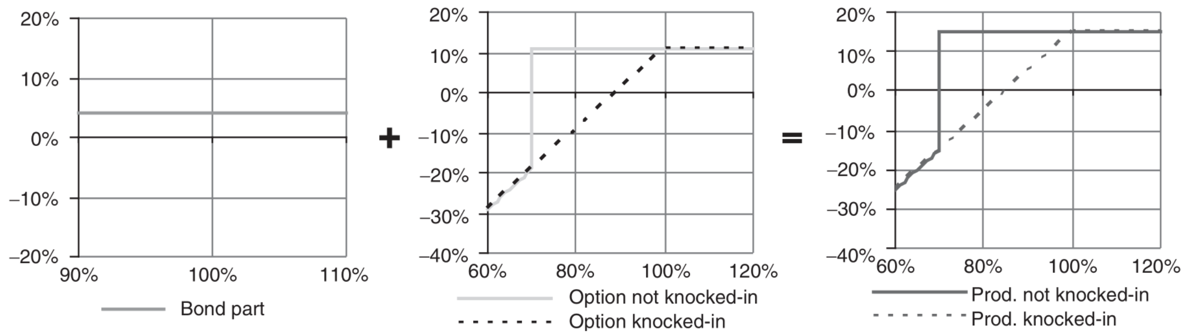


> **i** Example
>
> The zero-coupon bond yields 4%, and the short put option provides a premium of 8%. The combined coupon is 12%, allowing the underlying asset to decline by up to 12% before the investor experiences a loss.

- **Risk Considerations:** The maximum potential loss is limited to the coupon received. However, the risk of loss increases if the underlying asset's price falls significantly beyond the cushion provided by the coupon.

13.1.2 Worst-of Barrier Reverse Convertible

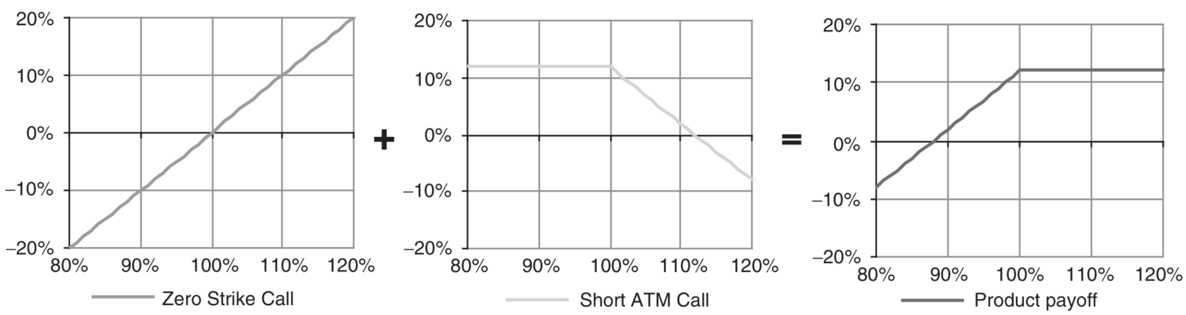
- **Enhanced Features:**
  1. **Multiple Underlyings:** The payoff depends on the performance of the worst-performing asset.
  2. **Knock-in Barrier:** Added to the short put option to provide conditional capital protection unless the barrier is breached.



- **Behavioral Dynamics:** The presence of a barrier can lead to a sharp price adjustment if the underlying reaches this level, activating the put option.

13.1.3 Discount Certificates

- **Structure:**
    1. **Long Zero-Strike Call Option:** Replicates the performance of the underlying asset.
    2. **Short Call Option:** Set at- or out-of-the-money to provide upfront income by reducing the purchase price.



- **Economic Rationale:** The premium from the short call lowers the effective purchase price of the zero-strike call, providing an immediate financial benefit, which contrasts with the deferred coupon payment structure of reverse convertibles.
- **Tax Efficiency:** In some jurisdictions, the structure of discount certificates can offer tax advantages over other investment products.

13.1.4 Strategic Variations and Recommendations

- **Barrier-Linked Coupons:** Linking the coupon payment to the breaching of one or more barriers can significantly increase the potential return but also adds to the risk.

- **Individualized Barriers:** In worst-of barrier products, setting specific barrier levels for each underlying can tailor the risk to more accurately reflect the volatility and performance expectations of each asset.
- **Barrier Perception:** Investors should not underestimate the risk associated with barriers, as these are high-risk products without capital guarantees.
- **Product Complexity:** Avoid investing in products based on too many underlying assets to manage risk more effectively. Preferably, limit the number of underlyings to two.
- **Barrier Style:** Opt for European-style barriers when market volatility is high and the medium-term market trajectory is uncertain, as these barriers typically offer a lower coupon but reduce the risk of early activation.

13.2 Behavior of Yield Enhancement Products

13.2.1 Knock-In Reverse Convertibles - Case Study

Knock-in reverse convertibles are complex financial instruments designed to enhance yields by incorporating a conditional protection feature, which activates under specific market conditions. Here's a detailed analysis of the behavior of such products, using a specific example for illustration.

Characteristic	Details
Underlying Risky Asset	Eurostoxx50 Index
Maturity	1 year
Implied Volatility	23%
Asset's Dividend Yield (p.a.)	4.0%
Interest Rate (4-year swap rate, p.a.)	4.5%
Barrier Level (% of spot)	75%
Coupon	10.4%

Factor (Increase)	Impact on Product's Price	Impact Level
Spot Price	Positive	Maximum
Implied Volatility	Negative	High
Implied Correlation	Positive	Medium-Low
Interest Rates	Negative	Low
Dividends	Negative	Low

13.2.2 Variations in Spot Price

- **At Issue Date:** The product exhibits bond-like characteristics when the spot price of the underlying asset increases, yet behaves akin to the asset itself with a decline in price, increasing the delta.
- **Near Maturity:** A significant gap develops; the delta can exceed 1, indicating increased sensitivity to spot price movements.
- **Delta Analysis:** The delta of approximately 34% suggests a 34% probability that the underlying's price decline will activate the knock-in feature (e.g., a 25% or greater drop in the Eurostoxx50).
- **Investor Perception:** While less risky than direct exposure to the underlying asset, the delta range (30%-40%) illustrates that these products are considerably more volatile than bonds.

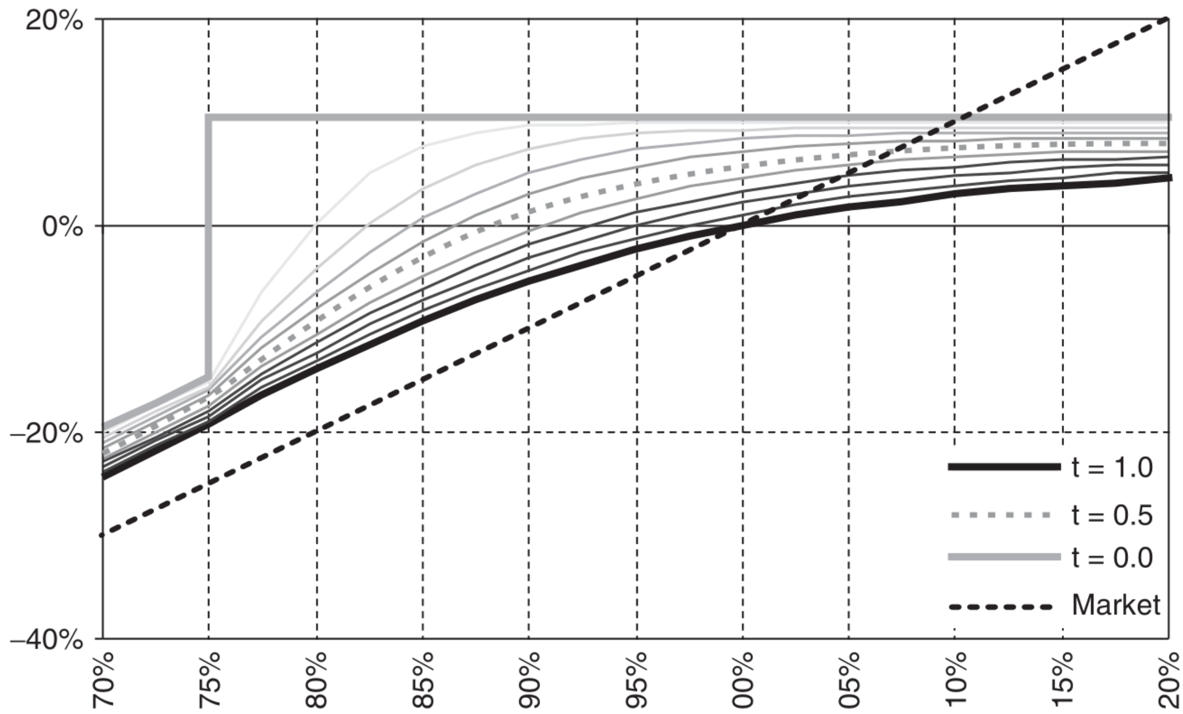


Figure 13.1: Reverse convertible as a function of spot and time to maturity

13.2.3 Impact of Implied Volatility Changes

- **Price Sensitivity:** Implied volatility significantly affects the product's price, particularly near the barrier. A lower volatility reduces price sensitivity to spot variations except near this critical threshold.

- **Investment Timing:** Higher initial implied volatility is advantageous as it allows for purchasing more protective options at a lower cost. Conversely, a drop in volatility after issuance benefits the mark-to-market valuation.
- **Product Lifespan:** Don't default to one-year products; sometimes shorter or longer maturities may align better with market conditions or portfolio strategies.

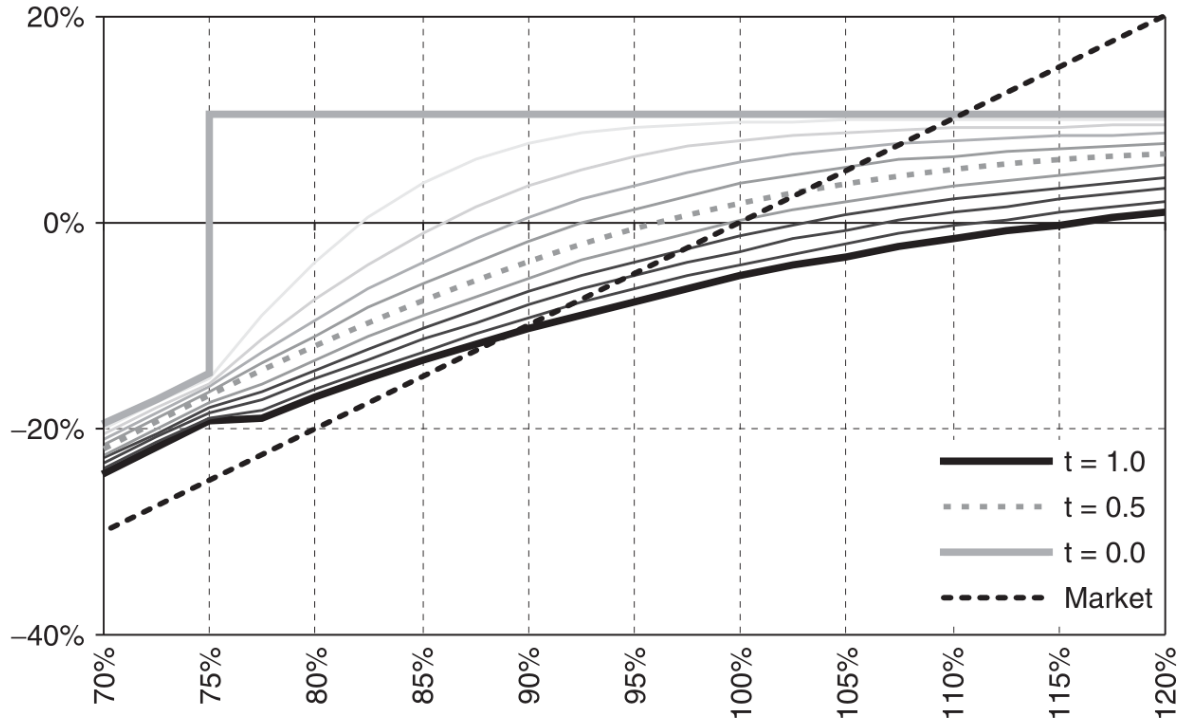


Figure 13.2: Increase in implied volatility (from 23% to 33%)

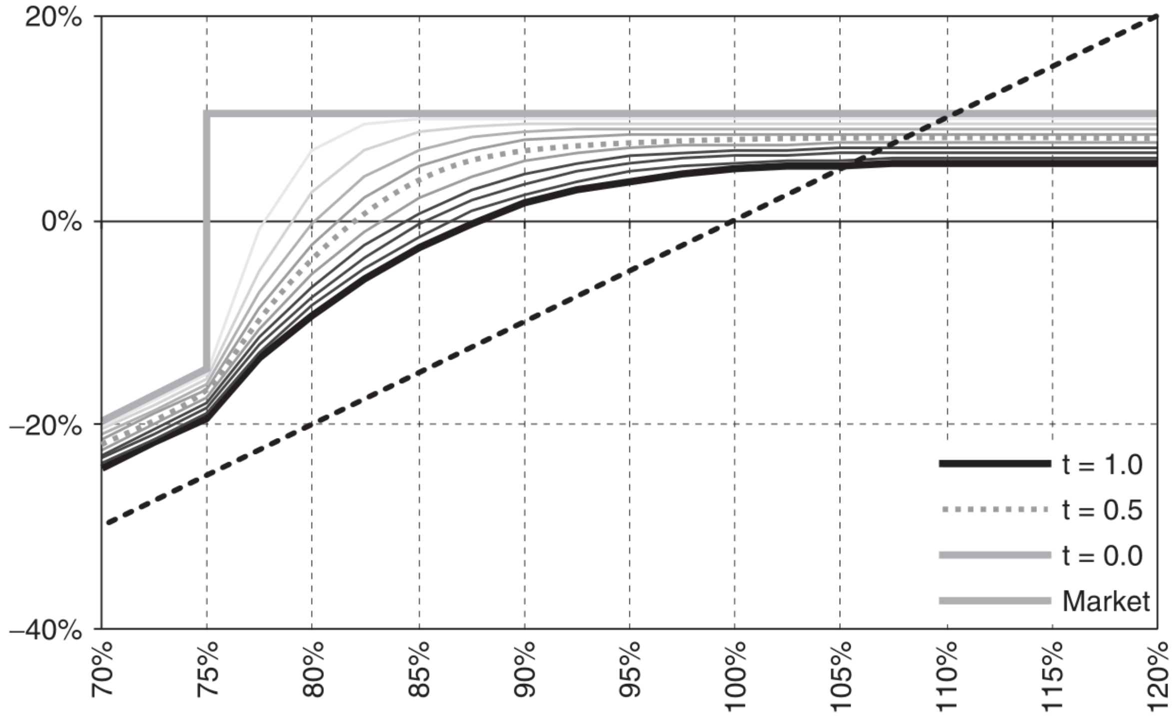


Figure 13.3: Decrease in implied volatility (from 23% to 13%)

13.2.4 Understanding Implied Correlation

Definition

Implied correlation reflects the expected linear relationship between two or more assets, as inferred from market prices.

- **Significance in Multi-Asset Products:** Essential for assessing the risk and potential return in products with multiple underlyings, such as 'worst-of' options, where a higher correlation can lead to enhanced coupon rates.
- **Asset Addition Risks:** Introducing additional underlyings for a modest increase in coupon (1%-2%) is generally not justified by the associated increase in risk, particularly for volatile stocks.
- **Risk Assessment:** Carefully evaluate the added value versus risk when including multiple assets. The incremental risk often outweighs the potential return enhancement, especially when the volatility of additional assets is high.

13.3 Leverage Products

Leverage products are financial instruments that provide a cost-effective way to gain exposure to various assets (shares, indices, commodities, currencies) with a smaller investment than purchasing the assets directly. These products amplify potential returns and increase the risk of loss, making them suitable for speculative purposes or for hedging against market movements.

13.3.1 Warrants

> Definition
>
> Warrants are financial instruments granting the holder the right, but not the obligation, to buy (Call Warrants) or sell (Put Warrants) an underlying asset at a predetermined price (strike price) on or before a specific date.

- **Characteristics:**
  - Warrants are similar to options but usually have longer maturities and cover a broader range of underlying assets.
  - Issued by financial institutions, warrants are bespoke products with terms set by the issuer.
  - Investors face the credit risk of the issuer since warrants are not backed by the assets themselves.
  - Warrants are highly liquid and can be used to leverage positions, hedge risks, or exploit arbitrage opportunities.

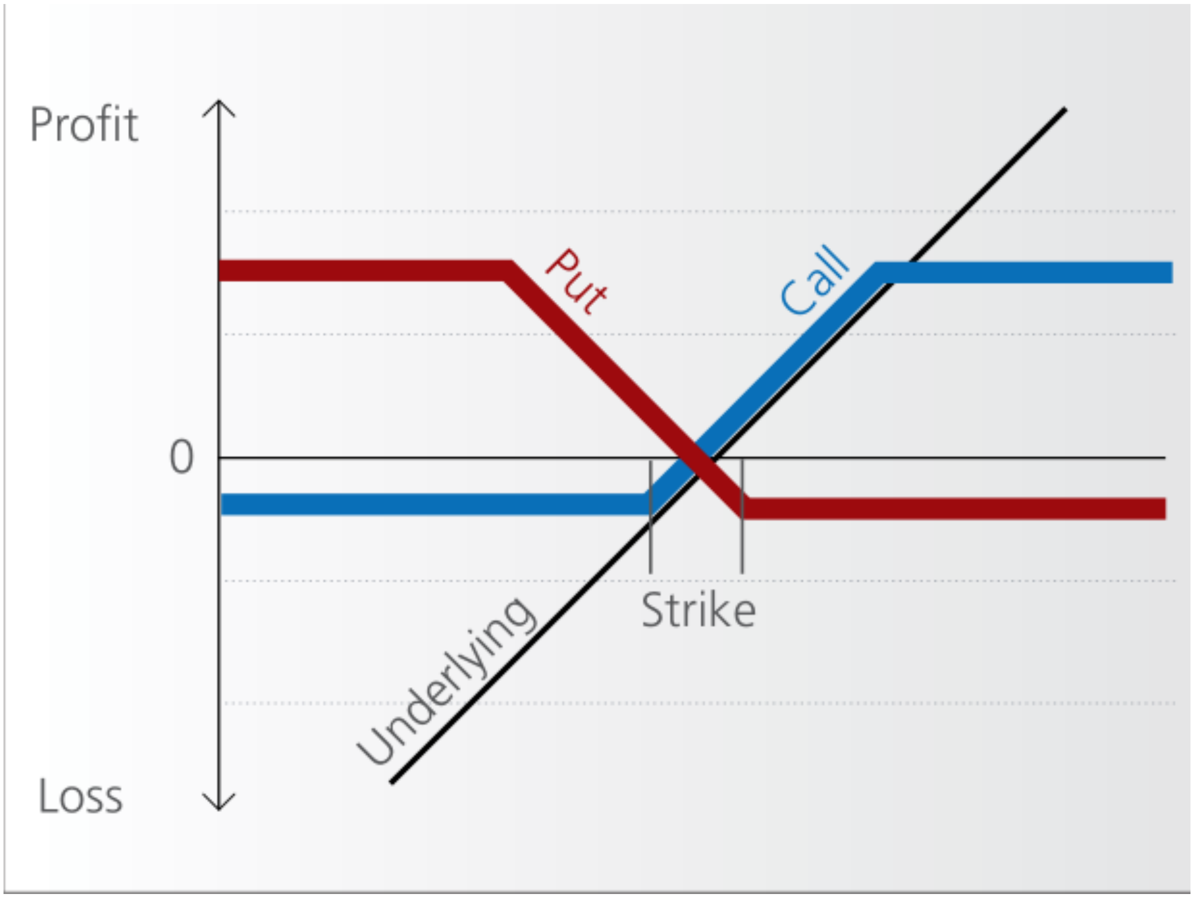
13.3.1.1 Types of Warrants

- **Traditional Warrants:** Issued by companies, often alongside bonds ("warrant-linked bonds") to raise capital at a lower interest cost. These warrants can be detached and sold separately.
- **Covered Warrants:** Issued by financial institutions and backed by assets the institution holds or can acquire. They can cover a wide range of assets beyond equities.
- **Naked Warrants:** Issued without backing assets; issuers hedge their exposure through other means.

13.3.1.2 Spread Warrants

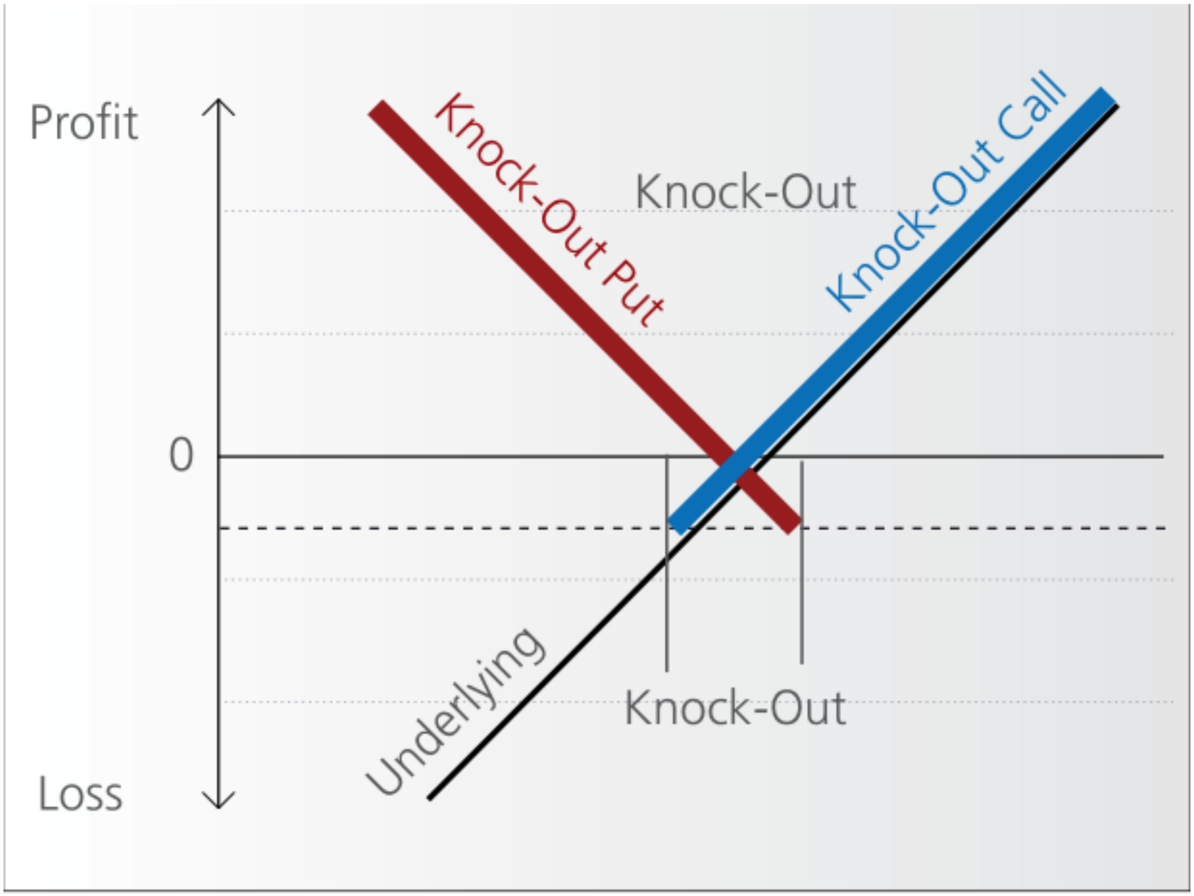
- **Purpose:** Designed to benefit from rising (bull warrants) or falling (bear warrants) markets.

- **Risks and Rewards:** While the initial investment is small and can yield significant leveraged gains, the maximum loss and gain are capped, and the value deteriorates as expiration approaches.



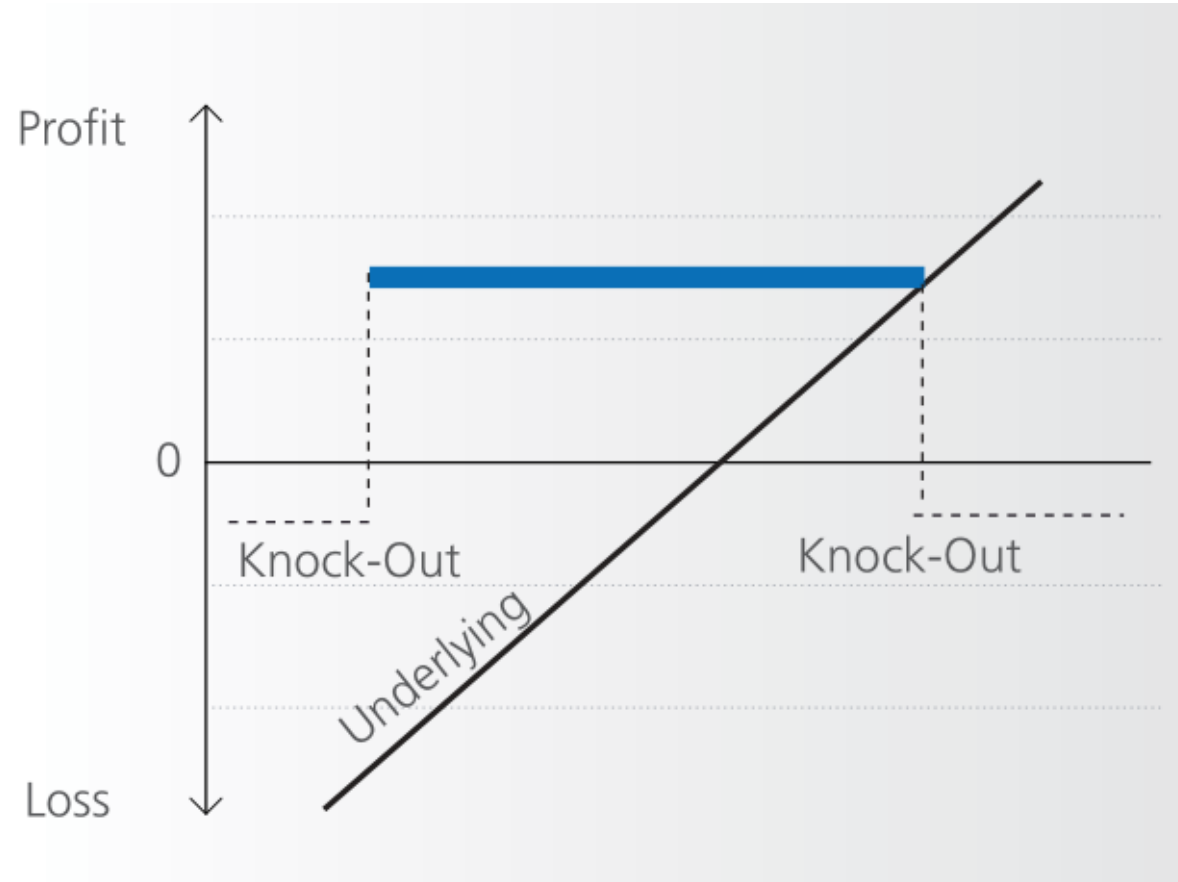
13.3.1.3 Knock-Out Warrants

- **Feature:** These products have an embedded barrier that, if reached, causes the warrant to expire worthless immediately.
- **Market Suitability:** Ideal for scenarios with clear directional market expectations and a significant buffer between the current price and the knock-out level.
- **Risk Profile:** Higher leverage compared to traditional warrants, with little to no time value, making them highly sensitive to market movements.



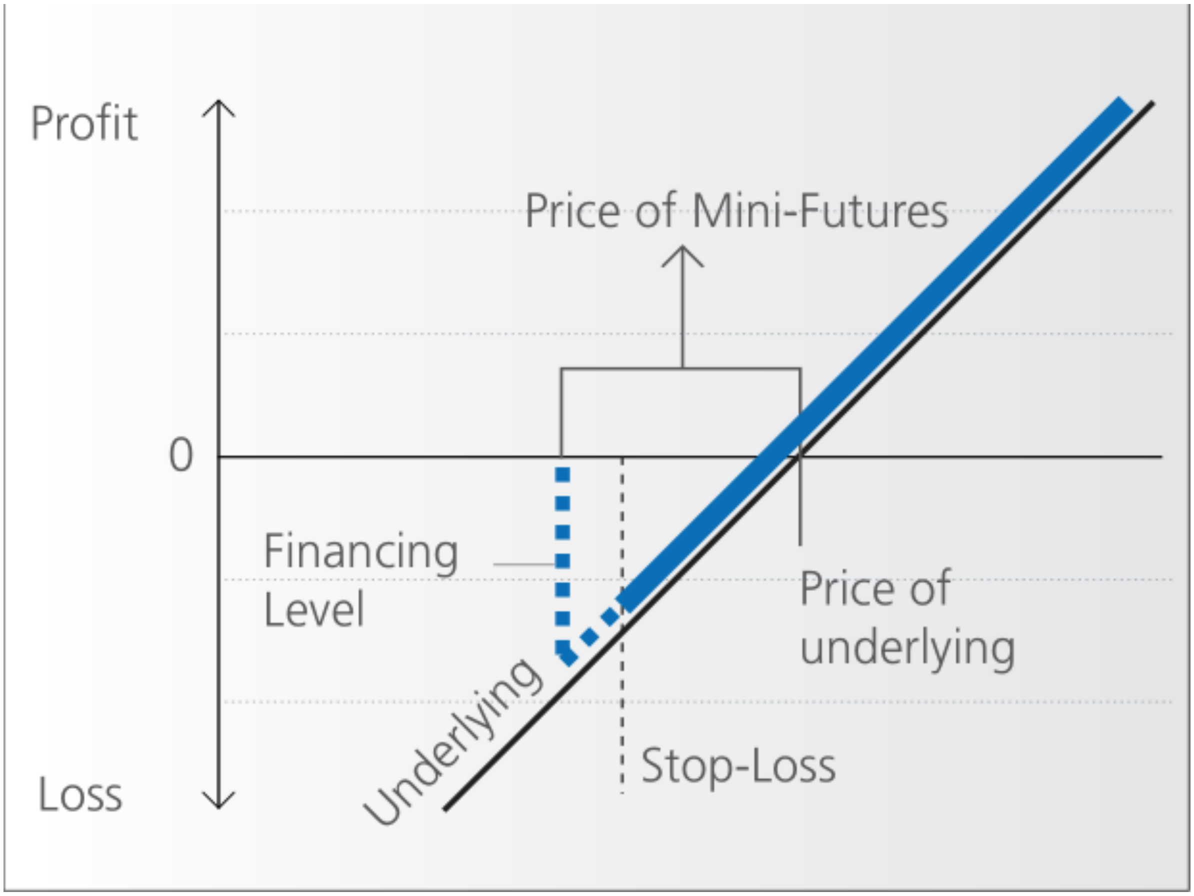
13.3.1.4 Double Knock-Out Warrants

- **Design:** These warrants have two barriers (upper and lower), offering a balanced approach to leveraging both upward and downward market movements.



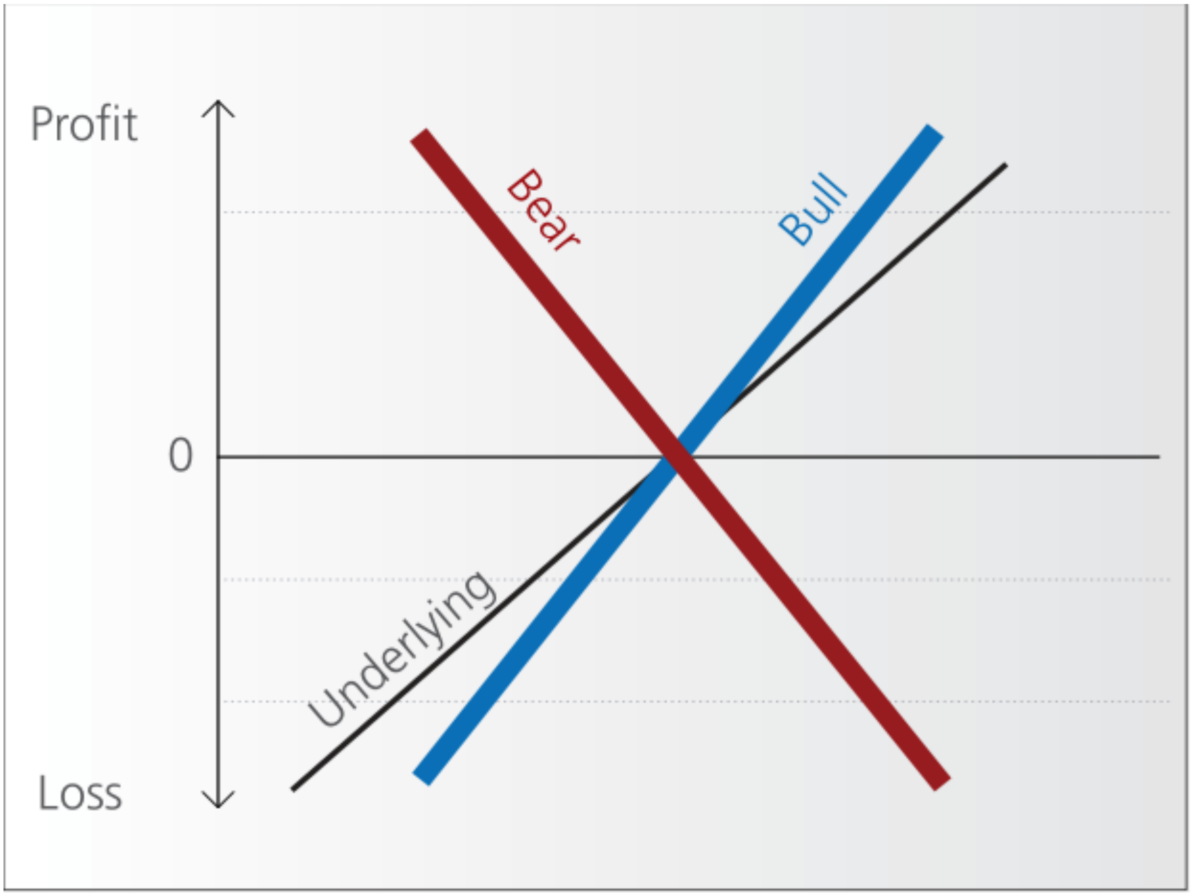
13.3.2 Mini-Futures

- **Comparison with Warrants:** Similar to knock-out products but open-ended, with no fixed expiration date unless the knock-out threshold is breached.
- **Investment Strategy:** Suitable for short-term speculative bets or hedging, with potential for residual value redemption after a stop-loss event.



13.3.3 Constant Leverage Certificates

- **Function:** These certificates provide fixed daily leverage on the price movements of an underlying asset, without any knock-out barriers.
- **Benefits:** They offer continuous exposure to price movements, unaffected by the underlying's volatility, and are not limited by time decay.



13.3.4 Additional Links

- https://www.investopedia.com/terms/w/warrant.asp
- https://en.wikipedia.org/wiki/Warrant_(finance)
- https://en.wikipedia.org/wiki/Covered_warrant
- https://www.six-structured-products.com/en/know-how/product-know-how/leverage-products-without-knockout/warrants
- https://www.six-structured-products.com/en/know-how/product-know-how/leverage-product-with-knock-out/knock-out-warrants

13.4 Common Special Features of Structured Products

Structured products often incorporate complex features to enhance their appeal to investors. These features, such as autocalls, callable options, and choices between physical or cash settlements, add flexibility and potentially higher returns under certain conditions, but also introduce specific risks and complexities.

13.4.1 Autocall and Callable Options

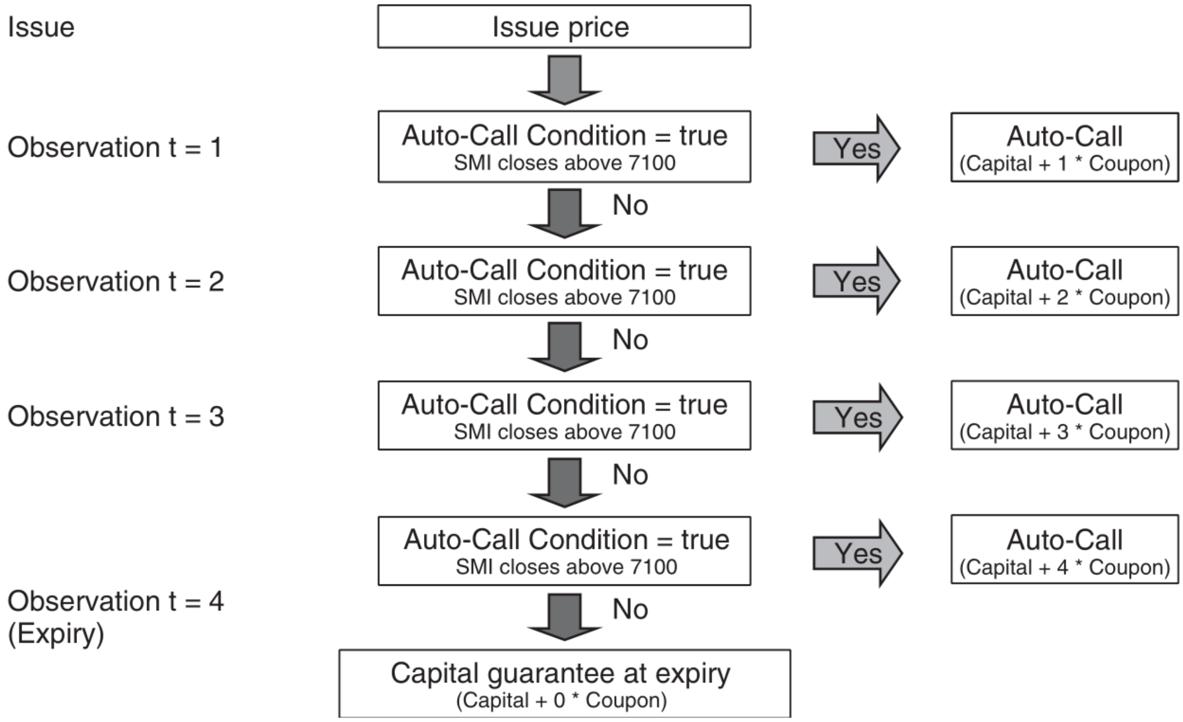
Definition

An autocall is a feature where the issuer is obligated to redeem the product when a predefined event occurs, which can be set to trigger at any time or on a specific date, at a predetermined price.

- **Callable Options:** Allow the issuer to decide whether to redeem the product on predetermined dates, providing more control over the product's lifecycle.

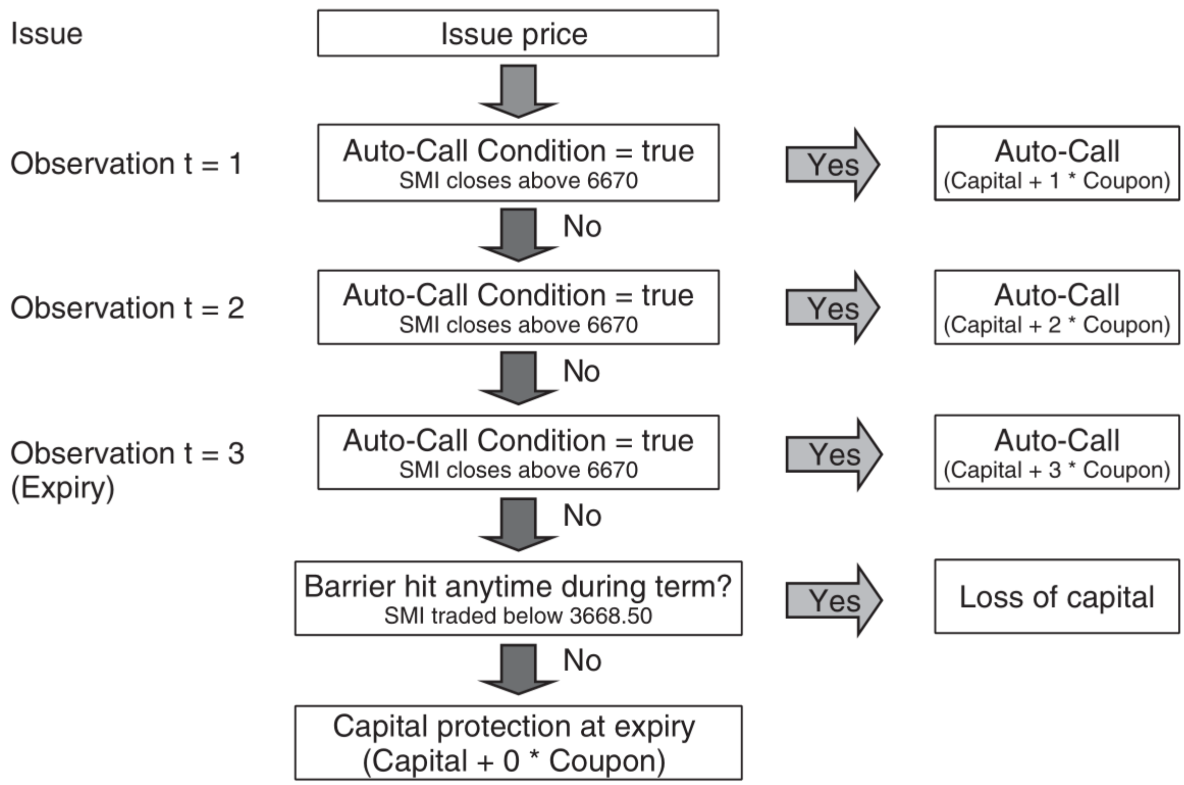
13.4.1.1 Autocallable Notes

- **Structure:** Typically includes full capital protection with the issuer redeeming the product if the underlying asset closes above a certain threshold (e.g., 100%) at predefined observation dates.
- **Returns:** Investors receive their capital plus a coupon, which is a percentage multiplied by the number of years the product has run.



13.4.1.2 Express Certificate

- **Features:** Combines an autocallable structure with conditional capital protection, usually involving a down-and-in barrier option.
- **Volatility Sensitivity:** High sensitivity to volatility due to the barrier; investors are effectively short volatility, losing out when it rises.



13.4.1.3 Autocallable Worst-Of Barrier Reverse Convertibles

- **Mechanism:** Functions like standard barrier reverse convertibles but includes an auto-call feature that triggers early redemption if all underlying assets exceed a certain level (e.g., 105% of initial spot).
- **Correlation Impact:** The correlation between the underlyings is crucial, as it affects the likelihood of all assets meeting the trigger condition simultaneously.

13.4.2 Settlement Types in Equity-Based Products

Structured products based on equities, such as reverse convertibles or bonus certificates, can offer different types of settlements:

1. **Physical Delivery:** Investors receive the actual shares at the redemption date, which might occur weeks after the expiry date of the embedded options.
2. **Cash Settlement:** Provides investors with a cash amount equivalent to the value of the shares at expiry.

- **Investor Considerations:** Cash settlements are often preferred as they provide immediate liquidity and avoid the market risk associated with holding the actual shares until the redemption date.
- **Market Preferences:** Retail products are typically settled physically, appealing to investors' behavioral tendencies to hold onto stocks rather than realizing losses.

13.4.3 Issue Minimum/Maximum Size and Liquidity Considerations

13.4.3.1 Minimum Size

- **Cost Implications:** The issuer will only launch a product if it can cover fixed and variable costs such as listing fees, back-office operations, and computational expenses.
- **Investor Strategy:** Prefer products from issuers with lower operational costs, as these savings can be passed on in the product's pricing.

13.4.3.2 Maximum Size

- **Risk Management:** Issuers limit the size of a product based on their capacity to hedge the associated risks effectively, particularly for less liquid underlyings.
- **Market Influence:** Large hedging operations can inadvertently affect the market price of the underlying, potentially impacting the performance of the product.

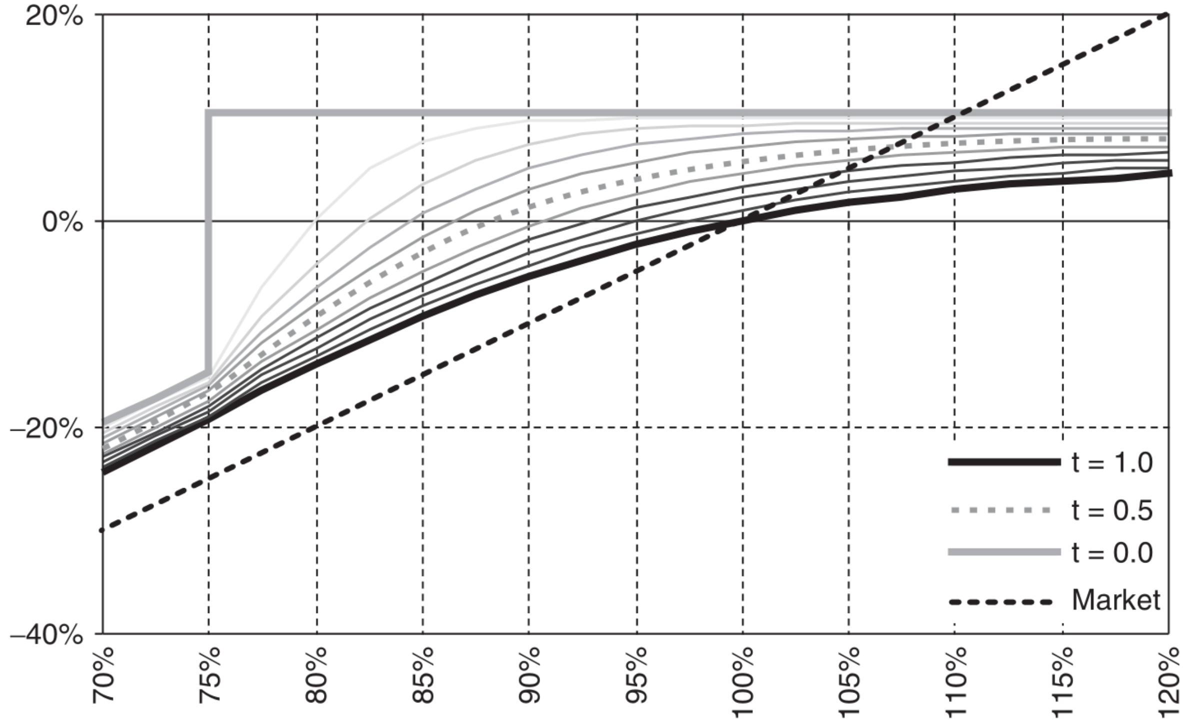


Figure 13.4: Reverse convertible as a function of spot price and time to maturity

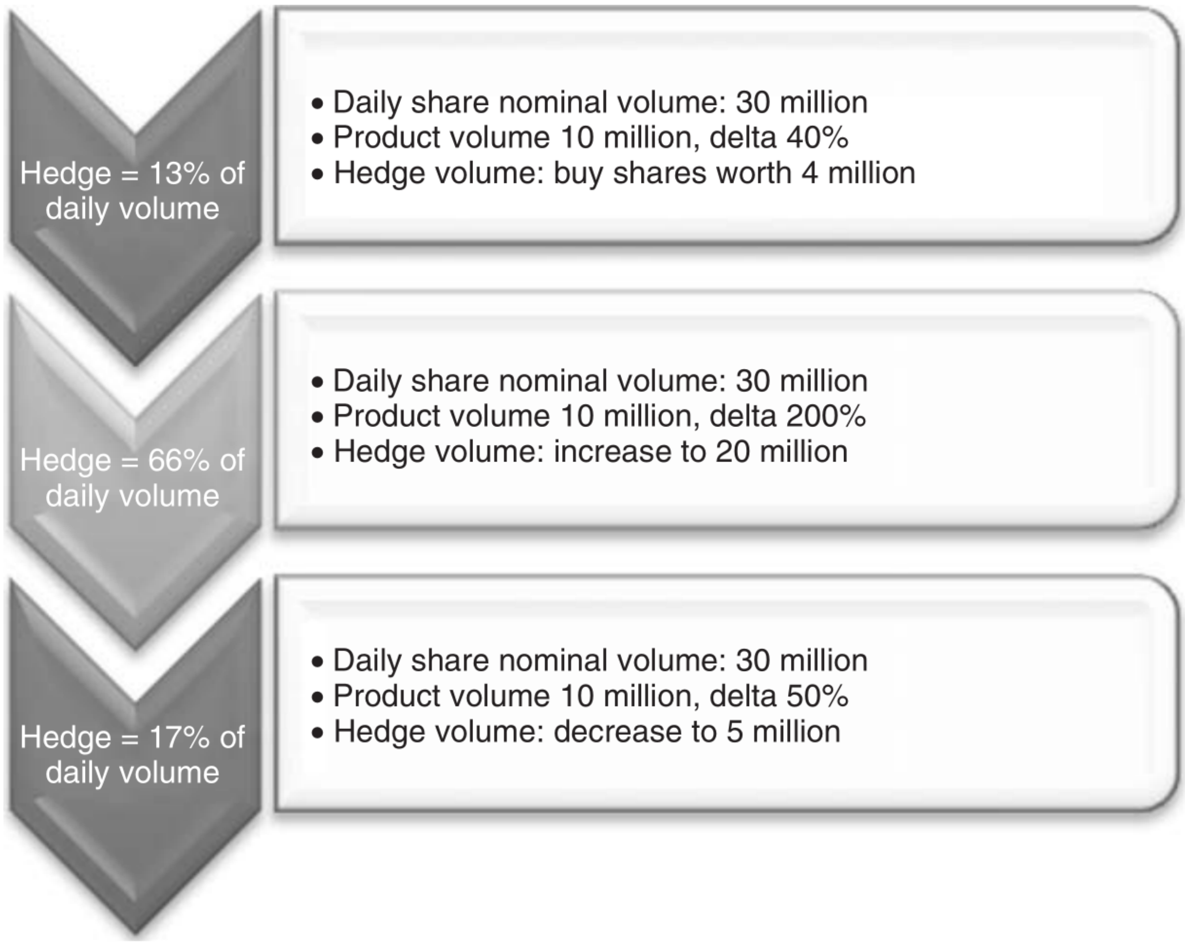


Figure 13.5: Reverse convertible hedging example

13.4.3.3 Liquidity

- **Trading Spreads:** Vary significantly by region and issuer; narrower spreads in markets like Germany enhance trading viability for smaller sizes.
- **Secondary Market:** Liquidity is typically provided by the issuer, making it essential for investors to consider the availability of market quotes, especially if they plan to trade the product before maturity.

13.5 Practice Questions and Problems

1. What is the value of the reverse convertible at maturity if the value of the underlying asset ends at 100, the strike price of the certificate is 80, and the coupon is 12%? The

strike price is equal to the initial spot price of the underlying.

Solution

89.6

2. What is the value of the reverse convertible at maturity if the value of the underlying asset ends at 45, the strike price of the certificate is 60, and the coupon is 9%? The strike price is equal to the initial spot price of the underlying.

Solution

50.4

3. What is the value of the barrier reverse convertible at maturity if the value of the underlying asset ends at 90, the strike price of the certificate is 100, the coupon is 7%, and the knock-out barrier is 70? The lowest value of the underlying asset during maturity was 65. The strike price is equal to the initial spot price of the underlying.

Solution

97

4. What is the value of the barrier reverse convertible at maturity if the value of the underlying asset ends at 90, the strike price of the certificate is 100, the coupon is 7%, and the knock-out barrier is 70? The lowest value of the underlying asset during maturity was 75. The strike price is equal to the initial spot price of the underlying.

Solution

107

5. What is the value of the barrier reverse convertible at maturity if the value of the underlying asset ends at 123, the strike price of the certificate is 100, the coupon is 7%, and the knock-out barrier is 70? The lowest value of the underlying asset during maturity was 65. The strike price is equal to the initial spot price of the underlying.

Solution

107

6. What is the value of the barrier reverse convertible at maturity if the value of the underlying asset ends at 137, the strike price of the certificate is 100, the coupon is 7%, and

the knock-out barrier is 70? The lowest value of the underlying asset during maturity was 75. The strike price is equal to the initial spot price of the underlying.

> **Solution**
>
> 107

7. What is the value of the autocallable note at maturity if the value of the invested capital was 1000, autocall occurs if the underlying price at the anniversary date is higher then initial asset price 4600, and the coupon is 6%? The price of the underlying each year ended at [4583, 4309, 5128, 5630].

> **Solution**
>
> 1180

8. What is the value of the autocallable note at maturity if the value of the invested capital was 1000, autocall occurs if the underlying price at the anniversary date is higher then initial asset price 4600, and the coupon is 4%? The price of the underlying each year ended at [4583, 4309, 3925, 4528].

> **Solution**
>
> 1000

9. What is the value of the autocallable note at maturity if the value of the invested capital was 1000, autocall occurs if the underlying price at the anniversary date is higher then initial asset price 4600, and the coupon is 8%? The price of the underlying each year ended at [4672, 4982, 5023, 5628].

> **Solution**
>
> 1080

10. What is the value of the express certificate at maturity if the value of the invested capital was 500, autocall occurs if the underlying price at the anniversary date is higher then initial asset price 2500, barrier is 1500, and the coupon is 14%? The price of the underlying each year ended at [2403, 2625, 2039, 2901]. The lowest value of the underlying asset during maturity was 1958.

> **Solution**
>
> 640

11. What is the value of the express certificate at maturity if the value of the invested capital was 500, autocall occurs if the underlying price at the anniversary date is higher then initial asset price 2500, barrier is 1500, and the coupon is 14%? The price of the underlying each year ended at [2200, 2350, 2194, 1830]. The lowest value of the underlying asset during maturity was 1820.

Solution

500

12. What is the value of the express certificate at maturity if the value of the invested capital was 500, autocall occurs if the underlying price at the anniversary date is higher then initial asset price 2500, barrier is 1500, and the coupon is 14%? The price of the underlying each year ended at [2200, 1800, 1630, 1775]. The lowest value of the underlying asset during maturity was 1420.

Solution

355

13. What is the value of the express certificate at maturity if the value of the invested capital was 500, autocall occurs if the underlying price at the anniversary date is higher then initial asset price 2500, barrier is 1500, and the coupon is 14%? The price of the underlying each year ended at [2200, 2431, 2700, 2398]. The lowest value of the underlying asset during maturity was 1420.

Solution

710

14. What is the value of the long constant leverage certificate if the value of the underlying asset moves from 100 to 120, and the leverage was 3?

Solution

160

15. What is the value of the long constant leverage certificate if the value of the underlying asset moves from 100 to 90, and the leverage was 5?

Solution

50

16. What is the leverage of the mini-future certificate if the value of the underlying asset is 245 and financing level is 195?

Solution

4.9

17. What is the leverage of the mini-future certificate if the value of the underlying asset is 350 and financing level is 290?

Solution

5.83

A Time Value of Money

References

This material was originaly published HERE by Department of Mathematics, Penn State University Park.

Learning Outcomes:

- Understand the concept of the time value of money (TVM).
- Calculate the present value (PV) of both single and multiple future cash flows using appropriate discount rates.
- Calculate the future value (FV) of both single and ongoing investments using given interest rates.
- Apply TVM formulas to various financial scenarios, including loans, savings, and investments, to make informed decisions.
- Understand the effects of compounding frequency on FV and PV calculations.
- Critically analyze TVM problems, taking into account the impact of rate, time, and cash flows on financial decisions.

A.1 Notation and Terminology

A.1.1 Basic Notation and Terminology

- P = Principal (i.e., value of initial deposit)
- A = Accumulated amount (i.e., sum of the principal and interest)
- r = Nominal interest rate
- m = Number of conversion periods per year, (a conversion period is the interval of time between successive interest payments)

Annually	Semiannually	Quarterly	Monthly	Weekly	Daily
$m = 1$	$m = 2$	$m = 4$	$m = 12$	$m = 52$	$m = 365$

- t = Term of investment (in years)

A.1.2 Simple Interest

Interest is always computed based on the original principal.

Interest Earned	Accumulated Amount
$I = Prt$	$A = P(1 + rt)$

A.1.3 Discrete Compound Interest

Interest payments are added to the principal at the end of each conversion period and therefore earn interest during future conversion periods.

Accumulated Amount	Present Value Formula
$A = P\left(1 + \frac{r}{m}\right)^{mt}$	$P = A\left(1 + \frac{r}{m}\right)^{-mt}$

A.1.4 Continuous Compound Interest

Continuous compounding of interest is equivalent to a discrete compounding of interest where m, the number of conversion periods per year, goes to infinity.

Accumulated Amount	Present Value Formula
$A = Pe^{rt}$	$P = Ae^{-rt}$

A.1.5 Effective Rate of Interest

The effective interest rate, r_{eff}, is the simple interest rate that produces the same accumulated amount in 1 year as the nominal rate, r, compounded m times a year.

$$r_{\text{eff}} = \left(1 + \frac{r}{m}\right)^m - 1$$

A.2 Future Value Examples

A.2.1 Example 1

Suppose \$1,000 is deposited into an account with an interest rate of 16% compounded annually. How much money is in the account after 3 years?

Step 1: Since interest is compounded annually, use the accumulated amount for discrete compound interest.

$$A = P\left(1 + \frac{r}{m}\right)^{mt}$$

Step 2: Plug in the given values: $P = 1000$, $r = 0.16$, $m = 1$, and $t = 3$.

$$A = 1000\left(1 + \frac{0.16}{1}\right)^{1 \cdot 3}$$

$$= 1000\,(1 + 0.16)^3$$

$$= 1000\,(1.16)^3 \approx \$1,560.90$$

Therefore, after 3 years of accumulating interest, the original investment of \$1,000 is worth \$1,560.90.

A.2.2 Example 2

Suppose \$1,000 is deposited into an account with an interest rate of 16% compounded quarterly. How much money is in the account after 3 years?

Step 1: Since interest is compounded quarterly, use the accumulated amount for discrete compound interest.

$$A = P\left(1 + \frac{r}{m}\right)^{mt}$$

Step 2: Plug in the given values: $P = 1000$, $r = 0.16$, $m = 4$, and $t = 3$.

$$
\begin{aligned}
A &= 1000\left(1+\frac{0.16}{4}\right)^{4\cdot 3} \\
&= 1000\,(1+0.04)^{12} \\
&= 1000\,(1.04)^{12} \approx \$1,601.03
\end{aligned}
$$

Therefore, after 3 years of accumulating interest, the original investment of \$1,000 is worth \$1,601.03.

> **Observation**
>
> Compare the accumulated amounts in the above two examples. Both examples have the same principal, interest rate, and term. But since interest is compounded more frequently in Example 2 (4 times a year) than in Example 1 (1 time a year), the accumulated amount is higher in Example 2.

A.2.3 Example 3

Find the interest rate required for an investment of \$3,000 to double in value after 5 years if interest is compounded quarterly.

Step 1: Since interest is compounded quarterly, use the accumulated amount for discrete compound interest.

$$A = P\left(1+\frac{r}{m}\right)^{mt}$$

Step 2: Plug in the given values: $P = 3000$, $A = 6000$ (since the investment is to double in value), $m = 4$, and $t = 5$.

$$
\begin{aligned}
6000 &= 3000\left(1+\frac{r}{4}\right)^{4\cdot 5} \\
&= 3000\left(1+\frac{r}{4}\right)^{20}
\end{aligned}
$$

Step 3: Solve for the interest rate, r.

Method 1

Divide both sides by 3000

$$2 = \left(1 + \frac{r}{4}\right)^{20}$$

Take the natural logarithm of both sides.

$$\begin{aligned} \ln(2) &= \ln\left[\left(1 + \frac{r}{4}\right)^{20}\right] \\ &= 20\ln\left(1 + \frac{r}{4}\right) && \text{since } \ln(m^n) = n\ln(m) \end{aligned}$$

Divide both sides by 20.

$$\ln(2)/20 = \ln\left(1 + \frac{r}{4}\right)$$

Take the exponential of both sides.

$$\begin{aligned} e^{\ln(2)/20} &= e^{\ln(1+\frac{r}{4})} \\ &= 1 + \frac{r}{4} && \text{since } e^{\ln(x)} = x \end{aligned}$$

Subtract 1 from both sides.

$$e^{\ln(2)/20} - 1 = \frac{r}{4}$$

And finally, multiply both sides by 4.

$$r = 4(e^{\ln(2)/20} - 1) \approx 0.1411.$$

Method 2

Here is an alternate method for solving for the interest rate r. We start with the following equation.

$$2 = \left(1 + \frac{r}{4}\right)^{20}$$

Instead of taking the natural logarithm of both sides as we did before, now take the 20th root of both sides (i.e., raise both sides to the power of 1/20).

$$2^{1/20} = 1 + \frac{r}{4}$$

Subtract 1 from both sides.

$$2^{1/20} - 1 = \frac{r}{4}$$

And finally, multiply both sides by 4.

$$r = 4(2^{1/20} - 1) \approx 0.1411$$

Note that this value of r is numerically equal in both methods since

$$\begin{aligned} e^{\ln(2)/20} &= e^{\ln(2^{1/20})} && \text{since } n\ln(m) = \ln(m^n) \\ &= 2^{1/20} && \text{since } e^{\ln(x)} = x \end{aligned}$$

Therefore, an interest rate of approximately 14.11% compounded quarterly is required for an investment of \$3,000 to double in value in 5 years.

A.2.4 Example 4

Find the interest rate required for an investment of \$3,000 to double in value after 5 years if interest is compounded continuously.

Step 1: Since interest is compounded continuously, use the accumulated amount for continuous compound interest.

$$A = Pe^{rt}$$

Step 2: Plug in the given values: $P = 3000$, $A = 6000$ (since the investment is to double in value), and $t = 5$.

$$6000 = 3000e^{5r}$$

Step 3: Solve for the interest rate, r.

Divide both sides by 3000.

$$2 = e^{5r}$$

Take the natural logarithm of both sides.

$$
\begin{aligned}
\ln(2) &= \ln(e^{5r}) \\
&= 5r && \text{since } \ln(e^x) = x
\end{aligned}
$$

Divide both sides by 5.

$$r = \ln(2)/5 \approx 0.1386$$

Therefore, an interest rate of approximately 13.86% compounded continuously is required for an investment of \$3,000 to double in value in 5 years.

> **Observation**
>
> Compare the last two examples. Since continuous compounding of interest earns interest faster than discrete compounding, a lower interest rate is needed for an investment to double in value over a fixed term if interest is compounded continuously. In our examples, an interest rate of 13.86% was needed for the investment with continuous compound interest to double in value in 5 years, while an interest rate to 14.11% was needed for the investment with quarterly compound interest.

A.2.5 Example 5

How long will it take for \$5,000 to grow to \$8,000 if the investment earns interest at 6% per year compounded monthly?

Step 1: Since interest is compounded monthly, use the accumulated amount for discrete compound interest.

$$A = P\left(1 + \frac{r}{m}\right)^{mt}$$

Step 2: Plug in the given values: $P = 5000$, $A = 8000$, $m = 12$, and $r = 0.06$.

$$
\begin{aligned}
8000 &= 5000\left(1 + \frac{0.06}{12}\right)^{12\cdot t} \\
&= 5000\left(1 + 0.005\right)^{12t}
\end{aligned}
$$

Step 3: Solve for the unknown term t.

Divide both sides by 5000.

$$8/5 = 1.005^{12t}$$

Take the natural logarithm of both sides.

$$\begin{aligned} \ln(8/5) &= \ln(1.005^{12t}) \\ &= 12t \ln(1.005) && \text{since } \ln(m^n) = n\ln(m) \end{aligned}$$

Divide both sides by $12\ln(1.005)$.

$$t = \frac{\ln(8/5)}{12\ln(1.005)} \approx 7.85$$

Therefore, it will take approximately 7.85 years for \$5,000 to grow to \$8,000 if the investment earns interest at 6% per year compounded monthly.

A.2.6 Example 6

How long will it take for \$5,000 to grow to \$8,000 if the investment earns interest at 6% per year compounded continuously?

Step 1: Since interest is compounded continuously, use the accumulated amount for continuous compound interest.

$$A = Pe^{rt}$$

Step 2: Plug in the given values: $P = 5000$, $A = 8000$, and $r = 0.06$.

$$8000 = 5000e^{0.06t}$$

Step 3: Solve for the unknown term t.

Divide both sides by 5000.

$$8/5 = e^{0.06t}$$

Take the natural logarithm of both sides.

$$\begin{aligned} \ln(8/5) &= \ln(e^{0.06t}) \\ &= 0.06t && \text{since } \ln(e^x) = x \end{aligned}$$

Divide both sides by 0.06.

$$t = \frac{\ln(8/5)}{0.06} \approx 7.83$$

Therefore, it will take approximately 7.83 years for \$5,000 to grow to \$8,000 if the investment earns interest at 6% per year compounded monthly.

> ⚠ Observation
>
> Compare the last two examples. Both examples have the same principal, accumulated amount, and interest rate. But since continuous compounding of interest earns interest faster than discrete compounding, it should take less time for the investment to grow to \$8,000 if interest is compounded continuously.

A.2.7 Example 7

Find the effective interest rate corresponding to a nominal interest rate of 10% compounded semiannually.

Step 1: Recall the formula for effective interest rate, r_{eff}.

$$r_{\text{eff}} = \left(1 + \frac{r}{m}\right)^m - 1$$

Step 2: Plug in the given values: $r = 0.1$ and $m = 2$.

$$\begin{aligned} r_{\text{eff}} &= \left(1 + \frac{0.1}{2}\right)^2 - 1 \\ &= 1.05^2 - 1 \\ &= 0.1025 \end{aligned}$$

Therefore, an investment earning interest compounded semiannually at 10% earns the same amount of interest after 1 year as an investment earning simple interest at 10.25%.

A.2.8 Example 8

Suppose you have \$12,000 in the bank earning interest at a rate of 12% compounded quarterly. Your cousin calls you and needs \$12,000 to buy a new car. You are willing him to loan him the money, but you'd hate to lose out on the interest you would gather by simply leaving your money alone. If you charge your cousin an interest rate compounded continuously, what rate should you charge in order to earn the same amount of interest you otherwise would have?

Step 1: Assume your cousin is prepared to pay you back after t years.

We'll use t as the term in each of the following calculations. Eventually, we'll see that the interest rate you charge does not depend on the specific value of t.

Step 2: Compute the accumulated amount of the \$12,000 after t years assuming you leave your money in the bank.

$$\begin{aligned} A &= P\left(1+\frac{r}{m}\right)^{m\cdot t} \\ &= 12000\left(1+\frac{0.12}{4}\right)^{4t} \\ &= 12000\left(1.03\right)^{4t} \end{aligned}$$

Step 3: Compute the accumulated amount of the \$12,000 after t years assuming you let your cousin borrow the money.

This would be the amount that your cousin repays you after t years.

$$\begin{aligned} A &= Pe^{rt} \\ &= 12000e^{rt} \end{aligned}$$

Step 4: Equate the two accumulated amounts and solve for r.

$$12000\left(1.03\right)^{4t} = 12000e^{rt}$$

Divide both sides by 12000.

$$1.03^{4t} = e^{rt}$$

Take the natural logarithm of both sides.

$$\ln(1.03^{4t}) = \ln(e^{rt})$$

Simplify using properties of logarithms ($\ln(m^n) = n\ln(m)$ and $\ln(e^x) = x$).

$$4t\ln(1.03) = rt$$

And finally, divide both sides by t. Here is where we see that the time it would take your cousin to repay you does not affect the interest rate you would charge.

$$r = 4\ln(1.03) \approx 0.1182$$

Therefore, charging your cousin 11.82% interest compounded continuously earns the same amount of interest as leaving your money in the bank earning interest at a rate of 12% compounded quarterly.

A.3 Present Value Examples

A.3.1 Example 1

How much money should be deposited in a bank paying a yearly interest rate of 6% compounded monthly so that after 3 years, the accumulated amount will be $20,000?

Step 1: Notice that this is a present value problem since we're given the accumulated amount and we're asked to find the principal. And since interest is compounded monthly, we'll use the present value formula for discrete compounding of interest.

$$P = A\left(1+\frac{r}{m}\right)^{-mt}$$

Step 2: Plug in the given values: $A = 20000$, $r = 0.06$, $m = 12$, and $t = 3$.

$$P = 20000\left(1+\frac{0.06}{12}\right)^{-(12)(3)}$$

$$= 20000(1.005)^{-36} \approx \$16,712.90$$

Therefore, $16,712.90 invested at 6% interest compounded monthly will be worth $20,000 in 3 years.

A.3.2 Example 2

Use the accumulated amount for discrete compound interest to solve the previous example.

Step 1: Start with the formula for accumulated amount for discrete compounding of interest.

$$A = P\left(1 + \frac{r}{m}\right)^{mt}$$

Step 2: Plug in the given values: $A = 20000$, $r = 0.06$, $m = 12$, and $t = 3$.

$$\begin{aligned} 20000 &= P\left(1 + \frac{0.06}{12}\right)^{(12)(3)} \\ &= P(1.005)^{36} \end{aligned}$$

Step 3: Solve for P.

$$P = \frac{20000}{1.005^{36}} \approx \$16,712.90$$

A.3.3 Example 3

Parents wish to establish a trust fund for their child's education. If they need \$170,000 in 7 years, how much should they set aside now if the money is invested at 20% compounded continuously?

Step 1: Notice that this is a present value problem since we're given the accumulated amount and we're asked to find the principal. And since interest is compounded continuously, we'll use the present value formula for continuous compounding of interest.

$$P = Ae^{-rt}$$

Step 2: Plug in the given values: $A = 170000$, $r = 0.2$, and $t = 7$.

$$P = 170,000e^{-(0.2)(7)}$$

$$= 170,000e^{-1.4} \approx \$41,921.48$$

Therefore, \$41,921.48 invested at 20% interest compounded continuously will be worth \$170,000 in 7 years.

A.3.4 Example 4

Use the accumulated amount for continuous compound interest to solve the previous example.

Step 1: Start with the formula for accumulated amount for continuous compounding of interest.

$$A = Pe^{rt}$$

Step 2: Plug in the given values: $A = 170000$, $r = 0.2$, and $t = 7$.

$$\begin{aligned} 170000 &= Pe^{(0.2)(7)} \\ &= Pe^{1.4} \end{aligned}$$

Step 3: Solve for P.

$$P = \frac{170000}{e^{1.4}} \approx \$41,921.48$$

A.4 Try It Yourself

A.4.1 Exercise 1

If \$6,00 is invested at 7% compounded continuously, what will be the accumulated amount after 6 years?

Show answer

$A = 6000e^{0.42}$

A.4.2 Exercise 2

If \$7,000 is invested at 16% compounded quarterly, what will be the accumulated amount after 3 years?

Show answer

$A = 7000(1.04)^{12}$

A.4.3 Exercise 3

Find the interest rate r needed for an investment of \$2,000 to grow to \$8,000 in 7 years if compounded continuously.

Show answer

$r = \ln(4)/7$

A.4.4 Exercise 4

Find the interest rate r needed for an investment of \$7,000 to grow to \$12,000 in 21 years if compounded monthly.

Show answer

$r = 12\left[(12/7)^{1/252} - 1\right]$

A.4.5 Exercise 5

Find the time it would take for an investment of \$1,000 to grow to \$100,000 if interest is compounded quarterly at an annual rate of 8%.

Show answer

$t = \frac{\ln(100)}{4\ln(1.02)}$

A.4.6 Exercise 6

Find the time it would take for an investment of \$2,500 to grow to \$6,000 if interest is compounded continuously at an annual rate of 24%.

Show answer

$t = \frac{25}{6}\ln(12/5)$

A.4.7 Exercise 7

Calculate the effective rate of interest corresponding to a nominal interest rate of 52% compounded weekly.

Show answer

$r_{eff} = 1.01^{52} - 1$

A.4.8 Exercise 8

Your grandma would like to establish a trust fund for your education. How much should she set aside now if she wants $50,000 in 9 years and interest is compounded monthly at an annual rate of 12%?

Show answer

$P = 50000(1.01)^{-108}$

A.4.9 Exercise 9

You are preparing to run for president and want to have $100,000 in 6 years to start your campaign. How much money do you need now if interest is compounded continuously at an annual rate of 15%?

Show answer

$P = 100000e^{-0.9}$

A.4.10 Exercise 10

You have $50,000 in the bank earning 7% interest compounded quarterly. However, your cousin needs a $50,000 investment to start up his new financial consulting business. In order to get the same total return as leaving your money in the bank, what interest rate r should you request from your cousin if interest is compounded continuously?

Show answer

$r = 4\ln(1 + 0.07/4)$

B Interest Rates

> **References**
>
> - HULL, John. Options, futures, and other derivatives. Ninth edition. Harlow: Pearson, 2018. ISBN 978-1-292-21289-0.
>   - Chapter 4 - Interest Rates

Learning Outcomes:

- Understand the different types of interest rates.
- Define the risk-free rate and its significance in financial derivatives.
- Explain the concept of continuous compounding and its importance in pricing financial derivatives.

B.1 Types of Rates

B.1.1 Treasury Rate

- Rate on instrument issued by a government in its own currency.

B.1.2 The U.S. Fed Funds Rate

- Unsecured interbank overnight rate of interest.
- Allows banks to adjust the cash (i.e., reserves) on deposit with the Federal Reserve at the end of each day.
- The effective fed funds rate is the average rate on brokered transactions.
- The central bank may intervene with its own transactions to raise or lower the rate.
- Similar arrangements in other countries.

B.1.3 Repo Rate

- Repurchase agreement is an agreement where a financial institution that owns securities agrees to sell them for X and buy them bank in the future (usually the next day) for a slightly higher price, Y.
- The financial institution obtains a loan.
- The rate of interest is calculated from the difference between X and Y and is known as the repo rate.

B.1.4 LIBOR (ICE LIBOR)

- Detailed information about LIBOR: https://www.theice.com/iba/libor
- LIBOR is the rate of interest at which a AA bank can borrow money on an **unsecured** basis from another bank.
- Based on **submissions** from a panel of contributor banks (16 for each of USD and GBP).
- It is calculated daily for 5 currencies and 7 maturities.
- There have been some suggestions that banks manipulated LIBOR during certain periods.
- Why would they do this?

B.2 Alternative Reference Rates

Country/Currency/CODE	IBOR Rate	New Reference Rate
USA/Dollars/USD	USD ICE LIBOR	SOFR
UK/Pounds Sterling/GBP	GBP ICE LIBOR	SONIA
Switzerland/Swiss Francs/CHF	CHF ICE LIBOR	SARON
Japan/Yen/JPY	JPY ICE LIBOR, Tibor	TONAR
EU/Euro/EUR	Euribor	ESTER

B.2.1 SOFR (Secured Overnight Financing Rate)

- CME Group Education
- Administered by Federal Reserve Bank of New York (link)
- Transaction-based, calculated from overnight US Treasury repurchase (repo) activity.
- SOFR is a broad measure of the cost of borrowing USD cash overnight, collateralized by U.S. Treasury securities.
- SOFR is a good representation of general funding conditions in the overnight Treasury repo market.

- As such, it will reflect an economic cost of lending and borrowing relevant to the wide array of market participants active in the market.

B.2.2 SONIA (Sterling Overnight Index Average)

- CME Group Education
- Administered by Bank of England (link)
- Unsecured transaction-based index, wholesale based (beyond Interbank)
- It has been endorsed by the Sterling Risk-Free Reference Rate Working Group (Working Group) as the preferred risk-free reference rate for Sterling Overnight Indexed Swaps (OIS).
- In January 2018, the Working Group added banks, dealers, investment managers, non-financial corporates, infrastructure providers, trade associations and professional services firms.
- In April 2018, the BOE introduced a series of reforms of the SONIA benchmark.

B.2.3 €STR (or ESTER, Euro Short-Term Rate)

- Administered by European Central Bank (link)
- It is based on the unsecured market segment.
- The ECB developed an unsecured rate, because it is intended to complement the EONIA.
- Furthermore, a secured rate would be affected by the type of the collaterals.
- The money market statistical reporting covers the 50 largest banks in the euro area in terms of balance sheet size.
- While the EONIA (link) reflects the interbank market, the €STR extends the scope to money market funds, insurance companies and other financial corporations because banks developed significant money market activity with those entities.

B.3 OIS Rate

- An **overnight indexed swap** is swap where a fixed rate for a period (e.g. 3 months) is exchanged for the geometric average of overnight rates (or overnight rate compounded over the term of the swap).
- The underlying floating rate is typically the rate for overnight lending between banks, either non-secured or secured (SOFR, SONIA, €STR).
- For maturities up to one year there is a single exchange (swap term is not overnight).
- For maturities beyond one year there are periodic exchanges, e.g. every quarter.
- The OIS rate is a continually refreshed overnight rate.
- The fixed rate of OIS is typically an interest rate considered less risky than the corresponding interbank rate (LIBOR) because there is limited counterparty risk.

B.3.1 The Risk-Free Rate

- The Treasury rate is considered to be artificially low because:
  - Banks are not required to keep capital for Treasury instruments
  - Treasury instruments are given favorable tax treatment in the US
- OIS rates are now used as a proxy for risk-free rates in derivatives valuation.

B.4 Time Value of Money

B.4.1 Compounding Frequency

- When we compound m times per year at rate r an amount P grows to $P(1+r/m)^m$ in one year.
- The compounding frequency used for an interest rate is the unit of measurement.
- The difference between quarterly and annual compounding is analogous to the difference between miles and kilometers.
- Effect of the compounding frequency on the value of $100 at the end of 1 year when the interest rate is 10% per annum.

Compounding frequency	Value of $100 at end of year ($)
Annually m = 1	110.00
Semiannually m = 2	110.25
Quarterly m = 4	110.38
Monthly m = 12	110.47
Weekly m = 52	110.51
Daily m = 365	110.52

B.4.2 Continuous Compounding

- **Rates used in option pricing are nearly always expressed with continuous compounding.**
- In the limit as we compound more and more frequently we obtain continuously compounded interest rates.
- Notation:
  - r: continuously compounded annual interest rate
  - T: time to maturity in years
  - e: Euler's number (mathematical constant)

$$\text{Future value} = P \times e^{rT}$$

$$\text{Present value} = P \times e^{-rT}$$

- USD 100 grows to $100 \times e^{rT}$ when invested at a continuously compounded rate r for time T.
- USD 100 received at time T discounts to $100 \times e^{-rT}$ at time zero when the continuously compounded discount rate is r.

B.4.3 Conversion Formulas

- r_c: continuously compounded rate
- r_m: same rate with compounding m times per year

$$r_c = m \ln(1 + \frac{r_m}{m})$$

$$r_m = m(e^{r_c/m} - 1)$$

Examples:

- 10% with semiannual compounding is equivalent to $2\ln(1.05) = 9.758\%$ with continuous compounding.
- 8% with continuous compounding is equivalent to $4(e^{0.08/4} - 1) = 8.08\%$ with quarterly compounding.